3GPP TR 22.840 V19.0.0 (2023-12)

Technical Report

3rd Generation Partnership Project; Technical Specification Group Services and System Aspects; Study on Ambient power-enabled Internet of Things (Release 19)





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Keywords
Ambient IoT

3GPP

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Foreword

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Introduction

The present document describes use cases, traffic scenarios, device constraints of Ambient Internet of Things and identify new potential service requirements as well as new KPIs. An Ambient Internet of Things (Ambient IoT) device is an IoT device, being either battery-less or with limited energy storage capability (i.e., using a capacitor) and the energy is provided through the harvesting of radio waves, light, motion, heat, or any other suitable power source.

1 Scope

The present document provides Stage 1 potential 5G service requirements for ambient power-enabled Internet of Things (i.e., Ambient IoT). In the context of the present document, an Ambient power-enabled IoT device is an IoT device powered by energy harvesting, being either battery-less or with limited energy storage capability (e.g., using a capacitor) and the energy is provided through the harvesting of radio waves, light, motion, heat, or any other suitable power source. An ambient IoT device is expected to have lower complexity, smaller size and reduced capabilities and lower power consumption than previously defined 3GPP IoT devices (e.g., NB-IoT/eMTC devices). Ambient IoT devices can be maintenance free and can have long life span (e.g., more than 10 years).

The aspects addressed in the present document include:

- Study use cases of ambient power-enabled IoT and identify potential service requirements, including:
    - Security aspects, e.g., authentication and authorization, etc.
    - Network selection, access control, connection, mobility and identification management
    - Charging (e.g., per data volume, per message)
    - Aspects related to stakeholder models (e.g., involving interactions in PLMNs, NPNs or other parties)
    - Positioning
    - Aspects on device life cycle management related to 3GPP system.
- Study traffic scenarios, device constraints (e.g., power consumption) and identify potential performance requirements and KPIs
- Gap analysis between the identified requirements for ambient power-enabled IoT and what is already defined by existing 3GPP requirements.

Note: How Ambient IoT device performs energy harvesting is out of scope of this technical report.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
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3 Definitions, symbols and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

Ambient IoT device: An ambient power-enabled Internet of Things device is an IoT device powered by energy harvesting, being either battery-less or with limited energy storage capability (e.g., using a capacitor).

Store-and-forward communication: in the context of this study, store-and-forward communication is an operation mode of a 5G system where a single UE can collect and store information from an Ambient IoT device before forwarding this information to the network.

Ambient IoT Direct network communication: represents communication between the Ambient IoT device and 5G network with no UE conveying information between the Ambient IoT device and the 5G network.

Ambient IoT Indirect network communication: represents communication between the Ambient IoT device and the 5G network where there is an Ambient IoT capable UE helping in conveying information between the Ambient IoT device and the 5G network.

Ambient IoT device to UE direct communication: represents communication between an Ambient IoT device and an Ambient capable UE with no network entity in the middle.

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

LPWA Low Power Wide Area

4 Overview

In today's IoT networks, the IoT devices are usually powered by conventional batteries with a limited lifespan. The usage of conventional batteries has influenced the way these IoT devices are deployed and used. The astronomical growth of IoT network together with the deployment of huge numbers of IoT devices, has pushed up the maintenance costs, including both labor and battery costs. Large numbers of conventional batteries have been disposed of every year and only a small part of them can be efficiently recycled. In some extreme environmental conditions, maintaining the operation of IoT devices and replacing the batteries can be quite challenging. In this regard, battery-free IoT devices are proposed and it as they have the potential to improve the network performance and sustainability, and expand the application scenarios. By removing the conventional battery, the device size and cost can be significantly reduced, thus paving the way to a variety of new applications.

In the 5G era, various LPWA technologies such as eMTC, NB-IoT andRedCap. have been developed to fulfil the increasing demand from verticals. These LPWA technologies have achieved low cost, low power and massive connections and can meet requirements of many applications. However, there are still many use cases and applications that cannot be addressed. For example, conventional battery-powered device cannot be deployed in extreme environmental conditions (e.g. high pressure, extremely high/low temperature, humid environment). Also, the use of conventional battery devices can be limited where maintenance-free devices are required (e.g. where the devices are inaccessible and it is not possible to replace the device battery). Finally, ultra-low complexity, very small device size/form factor (e.g. thickness of mm), longer life cycle, etc. are required for mass market use cases.

Ambient power enabled IoT is a promising technology to fulfil the unmet market requirements stated above. An Ambient power-enabled IoT device is an IoT device powered by energy harvesting, being either battery-less or with limited energy storage capability (e.g. using a capacitor) and the energy is provided through the harvesting of radio waves, light, motion, heat, or any other suitable power source. This study targets at describing the use cases and potential requirements in support of ambient IoT devices with lower complexity, smaller size, reduced capabilities and lower power consumption than previously defined 3GPP IoT devices (e.g. NB-IoT/eMTC devices) with an emphasis on improving network efficiency. Ambient IoT devices can be maintenance free and can have long life span (e.g. more than 10 years).

Therefore, a new kind of IoT service for the verticals will be enabled by combining ambient power-enabled IoT with cellular networks, vastly benefitting the 3GPP ecosystem.

5 Use cases

5.1 Use case on Ambient IoT on automated warehousing

5.1.1 Description

The automated warehouse inventory scenario includes multiple stages, as shown in the figure below, which are divided into verification and unloading, gate-in inventory, inventory, gate-out inventory and check & loading. Along with the transfer, storage and inventory of goods, a large amount of warehousing information will be generated. This information

generally has the characteristics of frequent data read operations and large data volumes. Ambient IoT devices are attached to items of different values and usage, such as pallet containers and individual product, and relevant communication equipment is deployed. Through the information interaction between communication equipment and tags, accurate and rapid inventory and efficient management of storage information can be realized in each stage.

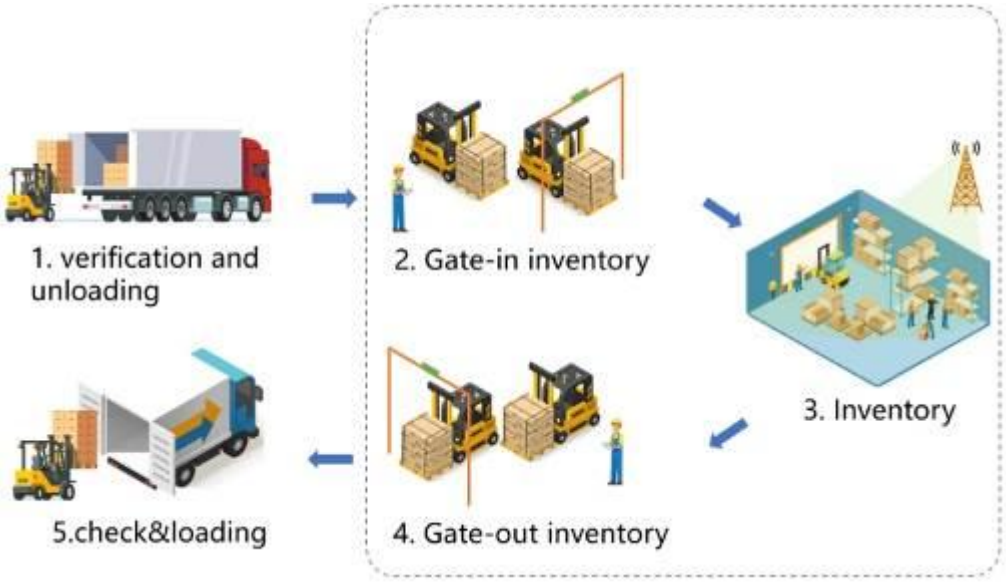


Figure 5.1.1-1: Automated warehouse inventory

5.1.2 Pre-conditions

The 5G network equipment used for inventory is deployed in the warehouse according to the needs of automated warehouse inventory scenario:

- Ambient IoT devices containing contain the assigned information, which can be read and written by the 5G network, get attached to different warehouse items, such as pallet containers, storage racks, forklift and individual product;

- The warehouse management platform is a trusted 3rd party and has subscribed the Ambient IoT services and could have interaction with the 5G network with necessary inventory information;

5.1.3 Service Flows

When some goods in one batch are delivered to a gate of the warehouse and are ready for the gate-in operation, the management platform selects a gate inventory mode.

There are two gate inventory modes supported by both the management platform and the 5G system.

- Manual-Triggered Mode: The inventory task is triggered by the command sent from the management platform, so the 5G system simply execute the command (e.g., start the inventory task).

- Auto-Triggered Mode: The inventory management system set the gate inventory mode to “Auto-Triggered”, the 5G network send discovery signal periodically to discover Ambient IoT devices within the gate area; if at least one Ambient IoT device is discovered, then the 5G system start inventory task automatically.

The goods arrive near the door of the warehouse, 5G network will perform gate inventory procedure according to the trigger from the management platform.

1. Ambient IoT devices establish communication with the 5G network, and send their own identity and corresponding goods information to the 5G network.
2. The 5G network sends the obtained information to the management platform;
3. The management platform generates a list of gate-in inventory results according to the inventory data and necessary information received from Ambient IoT devices by the continually inventory operation.

4. When the goods are placed on shelves in the warehouse and have been stored for a certain period of time, the management platform starts the indoor inventory task periodically for double check of the goods (total goods or per different batch/group), and generates a list of tags to be inventoried, sends the list to the 5G network;
5. The 5G network receives the list and sends large-scale/specified inventory signals;
6. The Ambient IoT device in the 5G network coverage establish communication with the network, and the 5G network interacts with the corresponding devices according to the inventory requirements to obtain goods information;
7. The 5G network sends the acquired goods information to the management platform;
8. The management platform summarizes the received information and generates a list of inventory results.
9. When the goods are ready for shipping and arrive the gate area, similar to gate-in operation, 5G network perform gate inventory procedure according to the platform's selection, interact with Ambient IoT devices and send the obtained information to the management platform.
10. The management platform generates a list of gate-out inventory results.

5.1.4 Post-conditions

With the support of 5G system, automated warehousing could be realized to improve the efficiency of goods management. If the inventory result list is consistent with the purchase/shipment list, the administrator can obtain the inventory result list, and update the inventory list in the management platform; If the result is inconsistent with the purchase or shipment order, the administrator would receive a notification/warning.

5.1.5 Existing features partly or fully covering the use case functionality

In previous releases, SA1 has finished several studies about IoT topic to introduce SA1 requirements in TS 22.011[9], TS 22.278[7], TS 22.368[6] and TS 22.261[8] to address requirement for IoT business about device lifetime, power consumption, data transmission and communication mechanism.

5.1.6 Potential New Requirements needed to support the use case

[P.R.5.1.6-001] The 5G system shall be able to support communication with Ambient IoT device which is battery-less or with limited energy storage (e.g., capacitor).

[P.R.5.1.6-002] The 5G system shall support to provide collected information from Ambient IoT devices to the trusted 3rd party.

[P.R.5.1.6-003] The 5G system shall support suitable security mechanisms for Ambient IoT devices, including encryption and data integrity.

[P.R.5.1.6-004] The 5G system shall be able to support the authentication and authorization mechanisms of Ambient_IoT devices.

[P.R.5.1.6-005] The 5G system shall be able to manage (e.g. provide service parameters, activate, deactivate) multiple Ambient IoT devices in bulk.

[P.R.5.1.6-006] The 5G system shall be able to provide Ambient IoT service with the following KPIs

Table 5.1.6-1: KPI Table of Ambient IoT for automated warehousing

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Automated warehousing	1s (note 3)	99 %	NA	<100/128bits/s (note 4)	96/128 bits (note 1)	$1\sim2/m^2$ [NA]	30m indoors	NA	5~10km/h	NA	NA	NA	2~3 m (note 2)
Note 1: Message size refers to the Ambient IoT device identifier used for goods identification in this use case; Note 2: Three-dimensional positioning (both horizontal and vertical) is considered; Note 3: End to end latency refers to the time taken for an Ambient IoT device to transmit the message; Note 4: User-experienced data rate is calculated as the message size (96/128bits) transmitted within 1s time period;													

5.2 Use case on medical instruments inventory management and positioning

5.2.1 Description

More and more medical instruments are utilized in hospital. They always need to be cleaned and sterilized, and shall withstand certain conditions e.g., high temperature, high pressure or humidity. Traditional inventory management for the medical instrument is usually operated manually, which is inefficient and even in some cases, causes serious accident e.g., lost or invalidity. To improve safe and efficient utilization of the medical instrument, remote medical instrument inventory and online maintenance are being developed.

For the remote inventory and online maintenance, the medical instrument is needed to be supplied with Ambient IoT device. Considering the working condition of the medical instrument, this kind of Ambient IoT device should be battery-less or with limited energy storage capability, maintenance-free and should have long service life time. Through 5G network and the IoT device, the medical instrument exchange inventory management and positioning information with medical instrument management platform.

Following is an example of service flow to illustrate an inventory and positioning operation for medical instruments.

5.2.2 Pre-Conditions

Network operator UU deploys a new service “Ambient IoT” through its 5G system. Hospital Z is subscribed to the new inventory management service for its orthopaedic instruments (e.g. orthopaedic knives, orthopaedic scissors, orthopaedic forceps, orthopaedic hooks, orthopaedic needles, orthopaedic scrapers, orthopaedic cones, orthopaedic drills, orthopaedic saws, orthopaedic chisels, orthopaedic files / shovels, orthopaedic active instruments, etc)

A number of Ambient-IoT devices recording different orthopaedic instrument information are stuck on these orthopaedic instruments. They are usually indoor stored in the instrument warehouse or medical instrument storage room or special storage cabinet.

The Ambient-IoT device is battery free or with limited energy storage capability.

These Ambient-IoT devices attached in the orthopaedic instrument are with very simple capability and not applications installed on them.

5.2.3 Service Flows

Belle is a nurse of Hospital Z. She has the authorization to remotely manage orthopaedic instrument through the inventory management platform of the hospital. She can operate this work in the hospital or out of the hospital.

1. Belle wants to acquire the inventory information of orthopaedic forceps. She uses her phone to send inventory request to the inventory management platform of the hospital. The platform requires the 5G network to collect inventory information of orthopaedic forceps in predefined duration, e.g., 10 seconds.

2. The 5G network then sends signals to the Ambient-IoT devices in Hospital Z. After receiving the signal, the Ambient IoT devices are active and can receive messages from 5G network.

3. The 5G network can help to transmit the “read” command from the inventory management platform transparently or to translate the command to ask the Ambient-IoT devices attached on the orthopedic forceps to report its status information which can be the serial number of the orthopaedic forceps (16 bits), usage status (2 bits), usage records (128 bits), years of use (6 bits), number of usage (18 bits), maintained status (2 bits), being stored at the predefined or indicated physical addresses of the Ambient IoT device.

4. Ambient-IoT devices on the orthopaedic forceps report the inventory information stored at the corresponding physical address to the inventory management platform of the hospital via 5G network.

5. The inventory management platform sends the inventory information to Belle's phone.

6. After receives the status list of the orthopaedic forceps, Belle selects a pair of orthopaedic forceps from the list of orthopaedic instruments and clicks on “positioning request”. The request is passed to the inventory management platform. The inventory management platform asks 5G network to collect the position of the selected orthopaedic forceps.

7. The selected orthopaedic forceps is now being transported in the hospital handcart from Hospital Outpatient building to Hospital Inpatient building. It’s moving speed is less than 6km/h.

8. The 5G network calculates the position of the selected orthopaedic forceps after receives response of the Ambient-IoT device attached on the orthopaedic forceps.

9. The 5G network then delivers the position information to the inventory management platform.

10. The platform responds the position information to Belle’s phone.Thus, Belle acquires the position of the selected orthopaedic forceps on her phone.

5.2.4 Post-Conditions

Hospital Z utilizes Ambient-IoT service to support the remote inventory management of medical instrument. Belle can read the information of medical instrument. She can also find a medical instrument through the positioning information provided by Ambient IoT service.

5.2.5 Existing features partly or fully covering the use case functionality

None

5.2.6 Potential New Requirements needed to support the use case

[PR 5.2.6-001] The 5G system shall be able to communicate with an Ambient-IoT device.

[PR 5.2.6-002] The 5G system shall be able to provide group communication for a group of Ambient-IoT devices.

[PR 5.2.6-003] The 5G system shall be able to provide a mechanism to expose the information collected from an Ambient-IoT device to a trusted 3rd party.

[PR 5.2.6-004] The 5G system shall be able to support positioning for an Ambient-IoT device.

[PR 5.2.6-005] The 5G system shall be able to provide communication service with KPIs listed in Table 5.2.6-1 for the Ambient IoT device/s.

Table 5.2.6-1: KPIs for use case of Medical Instrument Inventory management

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Medical instrument inventory management and positioning	Several seconds	99 %	NA	<2kbit/s (note 1)	176bit	≥1000/km^2 (note 2)	50m indoor 200m outdoor	NA	Stationary or walking speed <6 km/h	NA	NA	NA	3 m to 5 m indoor
Note 1: User experienced data rate is calculated based on inventory information (176 bits) within time period of e.g. 100 ms; Note 2: It refers typical medical instrument density condition in Chinese hospital.													

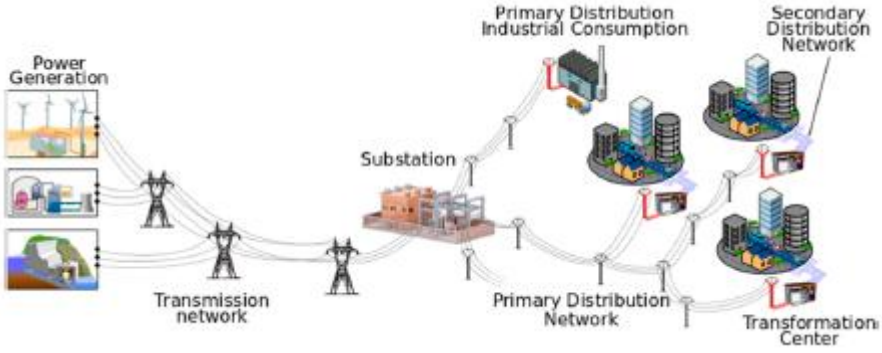
5.3 Use Case on Ambient IoT devices in substations in smart grids

5.3.1 Description

With around 80 million kilometres of transmission and distribution lines worldwide, electricity networks are the backbone of secure and reliable power systems. As stated in the World Energy Outlook 2020 [2], significant investment takes place in new network capacity between 2019 and 2030 as a result of growing demand for electricity, the addition of new renewable generation capacity and the need to develop smart grids. The expansion of electricity networks to 2030 is about 80% more than over the past decade. Around 30% of the increase in transmission lines and 20% of the increase in distribution network lines are attributable to the increase of renewables. Over the next ten years, around 16 million km of existing distribution lines and 1.5 million km of transmission lines need to be replaced or digitalised, together with switching equipment, transformers, meters and other crucial components. In regions with older power systems, such as the United States and the European Union, roughly one-fifth of current networks need to be replaced or digitalised; this corresponds to 2.7 million km in the United States and 3.7 million km in the European Union. More than 60% of global line replacements and new lines are in emerging market and developing economies, with China alone accounting for a third of what is needed (over 7 million km).

Smart grids with wide use of IoT devices have a vital role to play in supporting the penetration of variable renewables electricity sources. IoT offers a wide number of applications in the energy sector, i.e., in energy supply, transmission and distribution, and demand [3]. In particular substations are a significant part of the electrical power grid. Through these stations, the voltage level is converted from high voltage to low voltage using (transformer). The substation transfers power to distribution stations by the transmission lines (see (a) in Figure 5.3.1-1). Monitoring electrical substations are necessary to detect faults and treat them, because if left unattended, it may lead to electrical problems and cause long-term consequences. These problems not only cause energy losses but also lead to electrical outages and losses in expensive equipment, in addition to injuries and accidents such as fire. Therefore, monitoring of substations and their equipment is important to ensure safety, protection, and stability in the electric power networks. Different types of sensors (e.g.

temperature and humidity sensors) can be used in the outdoor ultra-high voltage substation (see (b) in Figure 5.3.1-1) to detect the anomaly and trigger predictive maintenance. In addition, various sensors can be used in other use cases in the power transmission and distribution networks (see (c) to (e) in figure F.3.1-1) for remote monitoring and protection purposes.



Overview



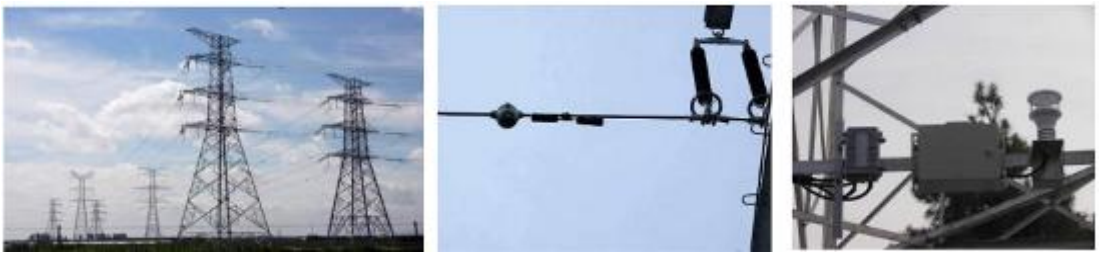
Outdoor ultra-high voltage substation
(Possible issues such as poor contact can cause faulty service.)



Indoor/outdoor shielding cabinet
(Problems such as poor wire connection can cause short circuit or even fire.)



Underground transmission and distribution lines
(Possible issues include that the underground cables can be cut or bitten by rats, and can be damaged due to the faulty drainage system.)



Aerial transmission and distribution lines
(Possible issues include that the aerial cables can be vibrated or displaced, and difficult to provide power sources for sensors to monitor these parameters.)

Figure 5.3.1-1: Transmission and distribution networks in smart grids

For these use cases, the data acquisition process is typically not latency-critical, but a large number of sensors have to be efficiently connected, especially considering many of these sensors have limited power source and relatively frequent data transmission (every 5-15 minutes) is expected in some cases (e.g., sensor data is collected once per several seconds for critical equipment monitoring and protection). Moreover, lifespans of the field IoT devices are expected to be one decade or longer, which is one of the main differences compared with consumer products. Many production systems are subject to regulatory approvals (e.g., safety certification), changes to a running production system have often to be avoided. Often sensors are deployed in locations that are inaccessible, where physical replacement would be unduly expensive. Research continues to develop efficient communication techniques to meet the requirements, among which Ambient IoT [4] is very promising to enable wireless communication with minimum energy consumption. The Ambient IoT devices typically are battery-less or with limited energy storage capability, and obtain energy from the environment. The communication power consumption of such Ambient IoT devices are expected to be a few hundred μW [80] [81] [82] [83]. Moreover, communication service availability with sufficient 5G coverage is important especially for remote monitoring of the critical equipment.

5.3.2 Pre-conditions

In this use case GreenGrid has service level agreement with GreenMobile to deploy 5G network to enable the communication of these Ambient IoT devices with the network. As part of the service level agreement, GreenMobile provides energy efficient communication and management services to GreenGrid including:

- interfacing with GreenGrid's grid monitoring and management platform;

- ensuring the lifespan of the Ambient IoT devices of 10-15 years;

- providing energy efficient device management for the Ambient IoT devices based on the instructions from the grid monitoring and management platform;

- providing energy efficient operation (e.g. inventory, read, write) the Ambient IoT devices based on the instructions from the grid monitoring and management platform;

- providing sufficient positioning information of the Ambient IoT devices;

- providing energy efficient security mechanisms for the communication between Ambient IoT devices and the network.

GreenGrid has installed wireless sensors, one form of Ambient IoT devices, in the outdoor ultra-high voltage substations of their power transmission and distribution networks to monitor the corresponding parameters to detect malfunctioning and broken elements in the surrounding environment. This environment is typically monitored using various types of sensors such as temperature sensors, humidity sensors, pressure sensors and vibration sensors. Remote monitoring along with a classification of the anomaly can help with predictive maintenance. For example, the sensor data (e.g. temperature, humidity) can be used to detect high-temperature problems (which is a typical indication of poor connection), excessive humidity (which could be due to floods), etc.

5.3.3 Service Flows

1. The 5G core network receives the request from the application function (in this case GreenGrid's grid monitoring and management platform) to operate on the Ambient IoT devices in a certain area. The 5G core network starts

to operate on these devices accordingly. Once detecting the signals from the 5G network these Ambient IoT devices can respond to the command.

2. These Ambient IoT devices send the identification information to the 5G core network and complete the authentication procedure.

3. The Ambient IoT devices (wireless sensors) measure the environmental parameters, such as temperature, humidity, pressure and vibration.

   For temperature, humidity and pressure measurement, typical sampling rate is 10 Hz with sample size of 32 bits, thus the data generation per Ambient IoT device is about 320 bit/s. For vibration measurement, typical sampling rate is 10 Hz with sample size of 96 bits, thus the data generation per Ambient IoT device is about 960 bit/s.

4. The 5G core network, based on the requests issued by the application function, performs operations such as "inventory", "read" and "write" on the Ambient IoT devices correspondingly. "Inventory" operation is to read the Ambient IoT device identifier. "Read" operation is to read sensor data. "Write" operation can be used to configure the Ambient IoT device.

5. The 5G core network then sends the results of the operations to the application function. The application function includes diagnostic functions that analyze the sensor data and detect the anomaly and trigger predictive maintenance when necessary. Typically, the sensor data is collected once per several seconds.

6. The 5G system is also expected to provide positioning service for these Ambient IoT devices, which can be for device management purpose (e.g., to automatically identify the locations of the Ambient IoT devices) or for security reason (e.g., to make sure the devices have not been removed). The positioning for these purposes is not expected to be performed frequently.

5.3.4 Post-conditions

The 5G system enables efficient communication, with enhanced security and tens of meter-level positioning accuracy, for the Ambient IoT devices installed in the power transmission and distribution networks for remote monitoring and protection purposes.

5.3.5 Existing features partly or fully covering the use case functionality

Service requirements for MTC (Machine-Type Communications) have been captured in TS 22.368 [6] since release 10, which specifies the service requirements for network improvements. In addition to the common service requirements, specific service requirements have also been defined corresponding to the following MTC Features:

- *Low Mobility;*
- *Time Controlled;*
- *Small Data Transmissions;*
- *Infrequent Mobile Terminated;*
- *MTC Monitoring;*
- *Secure Connection;*
- *Group Based MTC Features:*
  - *Group Based Policing;*
  - *Group Based Addressing.*

Resource efficiency is one of the key service requirements for IoT, for which there are a few requirements specified in 3GPP TS 22.278 [7] and 3GPP TS 22.261 [8]. The target IoT scenarios have been described as

> *As sensors and monitoring UEs are deployed more extensively, the need to support UEs that send data packages ranging in size from a small status update in a few bits to streaming video increases. A similar need exists for smart phones with widely varying amounts of data. Specifically, to support short data bursts, the network should be able to operate in a mode where there is no need for a lengthy and high overhead signalling procedure before and after*

small amounts of data are sent. The system will, as a result, avoid both a negative impact to battery life for the UE and wasting signalling resources.

The related service requirements are kept at a relatively general level, e.g.

The 5G system shall minimize control and user plane resource usage for data transfer from send only UEs.

The 5G system shall minimize control and user plane resource usage for stationary UEs (e.g. lower signalling to user data resource usage ratio).

The 5G system shall minimize control and user plane resource usage for transfer of infrequent small data units.

The 5G system shall optimize the resource use of the control plane and/or user plane for transfer of small data units.

Additional consideration needs to be given in support of Ambient IoT devices that are battery-less or with limited energy storage capability, which present new challenges to the 5G system.

5.3.6 Potential New Requirements needed to support the use case

[PR.5.3.6-001] The 5G system shall support energy efficient communication mechanisms (i.e. minimizing the overall and peak device communication power consumption) for Ambient IoT devices.

[PR 5.3.6-002] The 5G system shall be able to support energy efficient security mechanisms for Ambient IoT devices, including authentication, encryption and data integrity.

[PR 5.3.6-003] The 5G system shall support a mechanism to interface a 3rd party application to manage and operate on the Ambient IoT devices.

[PR.5.3.6-004] The 5G system shall be able to collect charging information for using Ambient IoT services on per Ambient IoT device basis (e.g., total number of communications per charging period).

[PR 5.3.6-005] The 5G system shall be able to collect charging information per application for using Ambient IoT services (e.g., total number of Ambient IoT devices per charging period).

[PR 5.3.6-006] The 5G system shall provide the network connection to address the KPIs for the use of Ambient IoT devices in substations in smart grids, see table 5.3.6-1.

Table 5.3.6-1: Potential key performance requirements for the use of Ambient IoT devices in substations in smart grids

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy

Remote monitoring of transmission and distribution networks in smart grids	1 s (note 4)	99%	NA	< 1kbit/s (note 5)	Typically < 100 bytes (note 1)	< 10,000 /km^2 (note 3)	Outdoor: typically 50-200 meters	[several km^2 up to 100 000 km^2] (note 2)	Stationary	5-15 min	NA	NA	several 10 m

NOTE 1: Electronic Product Code standard [5], this size is the payload size.
NOTE 2: The service are refers to the overall size of transmission and distribution networks. Typically, the size of the individual substations varies from 100m x 200m to 500m x 600m.
NOTE 3: The device density is calculated based on an individual substation, where typically several hundreds of Ambient IoT devices are required to monitor the environmental parameters.
NOTE 4: This is calculated based on assumption that the sensor data are collected once per several seconds.
NOTE 5: For temperature, humidity and pressure measurement, typical sampling rate is 10 Hz with sample size of 32 bits, thus the data generation per Ambient IoT device is about 320 bit/s. For vibration measurement, typical sampling rate is 10 Hz with sample size of 96 bits, thus the data generation per Ambient IoT device is about 960 bit/s.

5.4 Use case on supporting Ambient IoT in Non-Public Network for logistics

5.4.1 Description

The logistic chain is composed of different processes, such as warehouse inbound and outbound, etc. During the inbound, warehousing inventory needs to be done in order to track whether all the goods are inventoried. After the outbound, the cargo needs to be tracked to ensure that corresponding goods are moving to the right destination. This tracking that happens whenever the cargo moves across a specific area (e.g., equip with portals). The warehousing inventory and cargo tracking need the support of Ambient IoT in non-public network.

In the case of warehousing inventory, there are tons of goods waiting to be inventoried. A pallet is used for carrying goods (each pallet for maximumly 100 goods) and each pallet passes through a base station (e.g., integrated with a scanner) which can scan all the goods on the pallet and complete the inventory for those goods. The time interval for two neighboring pallets to pass through the base station is usually short (e.g., less than 3min). With the introduction of Ambient IoT devices, the logistics service provider can have its own NPN and perform the warehousing inventory using the NPN. All goods can communicate directly with the base station or send the data to the base station indirectly through a relay.

In the case of cargo tracking, A wagon carries many pallets with goods (more than 10 pallet per wagon) and travels to the destination. In order to track the goods within the wagon, several toll gates are set up among the route of the wagon. Whenever the wagon passes through the toll gate, the data of all goods are required to be sent to the toll gate which can then forward the data to the proper application server for tracking. The service provider of logistics may set up the tracking equipment on certain toll gate to track the data of goods carried by wagons. The efficiency for tracking the goods carried by wagon will be dramatically increased.

In the two processes described, the Ambient IoT devices used can be either battery-less or with limited energy storage capability (i.e., using a capacitor).

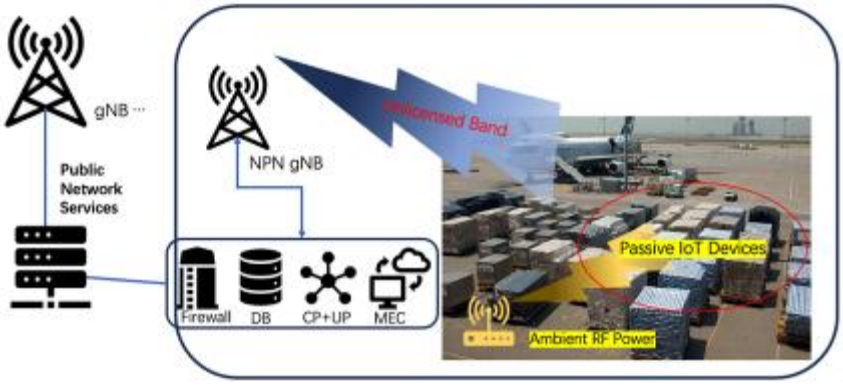


Figure 5.4.1-1: Ambient IoT in non-public network for logistics

5.4.2 Pre-conditions

The service provider has service agreement with the Network Operator. The service agreement includes the provisioning of NPN to the service provider.

The service provider set up its own NPN for managing the Ambient IoT devices in logistics.

The use case of logistics in 5.4 can be split into 2 key processes, which are warehousing inventory and cargo tracking.

Process A: Cargo warehousing inventory

A service provider of logistics has its NPN with the support of Network Operator for the access of Ambient IoT devices (e.g., tags) for tracking the good within each pallet (each pallet for more than 150 goods). Alternately, the service provider set up its own NPN for the purposes. The service provider of logistics uses licensed band or unlicensed band for accessing the NPN.

All the Ambient IoT devices conduct onboarding and provisioning for NPN credentials. Ambient IoT device registers with the onboarding NPN and obtains the NPN credential with low energy cost.

All the Ambient IoT devices register with the NPN.

Process B: cargo tracking

A service provider of logistics has its NPN with the support of Network Operator for tracking the Ambient IoT devices (e.g., tags) carried by wagon. Alternately, the service provider set up its own NPN for the purposes. The NPN is set up on the toll gate along the road for tracking the cargo.

All the Ambient IoT devices conduct onboarding and provisioning for NPN credentials. Ambient IoT device registers with the onboarding NPN and obtains the NPN credential with low energy cost.

5.4.3 Service Flows

Process A: Cargo warehousing inventory

1. When each pallet passes through the base station, all the Ambient IoT devices within the pallet complete the inventory by echoing the request for inventory from the base station.

2. After a short internal of time (less then 3min), the base station inventories the goods carried by the next pallet.

Process B: cargo tracking

1. All the Ambient IoT devices register with the NPN when the wagon carrying the cargo passes through the toll gate.

2. All the Ambient IoT devices within the wagon sends the response to the base station on the toll gate upon the receipt of the request from the base station.

5.4.4 Post-conditions

Process A: Cargo warehousing inventory

1. Inventory information is obtained by the service provider, who can proceed with the warehouse outbound.

Process B: cargo tracking

1. By receiving the tracking information of the cargo, the service provider knows that corresponding goods are moving to the right destination.

5.4.5 Existing features partly or fully covering the use case functionality

TS 22.261 has following requirements:

- The 5G system shall support operator-controlled alternative authentication methods (i.e., alternative to AKA) with different types of credentials for network access for IoT devices in isolated deployment scenarios (e.g., for industrial automation).

- The 5G system shall support the capability to operate in licensed and/or unlicensed bands.

Existing specification support alternative authentication method with different types of credentials for network access for IoT devices. For Ambient IoT device, the alternative authentication method should be light-weight in order make sure that the cost of power consumption caused by the authentication is strictly controlled.

Existing specification support the capability for 5G system to operate in licensed and/or unlicensed band. For Ambient IoT device, it should also be able to connect to the 5G network using licensed and/or unlicensed band. Service provider may decide to use licensed band or unlicensed band based on the situation of local spectrum and the SLA between Service provider and other parties (e.g., Network Operator, industry partner, etc.). Ambient IoT device operating in unlicensed band should have the same performance (e.g., transmission range) with the one operating in licensed band.

5.4.6 Potential New Requirements needed to support the use case

[PR.5.4.6-001] 5G system shall support network access for Ambient IoT devices while considering the constraint power consumption.

Note: The above requirement applies to both NPN and PLMN.

5.5 Use case on intralogistics in automobile manufacturing

5.5.1 Description

The automobile manufacturing industries are constantly looking for ways to increase productivity by improving inventory accuracy and material flows. Therefore, intralogistics for production facilities in automobile factories have been targeting at these goals by achieving timely visibility of inventories (a.k.a. precise materials scheduling) to achieve optimum efficiency in production. Specifically, it involves stocking, dispatching, and sorting. It is no surprise that there are competing technologies for supporting small-scale inventory, which at times would require human involvement. With the advent of Industry 4.0, manufacturing requires a much higher automatic intralogistics performance [10]. This means more key areas inside large manufacturing facilities would require highly-efficient and automated inventory of materials and parts. Ambient power-enabled IoT (Ambient IoT) service provided by 5G can be expected to meet the demanding needs by providing communication to Ambient IoT devices with good performance, with the Ambient IoT solely dependent on harvested ambient energy, being maintenance-and-battery-less or with limited energy storage capability, extremely small, thin, and of low complexity, very light-weight, and with a long lifespan. For this use case, communication service availability with sufficient 5G network coverage in the service area is important. Additionally, the 5G system can provide Ambient IoT devices with positioning services, as full automation would require AGVs or forklifts to fetch materials at accurate locations within the large storage areas. Moreover, the 5G system provides intrinsic core network functions to manage and authenticate Ambient IoT connections, enabling operators to build more interesting cases together with their business partners.

Automobile company A uses standardized load containers to realize flow of materials. The load containers are purchased, owned and managed by Company A. With its business growing, Company A is integrating the latest and advanced production and logistics management systems to ensure the production facilities are modernized for Industry 4.0. In quest of this, intralogistics at Company A is expected to automatically identify and track individual goods and materials throughout production facilities: not only at the dock where materials enter the production facilities, but also in the large floor-level storage areas, further in the sorting areas and down at production lines. Similar to the norm of the automobile manufacturing industry, a typical production facility of Company A covers a total area of around 600,000 square meters. Company A uses unique identifiers (e.g. 96 bit or 240 bit EPC codes) to distinctly identify load containers.



Fig. 5.5.1-1: Intralogistics in automobile manufacturing

Prior to deciding on a future-proof solution, a comprehensive analysis (e.g., ROI) is carried out by Company A to verify Ambient IoT's suitability for the long-term business objectives. By virtue of many of Ambient IoT's attractive wireless communication characteristics, Company A came to the conclusion of a positive case. To effectively meet their intralogistics demand, it entails installing around 1300 stationary "readers" strategically per typical production facility (600,000 square meters), where in total around 800,000 Ambient IoT devices would be physically present at the same time. To keep track of the production process, normally 20 Ambient IoT devices need to be read per second per "reader" (e.g., fast-moving AGVs passing a gate). After verifying with operator O, Company A adopts Ambient IoT by having a service contract with the operator, which is commissioned to design and deploy 5G network coverage within Company A's automobile production facilities and then to accordingly enable and manage Ambient IoT service. Wherever needed in the production facilities, sufficient network coverage is assumed.

It is in the huge floor storage area where the loaded containers are kept, accurate positioning service is needed by AGVs for pick-up, once certain materials or parts are needed by a production line. At Company A, the floor storage area is divided into blocks with the dimension of 18m by 18m, in each of which a stationary "reader" is deployed. Each block is further divided into grids of the size slightly larger than a load container, whose length is around 1.5m. To achieve uninterrupted material flow, Company A decide to use the space of two grids for parking load containers holding the same materials. In this way, as a load container holding certain material at a given grid is carried away by an AGV, the same material loaded in another container would always be available at the grid just next to it.

The personnel Sylvain at Company A attaches each of the load containers with an Ambient IoT device, where a unique code is written in its microchip. Ambient IoT devices are very thin non-battery powered or with limited energy storage capability IoT devices, which instead function using an energy harvesting mechanism that produces a limited amount of power. A typical energy harvesting device can produce up to a few hundred microwatts [11] (e.g., less than 250 micro-Watts). In the production facilities of Company A, an Ambient IoT device can be connected to the 5G network and the communication is enabled.

5.5.2 Pre-conditions

An Ambient IoT device can obtain energy by collecting energy sources such as solar and radio waves. Each load container is attached with an Ambient IoT device for supporting intralogistics. Ambient IoT devices have the capability of storing information needed by the inventory process. Base stations installed inside car manufacturing facilities provide network coverage. The communication service availability is achieved by providing seamless 5G network coverage inside the service areas. The 5G system delivers application data to the intralogistics management system of the car manufacturer.

5.5.3 Service Flows

1. Company A uses Ambient IoT devices attached to their load containers to support automated intralogistics process for improved productivity. They send load containers to various suppliers in order to bring back ordered goods, materials and parts.

2. By returning to Company A's automobile manufacturing facilities, large amount of load containers first enter the dock area. There the 5G network transmits signals intending to start inventory process. Since the Ambient IoT devices harvest power from the environment, once detecting the signals from the 5G network the Ambient IoT devices can respond by starting random access.

3. 5G network (i.e., core network) authenticates and authorizes Ambient IoT devices.

4. Based on the requests issued by the application function, the 5G network performs the automatic inventory operation (read out pre-written unique IDs in Ambient IoT devices) of the large number of incoming load containers entering the dock area fast and efficiently. The read-out information is sent by 5GC to Company A's inventory system (connected to ERP/APS), where the inventoried information is updated or saved.

5. Once registered in the inventory system, load containers are stored in the floor storage areas. The load containers (still loaded with various materials and parts) can be inventoried by Company A either periodically or when requested.

6. As automobile manufacturing process continues (almost non-stop), a picking list is automatically generated according to the production plan managed by the APS (Advanced Planning and Scheduling) system.

7. According to the information in the inventory system, AGVs (or forklifts) with their respective picking lists are sent to the floor storage areas to pick up the corresponding load containers (holding different materials or parts). As the floor storage area is extremely huge, Company A utilize the 5G network positioning service for AGVs to quickly fetch the target load containers holding the exact materials requested in the pick-up list. Since the floor storage area is divided into 18m by 18m blocks with each block being further divided into grids of 1.5m by 1.5m, the 5G system always provides the AGVs with the accurate positioning information of the target load containers.

8. When AGVs find the needed load containers in the floor storage area, convey them further to the sorting areas. It is in the sorting areas that different materials and parts brought in by AGVs are grouped in accordance with precise manufacturing schedules to be followed at the production lines.

9. At the production lines, materials and parts as logically grouped (by sorting machines at the sorting areas) are eventually fed into production. The empty load containers will be brought to the recycling area. These empty load containers await to be sent to suppliers at the next schedule determined by ERP/APS.

10. In all the key areas (floor storage areas, sorting areas, production lines and empty container recycling area) inside the manufacturing facility, the load containers need to be inventoried efficiently, timely, fast, accurately. When AGVs enter these areas, step 4 is repeated and the precise material scheduling is updated in the inventory system.

5.5.4 Post-conditions

Thanks to the Ambient IoT service provided by the 5G system, automobile manufacturing can enjoy automatic intralogistics, largely improve the efficiency and productivity.

5.5.5 Existing features partly or fully covering the use case functionality

SA1 has performed various studies on IoT in previous releases, where related normative stage 1 requirements are introduced in TS 22.011 [9], TS 22.278 [7], TS 22.368 [6], and TS 22.261 [8].

TS 22.011 introduces access control for MTC, examples of periodic network selection attempts are:

For UEs only supporting any of the following, or a combination of, NB-IoT, GERAN EC-GSM-IoT [18], and Category M1[13] of E-UTRAN enhanced-MTC, the UE shall interpret the interval value to be between 2 and 240 hours, with a step size of 2 hours between 2 and 80 hours and a step size of 4 hours between 80 and 240 hours.

In the absence of a permitted value in the SIM/USIM, or the SIM/USIM is phase 1 and therefore does not contain the datafield, then a default value of 60 minutes, shall be used by the UE except for those UEs only supporting any of the following, or a

combination of: NB-IoT, GERAN EC-GSM-IoT [18], and Category M1 [17] of E-UTRAN enhanced-MTC. For those UEs a default value of 72 hours shall be used.

NOTE: Use of values less than 60 minutes may result in excessive UE battery drain.

TS 22.368 addresses features of MTC communication and service requirements related to MTC device triggering, addressing, identifiers, low mobility, small data transmission, infrequent MT communication, security, remote MTC device management, group-based MTC features including policing and addressing, etc. Example requirements are:

The system shall provide mechanisms to lower power consumption of MTC Devices.

The system shall provide mechanisms for the network operator to efficiently manage numbers and identifiers related to MTC Subscribers.

TS 22.261 captures some important service requirements for IoT, e.g.

The 5G system shall support a secure mechanism for a home operator to remotely provision the 3GPP credentials of a uniquely identifiable and verifiably secure IoT device.

The 5G system shall support a secure mechanism for the network operator of an NPN to remotely provision the non-3GPP identities and credentials of a uniquely identifiable and verifiably secure IoT device.

An IoT device which is able to access a 5G PLMN in direct network connection mode using a 3GPP RAT shall have a 3GPP subscription.

The 5G system shall allow the operator to identify a UE as an IoT device based on UE characteristics (e.g. identified by an equipment identifier or a range of equipment identifiers) or subscription or the combination of both.

An IoT device which is able to connect to a UE in direct device connection mode shall have a 3GPP subscription, if the IoT device needs to be identifiable by the core network (e.g. for IoT device management purposes or to use indirect network connection mode).

The 5G system shall support operator-controlled alternative authentication methods (i.e. alternative to AKA) with different types of credentials for network access for IoT devices in isolated deployment scenarios (e.g. for industrial automation).

The 5G system shall support a suitable framework (e.g. EAP) allowing alternative (e.g. to AKA) authentication methods with non-3GPP identities and credentials to be used for UE network access authentication in non-public networks.
NOTE: Non-public networks can use 3GPP authentication methods, identities, and credentials for a UE to access network. Non-public networks are also allowed to utilize non-AKA based authentication methods such as provided by the EAP framework, for which the credentials can be stored in the ME.

The 5G system shall enable an NPN to be able to request a third-party service provider to perform NPN access network authentication of a UE based on non-3GPP identities and credentials supplied by the third-party service provider.

In these specifications, albeit the service requirements addressing traits for IoT in terms of low device power consumption, small and infrequent data transmissions, long service lifetime, and resource efficiently, the IoT devices considered in 3GPP have been assumed to be powered by at least batteries up till now. To enable extremely small, thin, light-weight, battery-less or with limited energy storage capability, or even disposable Ambient IoT devices that provide basic IoT data transaction at appropriate performance level suitable for the target scenarios, new challenges to the 5G system are foreseen and need to be addressed.

5.5.6 Potential New Requirements needed to support the use case

[PR 5.5.6-001] The 5G system shall support communication for an Ambient IoT device which is battery-less or with limited energy storage capability.

[PR 5.5.6-002] The 5G system shall support collection of charging information based on different charging policies for Ambient-IoT type of communication, i.e., total number of communication (e.g. data payload) per charging period, or total number of Ambient IoT devices per charging period.

[PR 5.5.6-003] The 5G system shall provide the network connection to address the KPIs for the use of Ambient IoT devices for intralogistics in automobile manufacturing, see table 5.3.6.1-1.

Table 5.5.6-1: Potential key performance requirements for the use of Ambient IoT devices for intralogistics in automobile manufacturing

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Automatic Intralogistics in automobile manufacturing	10 s (note 1)	99 %	NA	<1 kbit/s (note 2)	96 bits (note 3)	<1,5 Million/km^2 (note 4)	<30 meters Indoors	600 000 m^2 (note 5)	Up to 5 km/h	NA	NA	NA	3 m

NOTE 1: This value corresponds to peak reading rate of 100 tags per second. The average tag reading rate is lower.
NOTE 2: This value is calculated as the instant data rate for transmitting 96 bits within 100 ms time period. The need for data transmission is infrequent.
NOTE 3: EPC Tag Data standard [5], the length of the EPC number ranges from 96 bits to 496 bits. For intralogistics, EPC length of 96 bits is the most common EPC lengths to satisfy the use case.
NOTE 4: Daily around 1 million units of materials are used in the manufacturing area, but they are not used at the same time.
NOTE 5: A typical car manufacturing plant takes up to 600 000 m^2 in surface.

5.6 Use case on Ambient IoT sensors in smart homes

5.6.1 Description

Monitoring tasks in smart home scenarios can be roughly divided into two types:

- Monitoring of room environment. By deploying specific sensors that enable communication services, people can acquire real-time monitoring data (e.g. temperature, humidity) of their room environment no matter where they are.
- Monitoring of emergency situations. By deploying specific sensors that enable communication services, risk factors such as gas and smoke can be detected in the air and alarmed timely. When the sensor detects that the gas concentration in the home exceeds the threshold, it usually activates a very strong audio signal to alarm people at home. However, in case of people are out of home (e.g. while at work, or on shopping), it cannot reach people on this situation. So, it should be necessary to notify the family members through their phones as well.

In contrast with conventional battery-based sensors, Ambient IoT sensors can obtain and/or store energy from the environment, such as light, heat, wind, and radio waves, which can be converted to useable electrical energy [12]. Different ambient energy harvesting technologies have their own advantages and disadvantages, suitable for use in different environments. Energy harvesting technologies are not in the scope of the study.

Ambient IoT sensors can support diverse monitoring applications in smart home scenario with following advantages:

- Remove the demand for batteries, i.e. reduce the energy consumption of charging.
- Reduce the cost of maintenance, i.e. avoid human intervention for recharging or replacing.

- Increase device durability, i.e. devices can work continuously without charging and/or replacing the battery.
- Increase device portability, i.e., the locations of devices are no longer limited by electric cables or wires.

The Ambient IoT sensors can also consume very small amounts of energy. For example, the photovoltaic (PV) energy harvesting power density could provide 10μW~4mW, while the sensing power consumption of a gas sensor is only 1.2μW [13].

Considering the above advantages, Ambient IoT sensors can further improve the development of smart home applications.

5.6.2 Pre-Conditions

Tom's family lives in Home A. In Home A, there is a gas sensor in the kitchen. It's an Ambient IoT sensor, and the application server can obtain its sensor data through the network.

The gas sensor detects data continuously and allows the application server to request its data (e.g., a 32 bits packet) periodically. The application server stores the data and a pre-set methane concentration threshold (e.g., 5%).

5.6.3 Service Flows

1. The gas sensor monitors the methane concentration in the kitchen in real-time.

2. The application server acquires the sensor data and finds the data reaching half of the threshold (e.g., 2.5%), then it will update a shorter period to request data.

3. If the data exceeds the threshold, the application server can send an alarm message through the network.

4. The application server will distribute the alarm message to Tom's family through the network. If Tom's family members do not respond, the alarm message will be sent again after a preset time (e.g., 1 min).

5. Tom's family members receive the alarm message on their phones. Tom notices the alarm message and clicks on "Confirm" button in the app.

6. The application server receives the "Confirm" response and stops to send alarm messages to their phones until set time (e.g., 10min) passes and the data still exceeds the threshold.

5.6.4 Post-Conditions

Tom goes back to home, checks the kitchen range and closes the gas valve.

5.6.5 Existing features partly or fully covering the use case functionality

None.

5.6.6 Potential New Requirements needed to support the use case

[PR.5.6.6-001] The 5G system shall be able to support communication services for Ambient IoT devices.

[PR.5.6.6-002] The 5G system shall support suitable security mechanisms for Ambient IoT devices, including authentication, encryption and data integrity.

[PR.5.6.6-003] The 5G system shall be able to provide the required communication service according to KPI given in table 5.6.6-1.

Table 5.6.6-1: KPI for Ambient IoT devices in smart home

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Ambient IoT devices in smart home	20 s	99.9 %	NA	<1 kbit/s	8~96bits	<5 per 100 m^2	10-30m Indoors	NA	Stationary	NA	NA	NA	NA

5.7 Use Case on Ambient IoT for airport terminal / shipping port

5.7.1 Description

An airport terminal / shipping port manages a large inventory of different types of objects, including forklifts, trolleys, ramp leaders, pallet dollies, baggage carts, wheelchairs, among others (see Figure 1). Real-time tracking and management of such assets is an important part of efficient operation of an airport terminal/shipping port, including through (re)-deployment of assets based on time-varying demand in different locations (e.g., gates), prolonged asset life through timely maintenance, improved safety and travel experience, asset theft prevention.



Figure 5.7.1-1: Airport terminal / Shipping port requires real-time management of different types of assets.

Some key differences from other asset management/tracking use-case scenarios include the need for real-time location information, relatively large service area with a mix of indoor and outdoor deployment, and the need for mobility support.

5.7.2 Pre-conditions

An Ambient IoT device is attached to each asset (to be tracked) before deployment. The asset management system has subscription to the Ambient-IoT services with access to information about the Ambient IoT device, such as location, maintenance-related parameters;

The airport terminal / shipping port has public or private 5G network coverage to provide the Ambient-IoT services with support for a large number of Ambient-IoT devices. The Ambient IoT devices in this use case communicate to the network.

5.7.3 Service Flows

1. The asset management system requests the Ambient IoT service for regular/on-demand snapshot of the asset inventory information with a specified granularity, e.g., gate-level;
2. Based on the request, the 5G system queries the Ambient IoT devices to inventory different types of assets in the specified location(s).
3. The Ambient IoT service aggregates the responses from the Ambient IoT devices to respond to the application function's request at the specified granularity;
4. The Ambient IoT service provides additional agreed-upon information

5.7.4 Post-conditions

The airport terminal / shipping port utilizes the Ambient IoT service to obtain real-time inventory/location of the different types of assets, allowing for more efficient operation through (re)-deployment of assets based on time-varying demand in different locations, prolonged asset life through timely maintenance, improved safety and travel experience, asset theft prevention.

5.7.5 Existing features partly or fully covering the use case functionality

None.

5.7.6 Potential New Requirements needed to support the use case

[PR 5.7.6-001] The 5G system shall be able to support means to discover and locate Ambient-IoT devices in a certain geographical area, e.g. at cell level.

[PR 5.7.6-002] The 5G system shall be able to provide communication with Ambient-IoT devices with the following KPIs :

Table 5.7.6-1: KPI Requirements for Airport Terminal/Shipping Port Service

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Airport Terminal/ Shipping Port	1s to 10s	99%	NA	NA	256 bits (UL) (Note 1)	100 devices/1k m^2	50m (indoor)	1-10km^2 (Note 2)	3 to 10k m/h	NA	NA	90 %	cell level
Note 1: 128 bits for the Electronic Product Code (EPC) of the tracked object and additional 128 bits assumed for control / other data (e.g., location-related).													

Note 2: As an example, Newark Airport size ~8km^2.

5.8 Use case on Finding Remote Lost Item

5.8.1 Description

It is quite common for people to lose personal items sometimes, which includes keys, wallets, bags, phones, glasses, etc. In many of those scenarios, Ambient IoT devices, (with small form factor and low cost/complexity), could be attached to those items and be used to help finding their location via a 5G smartphone.

When a person loses his/her personal item in a remote place, far from the owner's location (e.g., in airport, subway, park, restaurant, etc), the large distance between the lost items (w/ Ambient IoT device attached) and the owner's 5G UE prevents the owner from using direct communication methods, between the UE and the Ambient IoT device, to perform e.g., local discovery/positioning/ranging.

This use case (shown in Figure 1) considers a scenario where Ambient IoT devices are used to locate lost items, which they are attached to, with the help of surrounding UEs/ RAN entities (supporting the tracking of Ambient IoT devices).

In terms of communication power availability, these Ambient IoT devices can operate based on intermittently harvested energy with energy storage or instantaneous energy provided on-demand. For devices with energy storage, we assume energy is continuously available during its communication. This use case covers scenarios without energy storage and with energy storage, assuming lower power consumption and complexity than typical/current IoT devices.

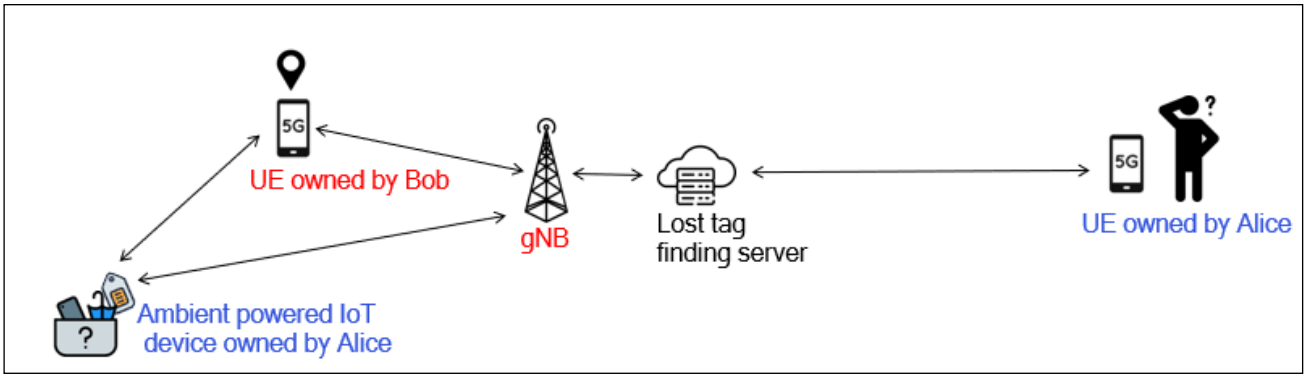


Figure 5.8.1-1: Remote tag finding service, using Ambient IoT devices and surrounding UEs/RAN entities

5.8.2 Pre-conditions

Alice subscribed to a "lost tag finding" service and bought an Ambient IoT tag for her baggage.

She attaches the tag to her baggage and associated the tag with her mobile phone in this service.

5.8.3 Service Flows

The case of remote lost item finding by crowdsourcing

1. The Alice is traveling from San Diego (SAN) to New York (JFK) using airplane with her tag (tag-A) attached to her baggage. When she arrives in JFK airport, she does not find her baggage in baggage claim area.
2. The tag-A attached to Alices' baggage identifies that it is lost and notify any nearby UEs/RAN entities supporting the lost tag finding service that the tag-A is currently lost.

3. As shown in Figure 1, the nearby UEs/RAN entities which is notified from the tag-A connects to server providing the lost tag finding service and reports the tag-A's current location and time. (For privacy, the tag-A's identity should not be known to other UEs/RAN entities other than Alice's UE or service provider.)

4. Alice, after finding that her baggage is missing in airport, opens an app in her mobile phone to find her tag-A's whereabout. The app communicates with the server and gets the recent information e.g., location/time/etc, related to the tag-A.

5. Alice can track the location of her baggage with the help of lost item finding service.

5.8.4 Post-conditions

Alice reclaims her lost baggage with the help of the lost item finding service.

5.8.5 Existing features partly or fully covering the use case functionality

SA1 has identified several IoT/MTC requirements that assume device type of higher complexity and higher availability of power/energy, higher processing power, etc, than the Ambient IoT device. Some excerpts are listed below:

3GPP TS 22.368 [6]: "Service requirements for Machine-Type Communications (MTC)"

TS 22.368 provides related requirements for identifiers for MTC subscribers, for handling large number of devices, feature addition, power consumption, security

- The system shall provide mechanisms for the network operator to efficiently manage numbers and identifiers related to MTC Subscribers.
- The network shall provide a mechanism to reduce peaks in the data and signalling traffic resulting from very large numbers of MTC Devices (almost) simultaneously attempting data and/or signalling interactions.
- The network shall provide a mechanism for the network operator to control the addition or removal of individual MTC Features to a subscription (e.g. based on matching or mismatching of MTC Features).
- The system shall provide mechanisms to lower power consumption of MTC Devices.
- The network operator shall be able to efficiently provide network security for connection between MTC Device and a MTC Server or between MTC Device and a MTC Application Server in case there is a direct connection with the MTC Application Server. This applies even when some of the devices are roaming i.e. connected via a VPLMN.

3GPP TS 22.278 [7]: "Service requirements for the Evolved Packet System (EPS)"

The 22.278 provides efficient data transmission in between core network and UE.

The 3GPP system shall support efficient transmission of IP data and non-IP data to/from a UE.

The 3GPP system shall support efficient transmission of small data to/from a UE.

3GPP TS 22.261[8]: "Service requirements for the 5G system"

TS 22.261 provides requirements on power, positioning.

The 5G system shall support UEs using small rechargeable and single coin cell batteries (e.g., considering impact on maximum pulse and continuous current).

The 5G system shall be able to make the position-related data available to an application or to an application server existing within the 5G network, external to the 5G network, or in the User Equipment.

NOTE 3: the position service latency can be tailored to the use cases.

The 5G system shall be able to manage and log position-related data in compliance with applicable traceability, authentication and security regulatory requirements.

The 5G system shall supply a method for the operator to configure and manage different positioning services for different users.

The 5G system shall allow the operator to identify a UE as an IoT device based on UE characteristics (e.g. identified by an equipment identifier or a range of equipment identifiers) or subscription or the combination of both.

5.8.6 Potential New Requirements needed to support the use case

[PR.5.8.6-001] The 5G system shall be able to assist an Ambient IoT device with discovery and communication with 5GS entities that can provide location related information.

[PR 5.8.6-002] Based on operator policy, the 5G system shall be able to support authorization of UEs communicating with an Ambient IoT device.

[PR.5.8.6-003] The 5G system shall be able to support means to support RAN entities and authorized UEs to communicate with Ambient IoT devices and transfer related information to other 5G system entities (e.g., core network) / servers.

[PR.5.8.6-004] The 5G system shall be able to provide a mechanism to protect the privacy of information (e.g., location and identity) exchanged during communication with an Ambient IoT device.

NOTE 1: This requirement refers to communication between Ambient IoT devices and 5G System entities (e.g., core network, RAN entities), application servers or authorized UEs.

[PR 5.8.6-005] The 5G system shall be able to support a UE to authenticate an Ambient IoT device.

[PR 5.8.6-006] The 5G system shall be able to support Ambient IoT devices with the following KPIs:

Table 5.8.6-1: KPI Requirements for "Finding Remote Lost Item" Service

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Remote lost item finding (Indoor)	5s	99% (Note 1)	NA	NA	256 bits (Note 2)	250 devices/100m^2 (Note 3)	10m	NA	NA	NA	NA	90%	~3m
Remote lost item finding (Outdoor)	5s	99% (Note 1)	NA	NA	256 bits (Note 2)	10 devices/100m^2 (Note 4)	100 m	NA	NA	NA	NA	90%	~10 m

Note 1: Service can be potentially provided by both multiple UEs and RAN entities.

Note 2: 64bits corresponds to 20 digits in decimal number. 20 digits is assumed for the length of tag ID. Additional 192bits were assumed for control and other data (e.g., location information, IP address of server).

Note 3: It assumes an $100m^2$ surface area inside an airport taken up by baggage.

Note 4: Considering moderately sized mobile panels or AGVs transporting baggages in an airport apron (an open area), given the limited load per mobile pannel or AGV, the density of baggages is very low per unit surface of $100m^2$.

5.9 Use case on LCS for Ambient IoT

5.9.1 Description

Ambient Internet of Things (Ambient IoT) is an IoT service with an IoT device powered by energy harvesting, being either battery-less or with limited energy storage capability (i.e., using a capacitor). It can enable communication with IoT devices without conventional power source and/or avoids human intervention for recharging or replacing. An Ambient IoT device can harvest energy from energy source from Radio, solar, light, motion/vibration, heat, pressure, or any other power sources.

Note: Energy harvesting is out of scope of this TR.

5.9.2 Pre-conditions

Tom buys an Ambient IoT device which has a 3GPP subscription and is equipped with 3GPP radio technology. The device registers to the 3GPP network based on its subscription and the network should record that the device can be found its location by Tom's UE.

Tom puts the Ambient IoT device on his pet dog collar.

5.9.3 Service Flows

1. In a morning, Tom gets up and takes his pet dog to the park nearby. In the park, the dog runs far away from Tom.
2. After some time, Tom needs to go home for the breakfast, he takes out his UE to ask the location of the Ambient IoT device.
3. The UE initiates LCS request to the network for the location of the Ambient IoT device.
4. The network finds the location of Ambient IoT device when the device has enough power. The power is from the ambient power source. The network sends the location result to the UE.
5. Tom can see the location result in the UE and walks towards his dog to take it home.

5.9.4 Post-conditions

Tom finds his pet dog with the help of 5G network and the Ambient IoT device.

5.9.5 Existing features partly or fully covering the use case functionality

None

5.9.6 Potential New Requirements needed to support the use case

[PR 5.9.6-001] 5G system shall optimize mobility management support for mobile Ambient-enabled IoT devices that are unable to constantly stay active.
[PR 5.9.6-002] 5G system shall be able to determine the location of Ambient IoT device, when it becomes active as triggered by the 5G network.

[PR 5.9.6-003] The 5G system shall be able to provide location services for Ambient IoT with the performances requirements reported in Table 5.9.6-1.

Table 5.9.6-1: Performance requirements for location service for Ambient IoT

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Absolute positioning	NA	NA	NA	NA	NA	NA	500 m	NA	Outdoor - up to 10 km/h	NA	10 s	95 %	Cell-level horizontal accuracy (NOTE 1)
NOTE 1: This KPI table is mostly from the positioning service levels 1 in Table 7.3.2.2-1 Performance requirements for Horizontal and Vertical positioning service levels, TS22.261[8].													

5.10 Use case on Relative positioning for Ambient IoT

5.10.1 Description

Ambient Internet of Things (Ambient IoT) is an IoT service with an IoT device powered by energy harvesting, being either battery-less or with limited energy storage capability (i.e., using a capacitor). It can enable communication with IoT devices without conventional power source and/or avoids human intervention for recharging or replacing. An Ambient IoT device can harvest energy from energy source from Radio, solar, light, motion/vibration, heat, pressure, or any other power sources.

Note: Energy harvesting is out of scope of this TR.

Relative positioning is to estimate position relatively to other network elements or relatively to other UEs.

5.10.2 Pre-conditions

Tom buys an Ambient IoT device which has a 3GPP subscription and is equipped with 3GPP radio technology. There is a relative positioning APP in Tom's UE. The relative positioning application server and clients records the UE and Ambient IoT device's relative positioning application layer ID.

Tom pastes the device on his key.

In a morning, Tom gets up and wants to walk in the park nearby.

5.10.3 Service Flows

1. Tom wants to take his key, but cannot find the key in the room. He takes out his UE to initiate a relative positioning request for the Ambient IoT device by using the relative positioning APP. The 5G system authenticates the UE and the Ambient IoT, and authorizes the UE and the Ambient IoT device to perform the relative positioning.

2. The UE sends the radio to activate the Ambient IoT device in the room and the relative positioning request with the relative positioning application layer ID of the UE as the requester and the Ambient IoT device as the target.

3. After receiving and storing enough energy, the Ambient IoT device responds the UE's relative positioning request with its relative positioning application layer ID. From the APP, Tom's UE displays that his key is still in the room.

4. After the relative positioning discovery procedure above, the UE continues sending the radio to the Ambient IoT device, and the Ambient IoT device receives and stores the energy for continuous relative positioning.

5. The UE performs relative positioning to find the direction and relative positioning to the Ambient IoT device.

5.10.4 Post-conditions

Tom finds his key under the bed.

5.10.5 Existing features partly or fully covering the use case functionality

None

5.10.6 Potential New Requirements needed to support the use case

[PR 5.10.6-001] 5G system shall be able to support an authorized UE to perform Ambient IoT relative positioning between the UE and specific Ambient IoT devices.

[PR 5.10.6-002] The 5G system shall be able to provide Relative positioning services for Ambient IoT with the performances requirements reported in Table 5.10.6.1-1.

Table 5.10.6-1: Performance requirements for Relative positioning service for Ambient IoT

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Finding Items in a home	NA	NA	NA	NA	NA	20 Ambient IoT devices/ ($100m^2$)	10m	NA	Static/ Moving (<1m/s)	500 ms	NA	95 %	1-3m (Note 1)
NOTE 1: This value depends partly on the actual communication range.													

5.11 Use case on online modification of medical instruments status

5.11.1 Description

More and more medical instruments in hospital need to be well stored, cleaned and sterilized to guaranteed normal reuse. They demand to withstand certain conditions e.g., high temperature, high pressure or humidity. Traditional information maintenance for the medical instrument status is usually operated manually, which is inefficient and even in some cases, causes serious accident e.g., lost or invalidity. To improve safe and efficient utilization of the medical instruments, online maintenance is being developed.

For the online maintenance, the medical instrument is needed to be equipped with Ambient IoT device. Considering the working condition of the medical instrument, this kind of Ambient IoT device should be battery-less or with limited energy storage capability, maintenance-free and should have long service life time and small size. Through 5G network and the IoT device, the medical instrument information (e.g., the serial number of the instrument, usage status, usage records, years of use, integrity, etc.) can be remotely read, modified and written by the medical instrument management platform.

Following is an example to illustrate online modification of medical instruments status.

Orthopaedic instruments generally refer to professional medical instruments specially used for orthopaedic surgery. According to the usage purpose, it can be classified as orthopaedic knives, orthopaedic scissors, orthopaedic forceps, orthopaedic hooks, orthopaedic needles, orthopaedic scrapers, orthopaedic cones, orthopaedic drills, orthopaedic saws, orthopaedic chisels, orthopaedic files / shovels, orthopaedic active instruments, etc.

A number of Ambient-IoT devices recording different orthopaedic instrument information (e.g., the serial number of the instrument, usage status, usage records, years of use, maintained status, etc.) are stuck on these orthopaedic instruments. They are usually indoor stored in the instrument warehouse or medical instrument storage room or special storage cabinet.

These Ambient-IoT devices attached in the orthopaedic instrument are battery-less or with limited energy storage capability. They are with very simple capability and not applications installed on them.

5.11.2 Pre-Conditions

Network operator UU deploys a new service “Ambient IoT” through its 5G system. Hospital Z has subscribed the new service for its orthopaedic instrument inventory management.

Bob is an instrument inventor manager of Hospital Z. He has the authorization to remotely maintain the orthopaedic instrument through the inventory management platform of the hospital. He can operate this work in the hospital or out of the hospital.

5.11.3 Service Flows

Same as the service flow in section 5.2.3, Bob can acquire the information list of the orthopedic instruments.

In the list, Bob can read that some orthopedic forceps are in "To be maintained" status. So, he selects them and asks engineer to repair them.

Part of the instruments are repaired well and returned back in cabinet, Bob wants to change the status information of the orthopedic forceps to "Normal". The status change request is delivered to the hospital inventory management platform. The platform informs 5G network to ask the Ambient-IoT devices attached on the orthopedic forceps to “modify” the status with “Normal”. The 5G network transmits “modify” command transparently or translate the command to the Ambient-IoT devices. In the “modify” command, it includes not only the updated information but also can be the physical address where the information needed to be stored. After receiving the “modify” command, the Ambient-IoT devices write the updated information into the storage information according to the corresponding physical address which is predefined or indicated. Then, Bob checks and finds the status of the orthopedic forceps have been updated with “Normal”.

The other part of the instrument is broken, so a new set of orthopedic forceps are purchased. Before the new instrument is put into use, related instrument information needs to be stored in the Ambient IoT devices which are stuck on the new

instrument. The hospital inventory management platform asks 5G network to send "write" command to write the status of the new orthopedic forceps. The "write" command includes the status information of orthopedic forceps which can include the serial number of the instrument (16 bits), usage status (2 bits), usage records (128 bits), years of use (6 bits), number of usage (18 bits), maintained status (2 bits), other potential information, which are transparent to 5G network. Considering the Ambient-IoT devices attached on the repaired orthopedic forceps are without application software, the 5G network translates and sends "write" commands to the Ambient-IoT devices attached on the new orthopedic forceps. In the status information of orthopedic forceps, each type of information is associated with a predefined or indicated physical address. After receiving the command, the Ambient-IoT devices store the status information at the corresponding physical address. Bob checks and finds the status of the new orthopedic forceps have been written.

Considering medical instruments density, the average end to end service latency is expected to be hundreds ms level to avoid excessive application delay. Further, the user experienced data rate can be calculated considering the status information less than 176 bits within time period e.g.100 ms.

5.11.4 Post-Conditions

Hospital Z utilizes Ambient-IoT service to support the online maintenance for its medical instrument. Bob can modify the status information of medical instrument or write status information for new medical instrument.

5.11.5 Existing features partly or fully covering the use case functionality

None

5.11.6 Potential New Requirements needed to support the use case

[PR 5.11.6-001] The 5G system shall be able to communicate with an Ambient IoT device.

[PR 5.11.6-002] The 5G system shall be able to provide communication service with KPIs listed in Table 5.11.6-1 for the Ambient IoT devices.

Table 5.11.6-1: KPIs for use case of Medical Instrument modification

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy

Medical instrument inventory management and positioning	Several seconds	99 %	NA	<2kbit/s (note 1)	176bit	≥1000/km^2 (note 2)	50m indoor 200m outdoor	NA	Static or walking speed <6km/h	NA	NA	NA	3 m to 5 m indoor
Note 1: User experienced data rate is calculated based on inventory information (176 bits) within time period of e.g. 100 ms; Note 2: It refers typical medical instrument density condition in Chinese hospital.													

5.12 Use case on Ambient IoT service for personal belongings finding

5.12.1 Description

For a smart home application scenario, discovery of personal item becomes one of the most important applications. A lot of personal items are in home, such as keys, passports, bank cards, wallets, children's toys, clothes etc. It is quite usual that people may forget where their items are so that they have to waste time to find them. Ambient IoT technology will help to find people's items at home much more efficiently.

Most of personal items which are easily to be lost are with a small size. For example, a key has a length of several centi-meters. A passport will have a size of 8.8*12.5 centi-meters. And typically, these things will be put in a storage box or a drawer. However, Ambient IoT devices can be easily attached to those small items.

Ambient IoT can provide a promising way for house asset management. Ambient IoT devices use energy harvested from heat or radio waves. Therefore, the device can work without a conventional battery and can work for a long-time duration, e.g., > 20 years.

However, the harvested energy would be very limited. For example, only tens of micro-watts power can be harvested if the energy is harvested from radio waves. Hence, it will put constraint on the maximum power consumption for Ambient IoT device(e.g. up to several hundred microwatts [81] [82] [83]). The device shall work with ultra-low power consumption.

For smart home application, the typical required communication distance would be less than 10 meters.

For home assert management application scenario, usually the device ID needs to be transmitted for discovery of personal item and the size of typical ID would be [96] bits [5]. Within a house of around 100m^2, 100~500 devices need to be deployed to manage most of the important items. A data rate of 10kbit/s is expected.

Usually, it needs to determine the position of the Ambient IoT device for home assert management application scenario. A positioning accuracy of around 1 meter is required.

Sometimes in daily life, Mickey cannot remember where he put his wallet, or his favourite pair of shoes. He may become crazy if he is in a hurry to go out.

With Ambient IoT service provided by 5GS, Mickey can attach Ambient IoT Devices to his wallet and shoes. Then he can easily find them using his mobile phone which supports Ambient IoT service. The Ambient IoT Device is solely dependent on harvested ambient energy, being maintenance-free, of extremely-low complexity, weight, and size.

5.12.2 Pre-conditions

Mickey bought a mobile phone supporting Ambient IoT service. He also obtains multiple Ambient IoT devices for personal belongings finding.

Bob is a neighbour to Mickey and Bob also has some Ambient IoT devices attached to his belongings.

Both licensed and unlicensed spectrum are applicable to Ambient IoT service.

5.12.3 Service Flows

1. Mickey's and his roommate Minnie's mobile phones may be authorized by their mobile operators to perform the Ambient IoT service. Both of their cell phones can send signal to Ambient IoT devices.

2. Mickey attaches one Ambient IoT device to his wallet, registers "Mickey's wallet" to the application server (e.g., Mickey's mobile phone obtains Ambient IoT device information including the device ID and transfer the information to the application server).

3. When Mickey wants to find his wallet, he opens the application in his mobile phone to search his wallet.

4. Mickey's mobile phone searches to the Ambient IoT device attached to his wallet.

   The Ambient IoT device attached to his wallet responds to the mobile phone thus it can be easily identified the wallet is nearby. Then, Mickey's mobile phone obtains the position of the wallet and it displays the position of the wallet (e.g., relative position of the mobile phone).

   Meanwhile the Bob's Ambient IoT device also receives the request but it does not respond to Mickey's cell phone.

5. The other day, Mickey goes out to the bank. When he arrives at the bank, he cannot find his wallet. He would like to check whether his wallet is left at home. Mickey opens the application in his mobile phone and authorizes Minnie's mobile phone to search the Ambient IoT device attached to his wallet. The application server requests Minnie's mobile phone to search the Ambient IoT device attached to Mickey's wallet. Minnie's mobile phone searches the Ambient IoT device and can determine it is at home when receiving from the Ambient IoT device attached to Mickey's wallet. In addition, Minnie's mobile phone can further determine the relative position of it using positioning service. Minnie's mobile phone sends the position information of the Ambient IoT device to the application server. Consequently, Mickey sees his wallet is at home through the application in his mobile phone.

   Mickey's mobile phone, Minnie's mobile phone and the Ambient IoT Device to Mickey's wallet may belong to one Personal IoT Networks as described in TS 22.261 [8] clause 6.38.

6. Mickey left his wallet at a bus stop. He can find his wallet with the help of the base station near the bus stop and UEs nearby. In order to implement this, the application server requests from the 5G system about the position of the Ambient IoT device attached to Mickey's wallet. The application server may indicate an area, where the wallet is possibly lost. Upon receiving the request, the 5G system can ask the RAN nodes or the UEs (which are within or near the area and are allowed to provide positioning service to Ambient IoT devices) to help searching the Ambient IoT device, identify its position in the bus stop and send the position information to the application server.

5.12.4 Post-conditions

Thanks to the Ambient IoT service provided by the 5G system, Mickey can find his wallet as soon as possible, both for indoor and outdoor cases.

The information of Bob's Ambient IoT device will not be exposed.

5.12.5 Existing features partly or fully covering the use case functionality

None.

5.12.6 Potential New Requirements needed to support the use case

[PR.5.12.6-001] The 5G system shall support to authorize a UE to obtain device information of an Ambient IoT device.

[P.R.5.12.6-002] The 5G system shall be able to collect information from a specific Ambient IoT device.

[P.R.5.12.6-003] The 5G system shall be able to provide information of a specific Ambient IoT device to the trusted 3rd party.

NOTE: The request from 3rd party can include the requested Ambient IoT device identity, the requested service area to find the Ambient IoT device, the requested information of an Ambient IoT device includes position information.

[PR 5.12.6-004] The 5G system shall be able to support indoor and outdoor positioning for Ambient IoT devices.

[PR. 5.12.6-005] The 5G system shall be able to support an Ambient IoT device to validate a UE which communicates with the device.

[PR. 5.12.6-006] The 5G system shall support to validate an Ambient IoT device.

[PR.5.12.6-007] The 5G system shall be able to provide Ambient IoT service with following KPIs:

Table 5.12.6-1: Ambient IoT service KPI for personal belongings finding

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Personal belongings finding (indoor)	1 s	99.9%	NA	<1 kbit/s	<1 kbits (Note 1)	<5 per 100 m^2	10 m	<200 m^2	Static	1 per hour	1 s	99%	1-3 m
Personal belongings finding (outdoor)	1 s	99.9%	NA	<1 kbit/s	<1 kbits (Note 1)	<10 per 100 m^2	100 m	Up to the whole PLMN	Static	1 per hour	1 s	99%	several 10m
NOTE 1: The payload includes Ambient IoT device information, e.g., Ambient IoT device ID [14] [5].													

5.13 Use case on Ambient IoT for Base Station Machine Room Environmental Supervision

5.13.1 Description

A cell site is composed of an outdoor cell tower and an indoor base station machine room, as shown in the figure below. A base station machine room (BSMR) includes the BBU cabinet, the power cabinet, the battery unit, the air conditioner and several cables. Unexpected network outages and electrical outages can be costly during base station operation.

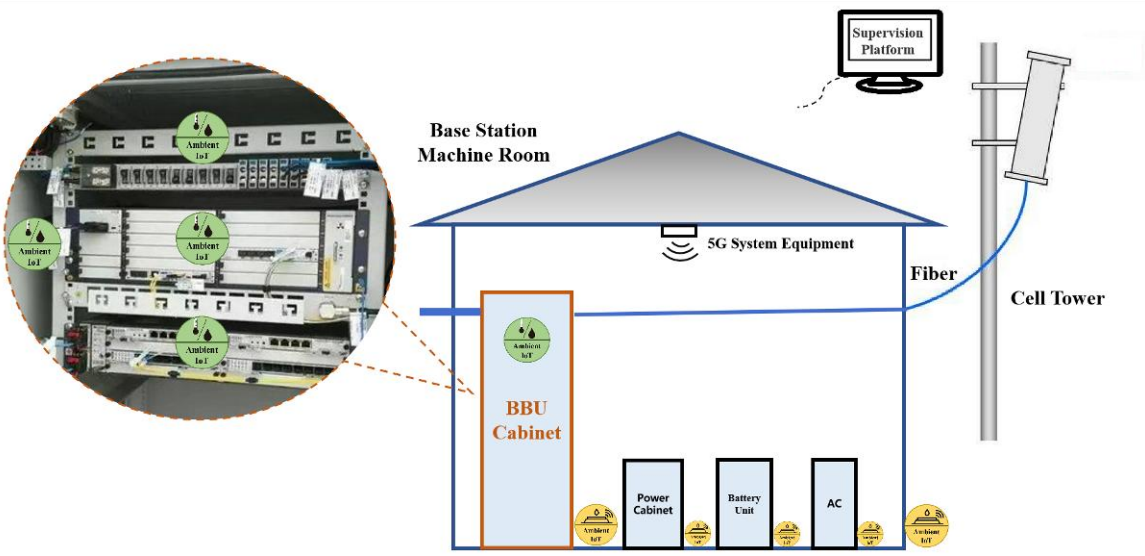


Figure 5.13.1-1: BSMR Environmental Supervision

Leaky air conditioners, groundwater, water leakage from underground pipes and severe weather such as rainstorms may cause the entire BSMR to shut down. Therefore, water leakage monitoring is an important part of BSMR environmental supervision. In addition, the temperature, humidity and other environmental parameters of BSMR also need to be monitored. Any abnormality of parameters will affect normal operation of equipment, which will lead to the deterioration of the network service quality or even lead to disconnection.

Since Ambient IoT devices are low-cost and maintenance-free, these devices can be deployed in the internal and external locations where water is easily flooded to monitor water leakage in BSMR. Ambient IoT Devices can also be deployed inside the BBU cabinet to monitor the temperature and humidity parameters of BSMR in time, which can help to detect potential problems early and reduce the probability of network outage. Meanwhile, we can deploy Ambient IoT Devices on other equipments in BSMR, for example, a backup BBU (which has been placed in the machine room but has not been powered up) can achieve periodical equipment inventory to avoid being stolen. In addition, periodical equipment inventory can help to monitor whether these cabinets have enough empty grids to meet the requirement of base station expansion.

5.13.2 Pre-conditions

1. Ambient IoT devices are deployed inside the BBU cabinet, outside the bottom of the BBU cabinet, the power cabinet, the battery unit, and the air conditioner, which can interact with 5G system.

2. Ambient IoT devices can collect energy from the environment through radio waves, vibration, light or other ways to realize message transmission and power supply of temperature sensors, humidity sensors and water-logging sensors.

3. Ambient IoT devices can per-set a threshold (such as a 50°C temperature threshold) based on the user's requirements.

4. The 5G system equipment which can interact with Ambient IoT devices and send the collected information to the supervision platform is deployed in the machine room according to the needs of BSMR environmental supervision;

5.13.3 Service Flows

1. The supervision platform starts a monitoring task request to 5G system.

2. The 5G system performs the monitoring task by transmitting signals periodically to activate Ambient IoT devices.

3. The Ambient IoT devices measure the environmental parameters and send the obtained information to the 5G system. For temperature and humidity monitoring, typically, the single packet size is 96bits and the sampling rate is 10Hz, therefore, the data generation per Ambient IoT device is about 960bits/s. For water-logging monitoring, data is generated only when water leaks occur and the data packet size is 96bits typically.

4. If the monitor data exceeds the per-set threshold, the Ambient IoT devices would report sensor information actively to the 5G system immediately.

5. The 5G system sends the acquired information to the supervision platform. The supervision platform analyzes this information to diagnoses the health status of all monitored equipment.

5.13.4 Post-conditions

With the support of 5G network, BSMR environment can be monitored more efficiently to reduce the risk of network outages, electrical outages and other failures.

5.13.5 Existing features partly or fully covering the use case functionality

In previous releases, SA1 has finished several studies about IoT topic to introduce SA1 requirements in TS 22.011, TS 22.278, TS 22.368 and TS 22.261 to address requirement for IoT business about device lifetime, power consumption, data transmission and communication mechanism.

5.13.6 Potential New Requirements needed to support the use case

[P.R.5.13.6-001] The 5G system shall be able to support communication with Ambient IoT devices.

[P.R.5.13.6-002] The 5G system shall be able to support suitable security mechanisms for Ambient IoT devices, including encryption and data integrity.

[P.R.5.13.6-003] The 5G system shall be able to support suitable mechanisms to authenticate and authorize Ambient IoT devices.

[P.R.5.13.6-004] The 5G system shall support energy efficient communication mechanisms for Ambient IoT devices (i.e., minimizing the device communication power consumption).

[P.R 5.13.6-005] The 5G system shall support transferring data collected from Ambient IoT devices to a trusted 3rd party.
[P.R 5.13.6-006] The 5G system shall be able to provide Ambient IoT service with following KPIs.

Table 5.13.6-1: KPI Table of Base Station Machine Room Environmental Supervision

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
BSMR environmental supervision	30s	99%	99.9%	<1kbit/s (Note 1)	96bits	1.5/ m^2 (Note 2)	30m indoors	NA	Stationary	NA	NA	NA	NA
NOTE1: The data rate generated by temperature, humidity, water-logging monitoring is typically less than 1kbit/s. NOTE2: The device density is calculated based on an individual BSMR, where typically about 20 Ambient IoT devices are required in one BSMR and the dimension of a typical BSMR is around 12 m^2.													

5.14 Use case on indoor positioning in shopping centre using Ambient IoT

5.14.1 Description

Shopping has always been an important part of our daily life and many giant shopping centres have been established all over the world. A shopping centre offers a wide range of services and products, including large supermarkets, a collection of retail stores, restaurants, banks, theatres, fitness and leisure facilities, underground parking areas, professional offices and other establishments. While enjoying various services in a giant shopping centre, customers sometimes find it troublesome to locate the target store or restaurant or find their own cars in the parking area due to lack of accurate indoor positioning and navigation system.

Ambient IoT can provide a promising solution for accurate indoor positioning. Ambient devices use energy harvested from ambient power, e.g., light, heat or radio waves etc. Therefore, such devices can work with limited energy storage capability (e.g., using a capacitor) or without any battery for extremely long time, e.g., > 10 years. Ambient device has other desired characteristics such as maintenance-free, extremely-low complexity, light weight, and small size.

The technology of Ambient IoT will provide positioning and navigation service to the customers in a shopping centre, and help them find the target shops and information much more efficiently, thus significantly improving customer satisfaction.

A shopping centre can occupy an area of tens to hundreds of thousands m^2, and it can be composed of one or multiple buildings, each of which has multiple storeys both over and underground. In the underground parking area, there can be hundreds to thousands of parking spots.

In order to fulfil the requirement, an indoor positioning accuracy of around [3] meters is required.

During weekends or holidays, Grace likes to go shopping. A big shopping centre near Grace's home, with an area of 200 thousand m^2, has opened recently. From the advertisement, Grace knows that the shopping centre has deployed a new indoor positioning and navigation system, which can help people quickly find the target places and items in the shopping centre.

5.14.2 Pre-conditions

With the help of an operator, an Ambient IoT system consisting of 50 thousand individual devices has been deployed across the entire shopping centre. Such devices are evenly distributed with 2-meter intervals in every room to help customers of the shopping centre realize locate target shops via indoor positioning. The position of each tag is measured and recorded in advance.

Grace understands that without a navigation system, it would be time-consuming for her to do shopping in a new shopping centre. So, she would like to try the new positioning and navigation system. For that, Grace bought a mobile phone supporting Ambient IoT service and subscribed the indoor positioning services.

5.14.3 Service Flows

1. Grace's mobile phones is authorized by the mobile operators to perform the Ambient IoT service. And the phone is allowed to send signal to Ambient IoT devices. Grace downloads the shopping Navigation APP and registers to the navigation service.

2. One day, Grace drives her car to the new shopping Centre for the first time. When the car arrives at the entrance of the parking area, the navigation APP in her cell phone reminds her that the indoor positioning and navigation system will provide service for her. After authorization, the mobile phone performs Ambient IoT communication service and Ambient IoT positioning service using Ambient IoT device(s), in order to get the Ambient IoT device's ID and relative position (i.e., relative distance and/or relative angle). Specifically, the system starts to work and Grace's smart phone begins to send triggering signals (the triggering signals may be sent continuously or intermittently) to the Ambient IoT devices that have been attached on the nearby walls or poles aside the road to the underground parking.

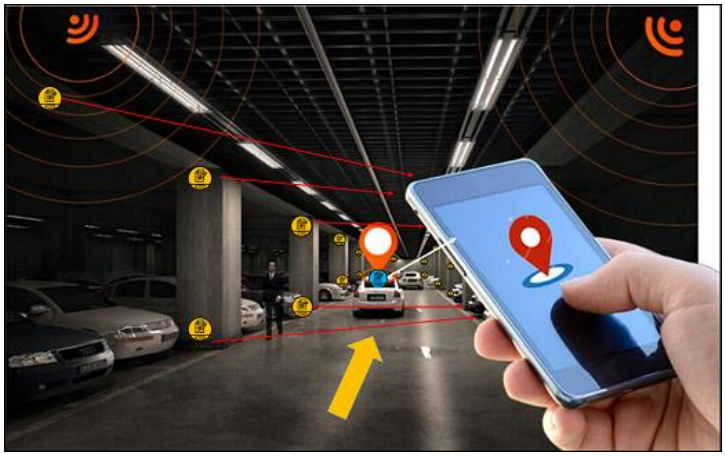


Figure 5.14.3-1: Positioning in Parking area using Ambient IoT

3. The device(s) near Grace's car is/are activated by the triggering signals. Then, the Ambient IoT device(s) responds to Grace's mobile phone. The device ID is sent to the phone, followed by a signal. Using the signal, the mobile phone device derives the relative distance and/or relative angle to the Ambient IoT Device. Using the ID and the derived relative distance and/or relative angle, the position of the car can be derived by the APP and be shown in the indoor navigation APP.

   While the car moves forwards, other Ambient IoT device(s) will sequentially respond to Grace's mobile phone and the position can be continuously updated.

   The APP navigates Grace to an empty parking spot using the navigation information and the real-time position of Grace's car. Finally, Grace parked her car in the target parking spot.

Note 1: At a moment, one or multiple Ambient IoT devices can be used for positioning. It will depend on further study in downstream groups.

Note 2: The Ambient IoT device(s) only respond to the UEs who have subscribed the indoor positioning services and authorized by the mobile operators.

4. Grace gets off her car. Today, she wants to shop for groceries and a new skirt. After checking the APP, she easily finds that the supermarket and the fashion shop are in the 1st floor the 3rd floor respectively. The APP further plans the route to the supermarket. With the help of Ambient IoT devices attached all over the parking area and the smart phone, she easily finds the elevator.

   When steps out the elevator in the 1st floor, the Ambient IoT device attached on the door of the elevator is activated by Grace's mobile phone and soon the mobile phone updates Grace's location in the navigation APP. The navigation APP switches to the 1st floor and it indicates to Grace that the entrance of the supermarket is 20 meters to the right.



Figure 5.14.3-2: Positioning in Shopping centre using Ambient IoT

5. The supermarket is very big, with an area of 20000 m^2. With the help of Ambient IoT devices, it only takes minutes for Grace to find the shelfs with the groceries that Grace wants to buy. In addition, when Grace walks near to the shelf and searches for the target items using the navigation APP, the Ambient IoT device attached on the shelf will respond to the smart phone the specific location of the target items: putting at which part of the shelf and at which layer.

6. After buying all the needed items in the supermarket, with the help of Ambient IoT devices, Grace also goes to the fashion shops and buy her preferred skirt.

5.14.4 Post-conditions

Thanks to the indoor positioning service provided by 5G Ambient IoT system, Grace has a great shopping experience in the shopping center.

5.14.5 Existing features partly or fully covering the use case functionality

Same as existing service requirements, both licensed and unlicensed spectrum are applicable to Ambient IoT device.

5.14.6 Potential New Requirements needed to support the use case of Positioning in shopping centre

[PR.5.14.6-001] The 5G system shall support be able to authorize a UE to perform Ambient IoT communication services with specific Ambient IoT devices.

[PR.5.14.6-002] The 5G system shall be able to support authorizing a UE to perform Ambient IoT positioning services with specific Ambient IoT devices.

[PR.5.14.6-003] Subject to user consent and operator's policy, the 5G system shall be able to expose the identities and positions of Ambient IoT devices to a 3rd party.

[PR. 5.14.6-004] The 5G system shall be able to support an Ambient IoT device to authenticate a UE triggering Ambient IoT services.

[PR. 5.14.6-005] The 5G system shall be able to support a UE to verify an Ambient IoT device's identity.

[PR.5.14.6-006] The 5G system shall be able to provide Ambient IoT service with following KPIs:

Table 5.14.6-1: Ambient IoT service KPI

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy

Parking area (e.g. in shopping centre)	0.5 s	99.9%	NA	<1 kbit/s	96 bits (Note1)	2500/ 10000m^2	10 m	NA	NA	NA	0.5 s	90%	3 m (Note 2, Note3)
Shopping area (e.g. in shopping centre)	0.5 s	99.9%	NA	<1 kbit/s	96 bits (Note1)	2500/ 10000m^2	10m	NA	NA	NA	0.5 s	90%	3 m (Note 2, Note 3)
NOTE 1: The payload includes Ambient IoT device information, e.g., Ambient IoT device ID[5]. NOTE 2: The positioning accuracy can be applied to horizontal NOTE 3: This value depends partly on the actual communication range.													

5.15 Use Case on Ambient IoT enablement for smart laundry

5.15.1 Description

Washing machines have been upgraded to enable automated washing process, freeing people from heavy laundry chores. Currently, washing machines are equipped with various functionalities and laundry modes, which targets to different types or materials of clothes. Therefore, there is increasing demand to determine an appropriate laundry mode based on the clothes to be washed by considering the colour, fabric, material, shape and stains in order to achieve intelligent laundry with water saving and electricity saving.

Ambient IoT service provided by 5G can be expected to meet the demanding needs of smart laundry. Ambient IoT devices installed with wireless sensors can be attached to clothes to monitor some parameter values such as temperature and humidity. Firstly, the Ambient IoT devices can store clothing information such as colour, fabric, materials and shape. Additionally, the parameter values can be used to detect sweat stains. The smart appliance application can request the 5G network to perform inventory of Ambient IoT devices to obtain this clothing information and recommend an appropriate laundry mode with suitable laundry parameters according to the clothing information. The recommended laundry mode with suitable laundry parameters will be transmitted to the washing machine so that the washing machine will wash the clothes with corresponding procedure, achieving better user experience and saving the water and electricity.

5.15.2 Pre-conditions

Cindy is a big fan of Company X's smart appliances and has bought various appliance from Company X. She downloads Company X's smart home application, registers her user account and provide the information of the smart appliances such as appliances ID and/or name on the application as well.

Recently, Cindy has bought a smart washing machine from Company X. Cindy registers the brand new washing machine on the Company X's smart home application. In order to using the intelligent function of the smart washing machine, Cindy bought clothes attached with Ambient IoT device that is installed with wireless sensors to detect the temperature and humidity of the clothes. When the clothes are produced, the Ambient IoT device attached on the clothes stores the clothing information such as colour, fabric, materials and shape. Cindy registers the clothes to the smart home application. During the registration, Cindy will provide the ID of the Ambient IoT device attached on the clothes to the application. Optionally, Cindy can also take a photo of the clothes and upload it to the application so that the application can display the clothes on the screen. Since the information of Cindy's smart washing machine and the clothes are registered to her user account, the application platform can associate the Cindy's clothes with the smart washing machines owned by Cindy.

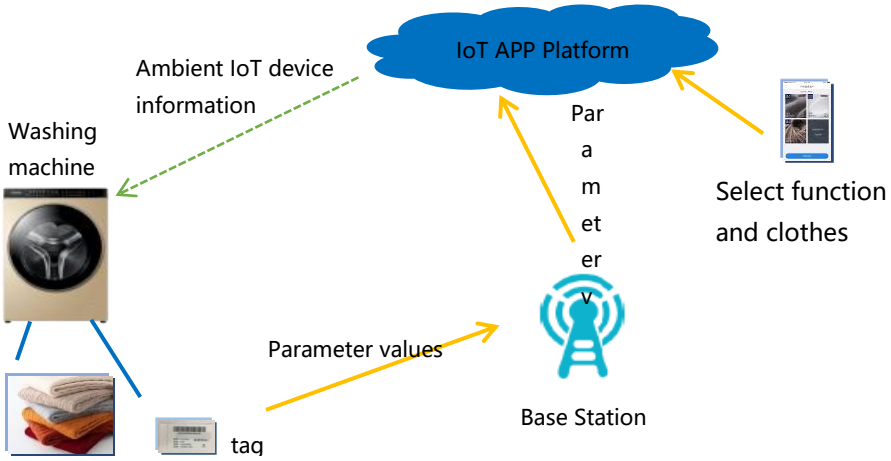


Fig. 5.15.2-1: Ambient IoT in Smart Laundry

5.15.3 Service Flows

1. Cindy is going to run in the park. Before leaving her home, she makes sure that information of the sportswear (i.e. the information of the Ambient IoT device attached on the sportswear) she wore has been registered to the application.

2. Cindy wore the sportswear to go for a run. After she finished running and went back home, she wanted to wash her sportswear.

3. Cindy opened the Company X's smart home application and chose the intelligent washing function. Additionally, she selected the clothes to be washed (i.e. the sportswear she just wore) on the application.

4. The Company X's IoT application platform can request the 5G network to inventory and report the Ambient IoT device attached to the sportswear based on the Ambient IoT device ID which has been registered on the application by Cindy. The Company X's IoT application platform can negotiate with 5G network about the frequency of inventory and report of Ambient IoT devices attached on the clothes so that the parameter values from the sensor can be obtained by Company X's IoT application platform with a certain frequency (e.g., every 5 minutes in a certain period or a single report). The negotiation between the Company X's IoT application platform and the 5G network can be based on the request from a user. For example, when Cindy is going out for a run, she can set the report frequency as every 5 minutes. After she finished running, she can stop periodically report and request for a single report before she washes her clothes.

5. Based on the request from the Company X's IoT application platform, the 5G network transmits signals intending to start inventory process. The Ambient IoT device harvests power from the environment. When detecting the signals from the 5G network, the Ambient IoT device can react by starting random access.

6. When the Ambient IoT device accesses to the 5G network successfully, the 5G network obtains the information of the Ambient IoT device (e.g., the color, fabric, material, shape and the detected parameter values such as temperature and humidity of the clothes) and send it to Company X's IoT application platform.

7. Company X's IoT application platform can obtain the Ambient IoT device information (e.g., the color, fabric, material, shape and the detected parameter values such as temperature and humidity of the clothes) via 5G network. Based on the analysis of the information reported by the Ambient IoT devices (e.g., the color, fabric, material, shape and the detected parameter values such as temperature and humidity of the clothes), the Company X's IoT application platform will determine an appropriate laundry mode. The recommended laundry mode will be sent to Cindy's smart washing machine, triggering the smart washing machine to wash the sportswear with a proper laundry procedure.

5.15.4 Post-conditions

Thanks to the Ambient IoT service provided by the 5G system, Company X's customers can enjoy smart washing machines and better user experience on laundry.

5.15.5 Existing features partly or fully covering the use case functionality

SA1 has performed various studies on IoT in previous releases, where related normative stage 1 requirements are introduced in TS 22.011 [9], TS 22.278 [7], TS 22.368 [6], and TS 22.261 [8].

TS 22.011 introduces access control for MTC, examples of periodic network selection attempts are:

For UEs only supporting any of the following, or a combination of, NB-IoT, GERAN EC-GSM-IoT [18], and Category M1[13] of E-UTRAN enhanced-MTC, the UE shall interpret the interval value to be between 2 and 240 hours, with a step size of 2 hours between 2 and 80 hours and a step size of 4 hours between 80 and 240 hours.

In the absence of a permitted value in the SIM/USIM, or the SIM/USIM is phase 1 and therefore does not contain the datafield, then a default value of 60 minutes, shall be used by the UE except for those UEs only supporting any of the following, or a combination of: NB-IoT, GERAN EC-GSM-IoT [18], and Category M1 [17] of E-UTRAN enhanced-MTC. For those UEs a default value of 72 hours shall be used.

NOTE: Use of values less than 60 minutes may result in excessive UE battery drain.

TS 22.368 addresses features of MTC communication and service requirements related to MTC device triggering, addressing, identifiers, low mobility, small data transmission, infrequent MT communication, security, remote MTC device management, group-based MTC features including policing and addressing, etc. Example requirements are:

The system shall provide mechanisms to lower power consumption of MTC Devices.

The system shall provide mechanisms for the network operator to efficiently manage numbers and identifiers related to MTC Subscribers.

TS 22.261 captures some important service requirements for IoT, e.g.

The 5G system shall support a secure mechanism for a home operator to remotely provision the 3GPP credentials of a uniquely identifiable and verifiably secure IoT device.

The 5G system shall support a secure mechanism for the network operator of an NPN to remotely provision the non-3GPP identities and credentials of a uniquely identifiable and verifiably secure IoT device.

An IoT device which is able to access a 5G PLMN in direct network connection mode using a 3GPP RAT shall have a 3GPP subscription.

The 5G system shall allow the operator to identify a UE as an IoT device based on UE characteristics (e.g. identified by an equipment identifier or a range of equipment identifiers) or subscription or the combination of both.

An IoT device which is able to connect to a UE in direct device connection mode shall have a 3GPP subscription, if the IoT device needs to be identifiable by the core network (e.g. for IoT device management purposes or to use indirect network connection mode).

The 5G system shall support operator-controlled alternative authentication methods (i.e. alternative to AKA) with different types of credentials for network access for IoT devices in isolated deployment scenarios (e.g. for industrial automation).

The 5G system shall support a suitable framework (e.g. EAP) allowing alternative (e.g. to AKA) authentication methods with non-3GPP identities and credentials to be used for UE network access authentication in non-public networks.
NOTE: Non-public networks can use 3GPP authentication methods, identities, and credentials for a UE to access network. Non-public networks are also allowed to utilize non-AKA based authentication methods such as provided by the EAP framework, for which the credentials can be stored in the ME.

The 5G system shall enable an NPN to be able to request a third-party service provider to perform NPN access network authentication of a UE based on non-3GPP identities and credentials supplied by the third-party service provider.

In these specifications, albeit the service requirements addressing traits for IoT in terms of low device power consumption, small and infrequent data transmissions, long service lifetime, and resource efficiently, the IoT devices considered in 3GPP have been assumed to be powered by at least batteries up till now. To enable extremely small, light-weight, battery-less or even disposable Ambient IoT devices that engage in basic IoT data transaction and appropriate level of operator management and charging suitable for the target scenarios, new challenges to the 5G system are foreseen and need to be addressed.

5.15.6 Potential New Requirements needed to support the use case

[PR.5.15.6-001] The 5G system shall support energy efficient communication for Ambient IoT devices while meeting the communication performance requirements.

[PR 5.15.6-002] The 5G System shall provide the network connection to address the KPIs for the use of Ambient IoT devices for preventive care in smart laundry, see table 5.15.6-1.

Table 5.15.6-1: Potential key performance requirements for the use of Ambient IoT devices for smart laundry

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Ambient IoT devices for smart laundry	>10 s	NA	NA	<100bit/s	Typically < 100 bytes	20 / 100m^2	NA	several m^2 up to 1000 m^2	Up to 6 km/h outdoor	NA	NA	NA	NA

5.16 Use case on Ambient IoT service for automated supply chain distribution

5.16.1 Description

Currently, there is an increasing demand for personalizing user requirements and customizing home appliance. Production enterprises customize products and achieve production-to-order according to different users' requirements can bring benefits. It not only enhances user experience, but also reduces stock costs and the risk of overstock. Production-to-order has high requirements for flexible digital and intelligent manufacturing. At the beginning of the production, each product has been targeted to its customer. Therefore, how to achieve the efficient management of whole process including parts supply, manufacturing, stocktaking, logistics, transportation and delivery is critical and essential. This means the entire logistics management process involves identification of products from production to delivery, including transportation across public area. Therefore, for this use case the communication service availability with sufficient 5G network coverage are important. The automated supply distribution enables enterprises to reduce management cost and improve the competitiveness of products.

Ambient IoT service provided by 5G can meet the demanding needs of efficient management of the whole process. Firstly, 5G system enables the communication of Ambient IoT devices with needed performance. Secondly, with the Ambient IoT devices operating solely depending on harvested ambient energy, the Ambient IoT devices are maintenance-free, which also eliminates replaceable batteries being discarded into the environment. And the feature of extremely-low complexity, weight, and size make the Ambient IoT devices suitable to use in practice in an affordable way. Thirdly, the 5G system can provide Ambient IoT device positioning services, which enables enterprises to monitor and track the products attached with Ambient IoT device from manufacturing to delivery, ensuring that the customized product is delivered to the right customer with right route. Last but not least, the 5G system can enable the enterprise to perform authentication and authorization of the Ambient IoT devices, ensuring that the information stored in the Ambient IoT devices would not be accessed by untrusted third party.

5.16.2 Pre-conditions

Company X is a well-known home appliance manufacturer. Apart from manufacturing various home appliances such as washing machine, television, intelligent wardrobe and so on, it also provides customized and personalized services to its customers such as sales agents. Different sales agents may order different types of home appliances with different quantities according to the demand of the region they sale to. Company X operates its own private warehouses to store manufactured goods until they are ordered. Owned by Company X, the private warehouses are customized according to its own needs, and they are typically located close to production plants. The warehouse management provides insight in inventory and by exchanging inventory information with other functional areas (e.g. logistics, trade) the overall business performance is improved. Compared with public warehousing, the typical storage area of a private warehouse is smaller, typically around 10000 square feet. This is representative according to known statistics of warehouse area dimensions [85]. Based on these typical dimensions, indoor communication range around 20m is sufficient to provide connectivity inside the warehouse storage area.

The southern regions in Country C have many economically developed cities, while western regions consist of some developing cities. Therefore, Company X has designed different types of washing machines to meet various demands. For example, type A washing machine is intelligent, equipped with several laundry modes according to the colour, texture, material and shape of the clothes. Type B washing machine is a power saving and water saving washing machine, which achieves washing clothes with high efficiency and low power and water consumption. Sale agent A from southern region may order 100 type A washing machines and 50 type B washing machines, while sale agent B from western region may order 30 type A washing machines and 90 type B washing machines. Based on the orders from different sale agents, Company X will package the corresponding types of washing machines with corresponding quantity together. Afterwards, different packages for different sale agents will be loaded to transportation vehicles accordingly for different destinations.

Company X has a service level agreement with service provider Y to deploy 5G network with sufficient coverage in the service area to enable the communication of Ambient IoT devices with the network.

Before the transportation vehicle enters the loading stack for product transportation, it is necessary to inventory the products to be transported in the warehouse in order to ensure that the products have been recorded. During the loading process, it is required to ensure that the products are loaded to the corresponding transportation vehicles. Moreover, during the transportation, Company X may also need to monitor the transportation route by tracking the products in order to ensure that the products are transported on the right route. Otherwise, the products will be transported to the wrong place which is time and cost consuming.



Fig. 5.16.2-1: Ambient IoT for automated supply chain distribution

5.16.3 Service Flows

1. Company X generates in its logistics management system a list of unique IDs for Ambient IoT devices to be used for products such as washing machines. Since these washing machines are going to be distributed to different sales agents for product sales across public areas, Company X may require an authentication and authorization mechanism before Ambient IoT devices start to communication to ensure that certain level of security could then be enforced to prevent the washing machines information from being obtained by any other

untrusted third-party companies. In this case, an authentication and authorization mechanism is needed for these Ambient IoT devices.

2. Ambient IoT devices are attached onto washing machines.

3. Before transporting the washing machines, Company X's inventory system can request the 5G network to perform inventory and report the information of Ambient IoT devices which were attached on the washing machines in order to confirm that the types and the number of washing machines to be transported are accurate.

4. Ambient IoT devices harvest power from the environment. Based on the request from the Company X's inventory system, the 5G network transmits signals intending to start inventory process. When detecting the signals from the 5G network the Ambient IoT devices can react by responding to connect to the 5G network.

5. When the Ambient IoT devices access to the 5G network successfully, the 5G network obtains the information of the Ambient IoT devices and send it to Company X's inventory system.

6. Based on the orders from several sale agents, different types of washing machines will be packaged together.

7. When the washing machines are loaded to the transportation vehicle, the Company X's inventory system can request the 5G network to perform positioning services. Since the transportation vehicles stop at fixed positions, with the positioning services, the Company X's inventory system can monitor whether the package of the washing machines is loaded to the corresponding transportation vehicles. Since the size of the transportation vehicles is around 10m*3m, the position service should support around 3m to 5m accuracy to meet the demand for location.

8. When the packages of washing machines are in transit, Company X's inventory system can request the 5G network to periodically perform inventory and location reporting (e.g. cell level granularity) of Ambient IoT devices which were attached onto the washing machines in order to make sure that the washing machines are still on the right route. As the transportation vehicles may go to different destinations via different highways, mobile network operator O can deploy Ambient IoT services at highway toll station to perform inventory and locate the route accordingly. Because of the request from the Company X's inventory system, the Ambient IoT devices perform authentication and authorization mechanism before sending the information in order to ensure that it is the trusted inventory system that performs the inventory of the Ambient IoT devices.

9. Based on the request from Company X's inventory system, the 5G network periodically transmits signals intending to start inventory process and record the relationship between the Ambient IoT devices and the location information. As the purpose of identifying the location during transportation is to crosscheck whether goods are on the correct routes, cell-level accuracy of location is sufficient. The 5G network reports the location information to the Company X's inventory system.

10. According to the report from the 5G network, Company X's inventory system can locate these washing machines with the location information. If the location information is not aligned with the right route, the Company X may confirm with the staff to check whether there are mistakes. In this way, Company X can find the mistakes as soon as possible in order to decrease the cost loss.

11. When the washing machines are transported to the corresponding sale agents, Company X's inventory system can request the 5G network to perform inventory and report the information of Ambient IoT devices which were attached on the washing machines in order to confirm that the washing machines are transported to the right place.

5.16.4 Post-conditions

Thanks to the Ambient IoT service provided by the 5G system, manufacturer Company X can distribute products with automatic management process, largely improve the product and service delivery efficiency.

5.16.5 Existing features partly or fully covering the use case functionality

SA1 has performed various studies on IoT in previous releases, where related normative stage 1 requirements are introduced in TS 22.011 [9], TS 22.278 [7], TS 22.368 [6], and TS 22.261 [8].

TS 22.011 introduces access control for MTC, examples of periodic network selection attempts are:

For UEs only supporting any of the following, or a combination of, NB-IoT, GERAN EC-GSM-IoT [18], and Category M1[13] of E-UTRAN enhanced-MTC, the UE shall interpret the interval value to be between 2 and 240 hours, with a step size of 2 hours between 2 and 80 hours and a step size of 4 hours between 80 and 240 hours.

In the absence of a permitted value in the SIM/USIM, or the SIM/USIM is phase 1 and therefore does not contain the datafield, then a default value of 60 minutes, shall be used by the UE except for those UEs only supporting any of the following, or a combination of: NB-IoT, GERAN EC-GSM-IoT [18], and Category M1 [17] of E-UTRAN enhanced-MTC. For those UEs a default value of 72 hours shall be used.

NOTE: Use of values less than 60 minutes may result in excessive UE battery drain.

TS 22.368 addresses features of MTC communication and service requirements related to MTC device triggering, addressing, identifiers, low mobility, small data transmission, infrequent MT communication, security, remote MTC device management, group-based MTC features including policing and addressing, etc. Example requirements are:

The system shall provide mechanisms to lower power consumption of MTC Devices.

The system shall provide mechanisms for the network operator to efficiently manage numbers and identifiers related to MTC Subscribers.

TS 22.261 captures some important service requirements for IoT, e.g.

The 5G system shall support a secure mechanism for a home operator to remotely provision the 3GPP credentials of a uniquely identifiable and verifiably secure IoT device.

The 5G system shall support a secure mechanism for the network operator of an NPN to remotely provision the non-3GPP identities and credentials of a uniquely identifiable and verifiably secure IoT device.

An IoT device which is able to access a 5G PLMN in direct network connection mode using a 3GPP RAT shall have a 3GPP subscription.

The 5G system shall allow the operator to identify a UE as an IoT device based on UE characteristics (e.g. identified by an equipment identifier or a range of equipment identifiers) or subscription or the combination of both.

An IoT device which is able to connect to a UE in direct device connection mode shall have a 3GPP subscription, if the IoT device needs to be identifiable by the core network (e.g. for IoT device management purposes or to use indirect network connection mode).

The 5G system shall support operator-controlled alternative authentication methods (i.e. alternative to AKA) with different types of credentials for network access for IoT devices in isolated deployment scenarios (e.g. for industrial automation).

The 5G system shall support a suitable framework (e.g. EAP) allowing alternative (e.g. to AKA) authentication methods with non-3GPP identities and credentials to be used for UE network access authentication in non-public networks.
NOTE: Non-public networks can use 3GPP authentication methods, identities, and credentials for a UE to access network. Non-public networks are also allowed to utilize non-AKA based authentication methods such as provided by the EAP framework, for which the credentials can be stored in the ME.

The 5G system shall enable an NPN to be able to request a third-party service provider to perform NPN access network authentication of a UE based on non-3GPP identities and credentials supplied by the third-party service provider.

In these specifications, albeit the service requirements addressing traits for IoT in terms of low device power consumption, small and infrequent data transmissions, long service lifetime, and resource efficiently, the IoT devices considered in 3GPP have been assumed to be powered by at least batteries up till now. To enable extremely small, light-weight, battery-less or even disposable Ambient IoT devices that engage in basic IoT data transaction and appropriate level of operator management and charging suitable for the target scenarios, new challenges to the 5G system are foreseen and need to be addressed.

5.16.6 Potential New Requirements needed to support the use case

[PR 5.16.6-001] The 5G system shall be able to support a mechanism to authenticate and authorize Ambient IoT devices.

[PR 5.16.6-002] The 5G system shall optimize mobility management support for non-stationary Ambient IoT devices that are unable to initiate communication towards the network.

[PR 5.16.6-003] The 5G System shall allow an operator to manage (e.g. provision, authenticate, authorise, etc.) Ambient IoT devices that have limited or no power source.

[PR 5.16.6-004] The 5G System shall be able to provide suitable and secure means to report to an authorized third-party the location of Ambient IoT devices.

[PR 5.16.6-005] The 5G System shall provide the network connection to address the KPIs for the use of Ambient IoT devices for automated supply distribution, see table 5.16.6-1.

Table 5.16.6-1: Potential key performance requirements for the use of Ambient IoT devices for automated supply chain distribution

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Ambient IoT devices for automated supply distribution	>10 s	99 %	NA	<100 bit/s	Typically, <100 bytes	<1,5 Million/km^2	30m indoor (note 1), 400m outdoor	600 000 m^2	NA	NA	NA	NA	[3] m (Indoor, 90% confidence level and in horizontal), cell-level outdoor
NOTE 1: The storage area of a private warehouse is typically around 10000 square feet, which is smaller than that of a typical public warehouse.													

5.17 Use case on Device Activation and Deactivation

5.17.1 Description

This use case illustrates the need to define capabilities that allows the end user or a third party to remotely manage the activation and deactivation of an Ambient IoT device.

The scenario describes an enterprise user who grows orchid plants for sale to end customers. The enterprise user utilises an Ambient IoT device with environmental sensors for each orchid plant to monitor its growing conditions.

5.17.2 Pre-conditions

The enterprise user has several inactive Ambient IoT devices with environmental sensors in storage.

5.17.3 Service Flows

Device activation

1. As new orchids are planted, the devices are taken out of storage and added to the new plants.

2. The enterprise user accesses an application that she uses to check the connectivity status of her Ambient IoT devices. Via this application, the enterprise user activates the devices to enable sensor data to be collected and the conditions of the new plants to be monitored.

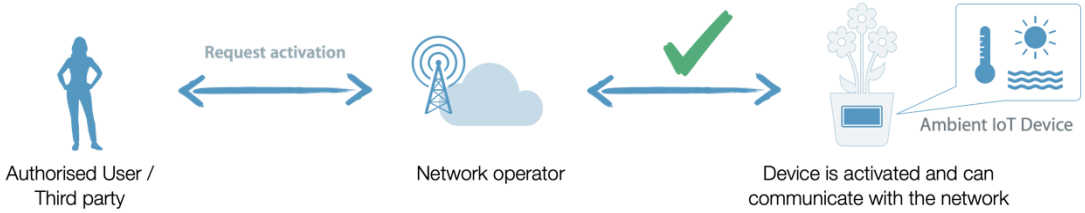


Figure 5.17.3-1: Device activation

Device deactivation

3. The enterprise user sold several mature orchid plants and wants to deactivate the associated Ambient IoT devices as she no longer needs them.

4. The enterprise user accesses an application that she uses to check the connectivity status of her Ambient IoT devices. Via this application, the enterprise user deactivates the devices associated with the sold plants.

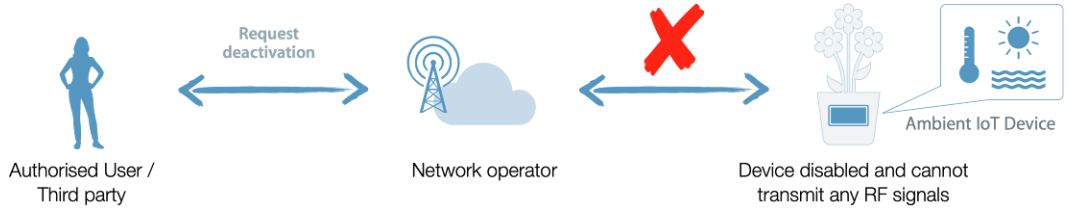


Figure 5.17.3-2: Device deactivation

5.17.4 Post-conditions

Device activation

The previously inactive Ambient IoT devices are activated and can transmit sensor data.

Device deactivation

The previously active Ambient IoT devices are deactivated and cannot transmit any RF signals.

5.17.5 Existing features partly or fully covering the use case functionality

TS 22.261 clause 6.14.1 describe the following:

During their life cycle these IoT devices go through different stages, ..., the activation of the IoT device by the preferred operator, a possible change of operators, etc. These stages need to be managed securely and efficiently.

Clause 6.14.2 defines the following requirement:

Based on operator policy, the 5G system shall provide means for authorised 3rd parties to request changes to UE subscription parameters for access to data networks, e.g., static IP address and configuration parameters for data network access.

The requirement above covers remote *UE subscription* activation and subscription suspension / deactivation.

If the subscription of an Ambient IoT device has been suspended or terminated, the device can still continually harvest energy and therefore may continue to attempt accessing the network. This can cause signalling overload as well as unwanted interference.

Therefore, there is a need to disable the device itself, and not only the subscription, so that the devices will not transmit any RF signals when suspended or terminated. Conversely, there is also a need to (re-)activate the devices.

5.17.6 Potential New Requirements needed to support the use case

[PR 5.17.6-1] Based on operator policy, the 5G system shall provide means for an authorised user or authorised third parties to request enable and disable an Ambient IoT device capability to transmit RF signals.

5.18 Use case on Fresh Food Supply Chain

5.18.1 Description

In the United States alone, food waste is estimated at between 30-40 percent of the food supply [26].

It is known that controlled environment for most of the fresh foods, like vegetables or meat, is critical for both the safety of the food [27] as well its shelf life expectancy [28-29].

In this use case, a large food supplier monitors its food supply chain by adding a simple and small form factor ambient IoT device (sticker) on to each of the Reusable Transport Item (RTIs) used for storing and transporting of the food. Example RTI can be seen in Figure 5.18.1-1. These RTIs are loaded with food at the post harvesting and packaging facilities. They are then transported to the fresh produce distribution center as seen in Figure 5.18.1-2. From there, the fresh products are routed to the local stores according to demand. After usage, these RTIs are either washed and sent back for more usage cycles or sent to a recycle center.



Figure 5.18.1-1: Example of an RTI with an ambient IoT sticker device

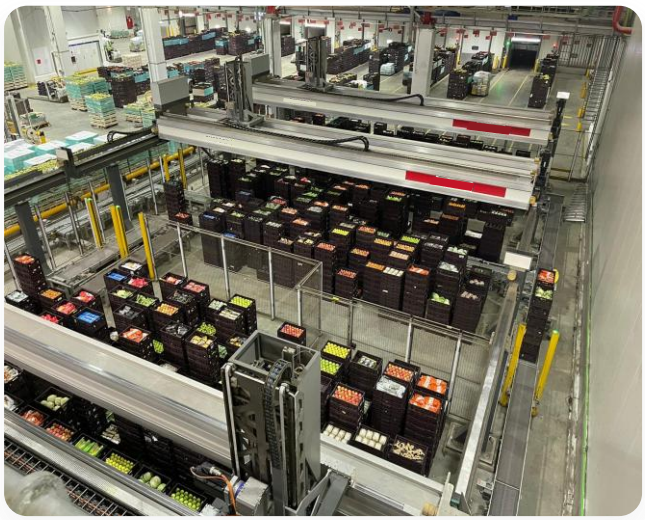


Figure 5.18.1-2: Distribution center facility for fresh food

At the harvester packaging facility, each RTI is attached with a simple, sticker form factor Ambient IoT device. The device ID is logged by the supplier using his internal records. The supplier can route the individual RTIs based on the combination of product expected longevity and real time demand from the stores. He can also use this data to alert transport company once temperature was compromised or once a specific RTI got mixed up.

This cycle can be seen on Figure 5.18.1-3



Figure 5.18.1-3: The use cycle of the RTI

The tags can operate based on intermittently harvested energy with energy storage like capacitor. This energy storage can be charged by an RF harvesting. For example, from a -30dBm received RF power with a 33% harvesting efficiency.

Since normal use is a reading once every 15 minutes, in order for the stored energy to last for 1 reading, each reading from the device should take less than 1uW*900*0.33 = 0.3 mjoules - in this use case. This number includes all energy consumed by the calculations done by the tag (like encryption/decryption), by reception, transmission, calibration and sensing.

The energy consumption of the tag in between readings is negligible.

The devices come in a sticker form factor, low complexity, and massive quantities. To be effective, they are distributed to the supplier in groups of hundreds, and the supplier expects to activate them all within few seconds from uploading them to the sticking gun. An example for their low complexity is to have their clock calibrated from the network, and in addition, this clock is less stable than larger form factor devices with higher cost.

Supplier is not interested in his devices' readings outside of the defined regions of operation (packaging facility, distribution center, stores, recycle center and the roads connecting them).

Location information is added as a meta-data by the network to the polling response as part of the service to the third party – the ambient IoT device is agnostic to this service.

Devices are expected to operate maintenance free - at least for a few years, until recycling the RTI.

5.18.2 Pre-conditions

- The ambient IoT devices are manufactured on a reel containing hundreds of stickers. The devices on the reel are configured during installation or manufacturing with a group ID configuration.
- The ambient IoT devices reel is registered at the supplier cloud server to enable secure connection through the network. The exact number of stickers per reel is not known in advance and can vary from reel to reel, as well as the order of the stickers on the reel. So all the stickers need to be accessible on the time of first use of any sticker from that reel.
- The group of ambient IoT devices are polled by the network once the server polls the network.
- Encryption and authentication of group of devices is used for the poll request and the replies.

5.18.3 Service Flows

1. Once in 15 minutes, the cloud server sends a polling request to the network with the group ID.
2. In this example, upon receiving the polling request, the network sends wakeup signals to allow calibrations by the ambient IoT devices.
3. Shortly after the wakeup signals, the network sends a polling request to the group of devices. It can either broadcast the request across its network or send it only to specific locations - based on operator policy.
4. Upon reception of the polling request, each ambient IoT device replies with its ID and temperature readings.

5. Upon reception of the replies, the network adds meta-data (e.g. which base station received the reply, what was the received power, or direction of arrival), and then forwards to the cloud server of the supplier.

6. The supplier accesses its cloud server and performs 3rd party processes on it, to optimize and control its supply chain.

7. Once RTI finished its usage, it is routed to the recycling center. In the recycling center, the ambient IoT devices are bulk deleted from the cloud server and logged out from the network.

5.18.4 Post-conditions

By using the ambient IoT devices, the entire supply chain is monitored and controlled. The ambient IoT can be used to monitor the location of each RTI, but it can also monitor its temperature, humidity or even the ethylene levels. This way, food waste is reduced to the minimum and food safety is significantly improved.

5.18.5 Existing features partly or fully covering the use case functionality

TS 22.261 in clauses 6.4.2.2, 6.4.2.3 and 6.4.2.4 adds requirement to manage control and operate bulk IoT devices in an efficient way, while minimizing signalling.

TS 22.368 in clauses 7.1.2 and 7.2.14 defines requirements for network triggering of MTC devices and also group based features such as policing and addressing.

5.18.6 Potential new requirements needed to support the use case

[PR 5.18.6-1] Based on operator policy, the 5G system shall provide means for an authorised third party to poll a group of multiple ambient IoT devices.

[PR 5.18.6.-2] The 5G system shall be able to provide ambient IoT service with following KPIs

Table 5.18.6-1: Potential key performance requirements for the use of Ambient IoT devices for food supply chain

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Ambient IoT devices for food supply chain	>1 minute	NA	NA	<0.12 bit/s (Note 1)	Typically, < 100 bits (Note 2)	1.5 Million devices/ km^2 (Note 3)	NA	30,000m^2	1 m/s	15 min (Note 1)	NA	NA	NA

NOTE 1: Based on sending 1 message of 100 bits once in 15 minutes

NOTE 2: If more sensors are used, like humidity or ethylene level, then longer message is required.

NOTE 3: This is the highest density inside the distribution center and is based on 50,000 RTIs inside a 30,000m^2 distribution center facility. See Figure 5.18.1-2.

5.19 Use case on Forest Fire Monitoring using Ambient IoT devices

5.19.1 Description

Ambient IoT is an attractive technology to adapt due to its low or zero maintenance due to its usage of emerging technologies such as Energy Harvesting, Energy Efficient (EE) communication and zero energy IoT technologies but unfortunately suffers faulty and time bound unreliable communication. Ambient IoT devices are usually small in size and operate on small computing and memory usage due to power limitation harvested from energy harvesting techniques such as RF-based, heat energy, photovoltaic, vibration energy, solar etc., Due to its limitation on the memory size and uncertainty in the power, timely communication to the Ambient IoT devices is required to avoid loss of critical data. Further, Ambient IoT devices use zero-energy technologies such as ambient backscatter communication (AmBC), compressed sensing (CS)-based random access techniques etc., These EE communications are unreliable due to drop in signal strength, data rate and drop in connection caused by radio frequency, electrical interference and environmental conditions such as rain, dampness, indoor, outdoor, buildings etc., Though some of these interference are intermittent and Ambient IoT might resume reliable communication but timely communication to and from the Ambient IoT devices is a must for mission critical systems. Due to unreliable power and communication of Ambient IoT can result in faulty and time bound unreliable communication [30-33].

Many applications such as Industrial automation, healthcare monitoring, traffic signal monitoring alert system, home monitoring, forest fire alert systems require fault-tolerant and time bound reliable systems.

Forest fire alert system. Early detection of forest fire can save animals and nature, which is required for human existence. Ambient IoT devices with smoke detectors are deployed in the forest as fire alert system as shown in the Figure-1d. Large number of Ambient IoT based sensors are required to monitor forest fire, such deployment has space, power and communication issues. Under fire these Ambient IoT based sensors can function faulty due to fire and can face intermittent power shortages due to poor signal coverage. Forest fire alert systems is a classic application, which urges for fault-tolerant and timebound reliable communication [35].



Figure 5.19.1-1: Forest Fire Monitor using fault tolerant and Reliable Ambient IoT communication [36]

5.19.2 Pre-conditions

1) California forest fire has endangered many animal and human lives, city council has decided to install Ambient IoT devices all across California forest to receive early detection.

2) The Ambient IoT devices are registered to the 5G system.

3) Forest fire monitoring using Ambient IoT devices are programmed to monitor forest fire and raise alarms to the subscribed authorities and users through 5G system.

5.19.3 Service Flows

1) On a hot summer in California, many areas in California forest have caught fire due to excessive heat.

2) Ambient IoT devices are installed in the forest and is able to detect forest fire and send a OnDemand communication to the 5G system.

3) Due to poor 5G signal coverage (bad weather conditions) and intermittent poor power harvesting, Ambient IoT devices are at the risk of faulty and jeopardize the fire monitoring capabilities.

4) Fault tolerant Ambient IoT communication helped Ambient IoT devices deployed all over forest to reliably detect forest fire and timely communicate with the 5G system so that forest fire could be putoff in a timely manner which saved lot of animal and human lives.

5) In case, if fault tolerant based Ambient IoT devices were not deployed then the forest fire monitoring system would have failed.

5.19.4 Post-conditions

1) Ambient IoT devices deployed for forest fire monitoring switches to normal mode, where it is programmed to send periodic status messages.

2) Ambient IoT devices deployed performs periodic built in tests to ensure it is able to communicate in a timely and faultless manner.

5.19.5 Existing feature partly or fully covering use case functionality

1) URLLC system design in clause 5.33 of 23.501 [6] has proposed dual redundant system to achieve ultrahigh reliability. Though - in this system design- there are dual RAN connections, PDU session established, SMF and UPF but it is designed for a UE with dual radios and not for Ambient IoT devices. URLLC system design is ultrareliable but not time bound ultra-reliable and fault tolerant. It also relies on the upper layer protocol such as IEEE 802.1 TSN (Time sensitive Network) FRER (Frame replication & elimination for reliability), to manage dual redundant systems such as replication and elimination of redundant packets or frames. Since Ambient IoT devices works on zero-power technologies, Ambient IoT devices cann't depend on such upper layer protocol.

2) Redundant user plane paths based on multiple UEs per device has been proposed in Annex F of 23.501[6]. In this system design, the device is expected to have two UE(s) and they independently connect to their RAN and have their own PDU sessions with a common DN. This system is not End-to-End fault tolerant since it has a common DN – a single point of failure- and requires dual UE(s) to achieve ultrahigh reliability. This architecture too assumes that some upper layer protocol (e.g. FRER) is used for replication and frame elimination, thus doubling the resources used over the radio. This architecture is for UE, which is not power savvy and not for an Ambient IoT devices, a fault tolerent and time bound reliable architecture for Ambient IoT is yet to be explored.

3) As per Multimedia Priority Service (MPS), mentioned in clause 5.16.5 of TS 23.501[6], allows service users priority access to the system resources under congestion, creating the ability to deliver or complete session of a high priority nature. Service users are priority users such as government officials, authorized users etc., This priority access is for special users – whose requirements are quite different from time bound reliable and fault tolerant Ambient IoT communications. Priority access as applied to Ambient IoT devices can be explored for time bound reliable and fault tolerant Ambient IoT communication.

4) RRC controls the scheduling of user data in the uplink by associating each logical channel with a logical channel priority, a prioritised bit rate (PBR), and a buffer size duration (BSD), mentioned in clause 10.5 of TS 38.300 [7]. Though these requirements for UE, these logical channels priority can be extended to Ambient IoT devices.

5) Massive Machine type Communication (mMTC) in TS 22.368 [8], defines KPI and protocol to communicate with a large number of IoT devices connected typically transmitting low volume of non-delay sensitive data. It also requires that the devices have long lasting battery. Though large number of Ambient IoT connected device communication is applicable but Ambient IoT devices operates on Energy Efficient technologies where power is unreliable. Hence these KPI and protocol may not be applicable directly but some of them can be extended to Ambient IoT communication.

5.19.6 Potential New Requirements needed to support the use case

[PR 5.19.6-001] The 5G system shall support on-demand access to/from Ambient IoT device.
[PR 5.19.6-002] The 5G system shall meet the following KPI table:

Table 5.19.6-1: Potential key performance requirements for Forest Fire Monitoring using Ambient IoT devices

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Forest Fire Monitor	> 10sec	99.9%	NA	NA	NA	100 per km^2 (NOTE 1)	[15-200] meters	[10000 – 400,000] km^2	Stationary	1hour	NA	NA	NA
NOTE 1: A typical forest fire monitors can detect fire covering up to 60 – 150 Square meters													

5.20 Use case on Smart Agriculture

5.20.1 Description

Ambient power-enabled IoT devices can be used in smart agriculture to monitor the environment and control the facilities such as irrigation system and temperature control system.

Farm A built a smart greenhouse for tomato planting. Some sensing Ambient power-enabled IoT devices are placed in the smart greenhouse to monitor the air temperature and humidity, carbon dioxide concentration, light, soil temperature, humidity and PH. Some operation Ambient power-enabled IoT devices are placed in the smart greenhouse to control the window and irrigation system. A pico cell or a reader UE (with subscription of Operator O) is placed in the smart greenhouse to communicate with the Ambient IoT devices.

These Ambient IoT devices power themselves by harvesting energy from the environment (e.g. solar, RF energy). The maximum power consumption for Ambient IoT device could be limited (e.g. several hundred micro-watts) [81] [82] [83].

Considering the greenhouse environmental control, crop growth characteristics and economic benefits, the optimal scale of greenhouse construction is 8 ~10 meters span, 80~100 meters length. The size of the greenhouse could be very huge, it is reported recently that a single greenhouse area reaches nearly 70,000 square meters, equivalent to ten standard football field size.



Figure 5.20.1-1: a picture for huge greenhouse
(https://www.holland.com/global/tourism/search.htm?keyword=greenhouse**)**

In charge of the technology is a team of eight people whose main task is to ensure that all the crops in the greenhouse grow well. To provide all-round technical support for a greenhouse of nearly 70,000 square meters, a "super brain" is needed. This computer monitors tens of thousands of sensors in the greenhouse, people can sit in front of the computer and know everything that's going on in the greenhouse. For example, the temperature and humidity in each area of the 70,000 square meters greenhouse, the temperature of the underground heating tube, the concentration of carbon dioxide are how much, whether the fan is opened, and whether the nutrition is enough for each tomato.

5.20.2 Pre-conditions

The pico cell(s) placed in the smart greenhouse is configured by Operator O to support the 5GS Ambient IoT service. The UE(s) is capable to and authorized by Operator O to directly communicate with the 5GS Ambient IoT device.

5.20.3 Service Flows

1. Operator O's 5GS authenticates Ambient IoT devices in farm A. The communication between the Ambient IoT device and 5GS are transferred through the pico cell. For example, the pico cell obtains Ambient IoT device ID and sends the ID to the 5GC for authentication.

2. Periodically, or upon request from the application server of farm A, the pico cell obtains the environmental monitoring information from the sensing Ambient IoT devices and sends the information to the application server of farm A.

3. Based on the received information and preconfigured logic, the application server requests 5GS to send control signaling to the operation Ambient IoT devices to control their operations, e.g. open or close the window of the greenhouse, start or close the irrigation system.

4. The 5GS transfers the control signalling from the application server to Ambient IoT devices through the pico cell.

5.20.4 Post-conditions

Thanks to the Ambient IoT service provided by the 5G system, operation Ambient power-enabled IoT devices automatically take care of the plants.

5.20.5 Existing features partly or fully covering the use case functionality

None.

5.20.6 Potential New Requirements needed to support the use case

[PR.5.20.6-001] The 5G system shall support communication with Ambient IoT device with 3rd party application server.

[PR.5.20.6-002] The 5G system shall be able to authenticate an Ambient IoT device.

[PR.5.20.6-003] The 5G system shall be able to provide Ambient IoT service with following KPIs:

Table 5.20.6-1: Ambient IoT KPI for Smart Agriculture

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Smart Agriculture	>1 s	99.9%	NA	<1 kbit/s	<1000 bits	1 per m^2	30-100m	500-70000 m^2 per greenhouse	Stationary	1 hour	NA	NA	NA

NOTE: There is no requirement for positioning of the Ambient IoT devices for this use case.

5.21 Use Case on Ambient IoT for Museum Guide

5.21.1 Description

Museums are a popular choice for people to spend their leisure time. With so many exhibits in museums, it is difficult for all visitors to know about the artistic value of each exhibit. A qualified museum guide can assist visitors in better understanding the information behind the exhibits.

Hiring a docent and renting an explanation device are two common methods for museum guides today. A docent usually serves multiple visitors at once, and it is difficult to provide personalized guidance based on each visitor's preferences. Moreover, for foreign visitors, docents who can speak their language are not always available. If visitors choose to rent an explanation device, they will need to manually enter the exhibit number each time they use the device, and the device usually only provides audio explanations, which is not very convenient for the visitors. Recently, some museums also offers other solutions, such as attaching the QR codes on the exhibit cases to provide the introduction information. By scanning the QR codes, visitors can also get information related to the exhibits. But for visitors, they still need to manually open the camera on their mobile phone and scan the QR code for each exhibit, which may be repeated lots of times during the visit.

Ambient IoT devices are a promising solution for museum guide. An exhibition hall in a museum can be thousands of square meters in size and have thousands of exhibits. Ambient IoT devices can work with limited energy storage capability or without any battery for an extremely long time, so the Ambient IoT devices are maintenance-free,

lightweight, and small-size. In the museum, the Ambient IoT devices can be attached to the glass of the exhibit case or placed in the exhibit case with the exhibit. The introduction information of the exhibits corresponding to each Ambient IoT device is uploaded to the application server in advance.

Abby enjoys going to museums, and her mobile phone supports Ambient IoT service and is able to send signals to Ambient IoT devices. She also downloads the museum guide application and subscribes to the guide services.

The exhibit hall in the museum that Abby intends to visit covers 6,000 square meters and has 3,000 exhibits. Abby learned that this museum has deployed a museum guide system using ambient IoT devices to provide the corresponding introduction information of the exhibit that will help the visitors to have a deep understanding of each exhibit.

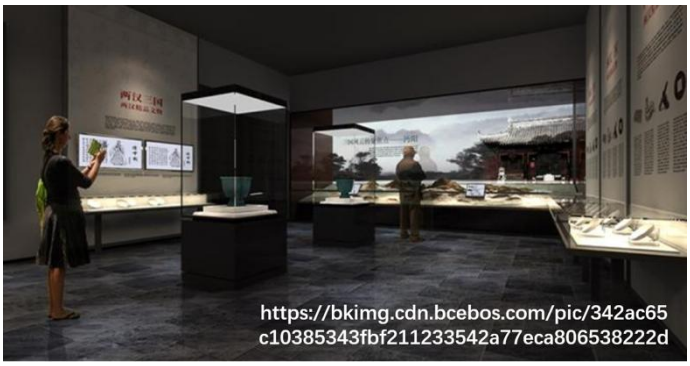


Figure 5.21.1-1: Ambient IoT for Museum Guide

5.21.2 Pre-conditions

In museums, the Ambient IoT devices are attached to the glass of the exhibit case or placed in the exhibit case with the exhibit. The introduction information of the exhibits corresponding to each Ambient IoT device is uploaded to the application server in advance.

Abby's mobile phone supports Ambient IoT service and is able to send signals to Ambient IoT devices. She also downloads the museum guide application and subscribes to the guide services.

The museum has public or private 5G network coverage to provide Ambient IoT services with support for a large number of Ambient-IoT devices.

The Ambient IoT services could have interactions with the 5G network with necessary information.

5.21.3 Service Flows

1. Abby arrives at the museum and walks into the exhibition hall. She opens the guide application on her mobile phone to get more information. She also authorizes her mobile phone for the Ambient IoT communication service and Ambient IoT positioning service, to obtain the relative positioning results of Ambient IoT devices.

2. Abby taps the button on the application to get more information about nearby exhibits, and her mobile phone sends the signal (continuously or intermittently) to wake up and trigger the Ambient IoT devices, and the Ambient IoT devices are attached to the glass of the exhibit case or placed in the exhibit case with the exhibit in advance.

3. The Ambient IoT devices close to Abby receive the signal and are activated. The Ambient IoT devices respond to Abby's mobile phone with their Ambient IoT device IDs.

4. Abby's mobile phone receives the response signals from the Ambient IoT devices with their IDs. Abby's mobile phone can also derive the relative positioning results of each Ambient IoT device using the response signals.

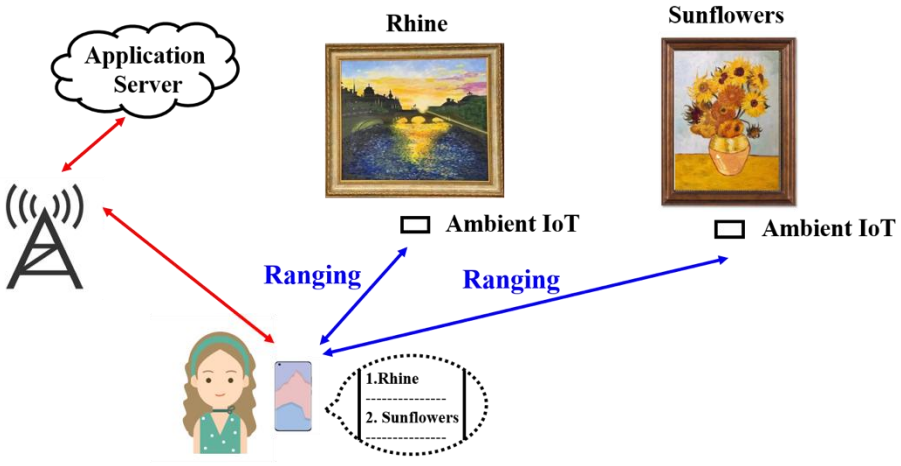


Figure 5.21.3-1: Ambient IoT for Museum Guide

5. Abby's mobile phone sends the acquired relative positioning resultss and Ambient IoT IDs to the application server with the help of 5G network.

6. The application server transmits the introduction information corresponding to the Ambient IoT IDs to Abby's phone. With the relative positioning results derived in step 4, Abby's phone can give different priorities to the introduction information based on the relative positioning results. The information about the exhibit closest to her is displayed at the top of the screen.

7. As Abby moves, other Ambient IoT devices will be activated by the signal and respond with their Ambient IoT device IDs to Abby's mobile phone. With these new response signals, Abby's mobile phone can derive a new list of relative positioning results, and the exhibit information on her mobile phone can be automatically updated.

5.21.4 Post-conditions

Thanks to the Ambient IoT service provided by the 5G system, Abby can better enjoy her museum trip.

5.21.5 Existing features partly or fully covering the use case functionality

None.

5.21.6 Potential New Requirements needed to support the use case

[PR.5.21.6-001] The 5G system shall support to authorize a UE to perform Ambient IoT communication services with specific Ambient IoT devices.

[PR.5.21.6-002] The 5G system shall be able to support to authorize a UE to perform relative positioning operations with specific Ambient IoT devices.

[PR.5.21.6-003] The 5G system shall support to authorize a UE to obtain device identity information of an Ambient IoT device.

[PR.5.21.6-004] The 5G system shall be able to expose the identities and relative positioning results of Ambient IoT devices to a trusted third party.

[PR. 5.21.6-005] The 5G system shall be able to support suitable security mechanisms for Ambient IoT devices, including encryption and data integrity.

[PR.5.21.6-006] The 5G system shall be able to provide Ambient IoT service with the following KPIs:

Table 5.21.6-1: Ambient IoT service KPI for museum guide

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Museum guide (indoor)	2 s	99.9%	NA	< 1 kbit/s UL (NOTE 1)	96 bits	<10,000 /km²	30m	20,000m² (NOTE 2)	3 km/h	NA	NA	90%	3 m
NOTE 1: The payload includes Ambient IoT device information, e.g., Ambient IoT device ID. NOTE 2: For a relatively large-sized museum, the typical size is about 20,000 m².													

5.22 Use case on smart grazing dairy farming enabled by Ambient IoT

5.22.1 Description

The global sensor market is predicted to grow from $193.9 billion in 2020 to $332.8 billion in 2025 at a CAGR of 11.4% [40]. Globally, for tracking and monitoring the IoT market size is forecast to grow from US$ 8,575 million in 2021 to US$ 18,525.0 million at a CAGR of 8.01% in 2031 [41].

It is certainly not new that connected sensors and IoT can play a role in animal husbandry. Precision livestock farming (PLF) as a trendier term adopts an innovative production system approach [42] playing a key role in Industry 4.0 [43]. More efficient production of quality food at lower cost will be an important tool to improve sustainability and respond to the imminent energy crisis and food shortage we are facing today. Physical vitals of livestock such as temperature are monitored for farmers to take early actions before potential diseases cause severe economic loss. The body temperature of livestock is a precise health indicator and changes in body temperature are often the first sign of an acute illness. For these purposes, the target data acquisition process of animal body temperatures is not latency-sensitive. Also, the needed network coverage is usually local (e.g. outdoor dairy cow paddocks), which is different from existing NB-IoT/(e)MTC targeting at providing long-range communication while achieving long battery life time.

As the adoption of PLF continues, feedback is often received from livestock farmers at industry conferences. In EU the EU-PLF conference was held as early as 2016. Feedback includes that dairy cow farmers in countries like the Netherlands (one of the global top 5 dairy exports) have been considering replacement of active monitoring IoT devices (with battery-powered transponders) with cheaper ear tags. This is partly because of economical drawbacks with increasing herd sizes. More importantly, the thick and heavy neck- or leg-mounted devices can cause discomfort to livestock, so that they are often scraped off against walls (of the pen) or damaged by animals involuntarily.

A more recent GSMA publication [10] explicates disadvantages as battery drain rendering the loss of asset visibility in only a few months. For these technologies the frequency of data transmission would impact the battery life time. For smart livestock farming, animal's physical vitals need to be monitored several times a day, and it is preferred to have the IoT device serve livestock's lifespan. This additionally implies the energy storage component in the IoT device should operate autonomously over a long period comparable to the lifespan of the IoT device excluding this component. In the same GSMA paper, another drawback of high CAPEX in dense deployment of the "tag readers" is highlighted, primarily due to the poor communication range supported by these alternative technologies.

These above aspects in smart livestock farming can be better addressed by Ambient IoT, particularly in smart dairy farming. A publication in Dairy Science Journal explains pasturing benefit for milk yield and dairy cow udder health [44], compared with primary intake of silage or concentrate associated with keeping animals indoors. Figure 5.22.1-1 illustrates dairy cows grazing on pasture.



Photo by: Sophie Nuytten, bio melkveebedrijf, Maarten Cerpentier

Figure 5.22.1-1: Dairy cows grazing on pasture

In fact, because of the per-country variations in pasture quantity and quality different dairy farms can achieve different grazing percentage [45]. Table 5.22.1-1 shows data related to percentage of grazing dairy cows in major dairy producing European counties (year 2015). Countries like the Netherlands have explicit ambition to further increase the percentage of dairy cow grazing [45].

Table 5.22.1-1: Grazing and automated milking in Europe, from members of European Grassland Federation (2015) [45]

Country	% grazing	% AMS farms	% grazing AMS
Belgium	65-95	5	5-30
Denmark	25	25	n.a.*
France	>90	10	n.a.
Hungary	0	0	0
Ireland	98-99.9	1-2	98-99.5
Luxembourg	75	>20	43
Netherlands	70	20	40-60
Norway	90	30	80
Poland	20	0.1	0
Slovenia	30-40	n.a.	<5
Spain	30	2	n.a.
Sweden	100	25-35	100
Switzerland	96	n.a.	n.a.
UK	70-90	1-20	5-50

*n.a. = not available

As grazing is important for dairy production, various grazing methods are possible (e.g. continuous grazing, strip grazing, rotational grazing, etc.) [46]. For strip or rotational grazing, a large pasture is subdivided into a number of smaller paddocks, so that grazing is managed in a planned sequence. The dimension of paddocks could be calculated by multiplying the number of cows by their total daily intake by days in the paddock and then divided by the ideal pre-grazing yield (PGY). There is a physical limitation of the paddock dimension. For instance, for 80 cows assuming regular values for parameters previously mentioned, the paddock size comes to 1.54 hectares (around 124m by 124m) [47].

This use case primarily proposes to support data acquisition process of dairy cows' physical vitals on grazing dairy farms. In terms of the total size of pasture on grazing dairy farms, data from the Netherlands by University of Wageningen [48] reveals the pasture area for grazing in practice. Table 5.1 in [48] demonstrates among the various Dutch dairy farms the pasture surface area for grazing ranges from 4.9 hectares (49000 m^2) to 34.3 hectares (343000 m^2).

Another publication summarizing Wisconsin dairy grazing practice [49] shows the farm count distribution versus herd size and distribution of average acres per cow versus herd size. Based on these statistics, it shows majority of Wisconsin farms have herd size ranging from 50 to 150, and respectively the average acres per cow ranges from 1.2 to 0.7. Therefore, the total pasture surface of the majority of farms ranges from 60 acres (around 250000 m^2) to 105 acres (around 430000 m^2).

Australian data additionally shows for grazing dairy farming, the total pasture size can be influenced by bay length. Publication by Rural Water Commission of Victoria [50] explains practice on designing paddocks for irrigated dairy

farms. Per requirement on bay length for economic reasons (i.e. short bay lengths leading to more spending on crossings, outlets, and drains, and overly long bays resulting in cow access problems), the ideal length of bay is between 300m and 500m [50]. The resulted pasture size is within an area of 600m by 600m.

To efficiently connect dairy cows, attached to each of them is a small, thin and light-weight tag (a type of Ambient IoT device) that has limited power source and includes a basic temperature sensor. These Ambient IoT devices power themselves by harvesting energy from the environment (e.g. solar, movement). The dairy cow health management system collects cows' temperature several times a day, usually once every 15 minutes.

The base stations provide to the tags random access and data transmission over the radio interface. The tags are capable of storing tags' identifiers and small sized data captured by sensors. The 5G system provides base station capability (e.g., "tag reading" functionality), tag operation and management. The monitored data is collected remotely according to the health-analyzing applications.

In this use case, grazing dairy farm BIO-DuurzameBeweiding is modernizing their dairy cow health management process to improve efficiency and productivity. They attach to dairy cows are wireless temperature sensors (tags), a form of Ambient IoT device, for the remote livestock health management application to retrieve dairy cow temperature to detect early signs of ailment that could take days or weeks to develop. The remote livestock health management application analyses the collected sensor values to identify potential illness of certain animals prior to symptoms appearing.

5.22.2 Pre-conditions

BIO-DuurzameBeweiding has a service level agreement with GroenTEL to deploy Ambient IoT service within 5G network coverage to enable the communication of Ambient IoT devices with the network. The communication service availability is achieved by providing seamless 5G network coverage. As part of the service level agreement, GroenTEL provides 5G coverage of the entire grazing pasture and efficient communication of Ambient IoT devices with the network. This includes:

- Interfacing with BIO-DuurzameBeweiding remote livestock health management system;
- Providing energy-efficient mechanisms for Ambient IoT devices' network access
- Providing efficient communication between the network and Ambient IoT with the required communication performance
- Providing energy efficient security mechanisms for the communication between Ambient IoT devices and the network.

5.22.3 Service Flows

1. Upon the request from the livestock health management application, the 5G network starts inventory process via the selected gNB(s) This operation is associated with a certain area (e.g. grazing pasture of the dairy farm). Triggered by the 5G network, tags detect the signals from the gNB and respond to the command.

2. These Ambient IoT devices send the identification information to the 5G network and 5G core network complete the authentication procedure.

3. The Ambient IoT devices (wireless sensors) measure dairy cow physical vitals (i.e. body temperature). These temperature sensors are very simple, typical sampling rate is less than 10 Hz with sample size of 32 bits [39], thus the sensor data rate generated per tag is less than 320 bit/s. Assuming tag ID length is 96 bits, and it is transmitted together with sensor data, then the total throughput is < 500 bit/s.

4. The 5G network, based on the requests issued by the application function, performs operations (i.e. "inventory", "read", etc.) on tags correspondingly. "Inventory" operation is to read the tag identifier. "Read" operation is to read temperature sensor data.

5. The 5G core network then sends the results of the operations to the livestock health management application. The application function includes analytics functions that detect the anomaly and notifies the farmers of BIO-DuurzameBeweiding when necessary.

6. This data acquisition by the livestock health management application takes place once every 15 minutes.

7. In some additional situations, BIO-DuurzameBeweiding livestock management application requests the 5G network to perform sensor data read-out operation on specific tags attached to particular individual livestock (e.g. pregnant sows, lactating cows). The corresponding tags respond to the operation and report the temperature data.

5.22.4 Post-conditions

The 5G network enables efficient communication for Ambient IoT devices, the livestock management application is enabled by the 5G system to retrieve temperature sensor data from Ambient IoT devices. Depending on the needs, the livestock management application is enabled by 5G system to obtain the sensor data from an entire herd, a subset of herd, or an individual dairy cow.

5.22.5 Existing features partly or fully covering the use case functionality

SA1 has performed various studies on IoT in previous releases, where related normative stage 1 requirements are introduced in TS 22.011 [9], TS 22.278 [7], TS 22.368 [6], and TS 22.261 [8].

TS 22.011 introduces access control for MTC, examples of periodic network selection attempts are:

> *For UEs only supporting any of the following, or a combination of, NB-IoT, GERAN EC-GSM-IoT [18], and Category M1[13] of E-UTRAN enhanced-MTC, the UE shall interpret the interval value to be between 2 and 240 hours, with a step size of 2 hours between 2 and 80 hours and a step size of 4 hours between 80 and 240 hours.*
>
> *In the absence of a permitted value in the SIM/USIM, or the SIM/USIM is phase 1 and therefore does not contain the datafield, then a default value of 60 minutes, shall be used by the UE except for those UEs only supporting any of the following, or a combination of: NB-IoT, GERAN EC-GSM-IoT [18], and Category M1 [17] of E-UTRAN enhanced-MTC. For those UEs a default value of 72 hours shall be used.*
>
> *NOTE: Use of values less than 60 minutes may result in excessive UE battery drain.*

TS 22.368 addresses features of MTC communication and service requirements related to MTC device triggering, addressing, identifiers, low mobility, small data transmission, infrequent MT communication, security, remote MTC device management, group-based MTC features including policing and addressing, etc. Example requirements are:

> *The system shall provide mechanisms to lower power consumption of MTC Devices.*
>
> *The system shall provide mechanisms for the network operator to efficiently manage numbers and identifiers related to MTC Subscribers.*

TS 22.261 captures some important service requirements for IoT, e.g.

> *The 5G system shall support a secure mechanism for a home operator to remotely provision the 3GPP credentials of a uniquely identifiable and verifiably secure IoT device.*
>
> *The 5G system shall support a secure mechanism for the network operator of an NPN to remotely provision the non-3GPP identities and credentials of a uniquely identifiable and verifiably secure IoT device.*
>
> *An IoT device which is able to access a 5G PLMN in direct network connection mode using a 3GPP RAT shall have a 3GPP subscription.*
>
> *The 5G system shall allow the operator to identify a UE as an IoT device based on UE characteristics (e.g. identified by an equipment identifier or a range of equipment identifiers) or subscription or the combination of both.*
>
> *An IoT device which is able to connect to a UE in direct device connection mode shall have a 3GPP subscription, if the IoT device needs to be identifiable by the core network (e.g. for IoT device management purposes or to use indirect network connection mode).*
>
> *The 5G system shall support operator-controlled alternative authentication methods (i.e. alternative to AKA) with different types of credentials for network access for IoT devices in isolated deployment scenarios (e.g. for industrial automation).*

In these specifications, albeit the service requirements addressing traits for IoT in terms of low device power consumption, small and infrequent data transmissions, long service lifetime, and resource efficiently, the IoT devices considered in 3GPP have been assumed to be powered by at least batteries up till now. To enable extremely small, light-weight, battery-less Ambient IoT devices that engage in basic IoT data transaction and appropriate level of operator management and charging suitable for the target scenarios, new challenges to the 5G system are foreseen and need to be addressed.

5.22.6 Potential New Requirements needed to support the use case

[PR 5.22.6-1] The 5G system shall support energy efficient communication mechanisms (i.e. minimizing the device communication power consumption) for Ambient IoT devices, while meeting the communication performance requirements.

[PR 5.22.6-2] The 5G system shall provide a mechanism for a 3rd party application to write user data to and to read user data from an Ambient IoT device.

[PR 5.22.6-3] The 5G system shall be able to collect charging information for a large group of closely located Ambient IoT devices in an efficient way.

NOTE: for example, the efficiency could be reduced total number of charging data related to a group of Ambient IoT devices, the reduction is compared with already specified 3GPP technologies.

[PR 5.22.6-4] The 5G system shall provide the network connection with the following KPIs for the use of Ambient IoT devices for smart dairy farms, see table 5.22.6-1.

Table 5.22.6-1: Potential key performance requirements for the use of Ambient IoT devices for smart grazing dairy farming

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Smart dairy farm	>1 s (note 1)	99%	NA	<50 0 bit/s	Typically, [< 100 bytes] (note 2)	<5200 devices / km^2 (note 4)	[300 m - 500 m] outdoor (note 6)	430000 m^2 (note 5)	Up to 3 km/h outdoor	15 min (note 3)	NA	NA	NA

NOTE 1: Latency is not critical.
NOTE 2: Electronic Product Code standard [5], this size is the payload size.
NOTE 3: The livestock health management application monitors dairy cow body temperature many times daily, typically two consecutive transfers of the application data have an interval of 15 minutes.
NOTE 4: Calculated from 80 dairy cows assuming regular values for parameters (e.g. daily intake, pre-grazing yield) previously mentioned, the paddock size comes to 1.54 hectares [47] (about 124m by 124m).
NOTE 5: For a relatively large-sized industrialized smart dairy farm, the surface area of pasture for grazing is typically 430000 m^2.
NOTE 6: Based on the statistics from the Netherlands [55], Wisconsin [49] and Australia [50], the total pasture is smaller than an area of 650 m by 650 m. Assuming the coverage by one base station, the communication range between the Ambient IoT device and the base station is smaller than 500m.

5.23 Use case on smart pig farm enabled by Ambient IoT

5.23.1 Description

The global sensor market is predicted to grow from $193.9 billion in 2020 to $332.8 billion in 2025 at a CAGR of 11.4% [40]. Globally, for tracking and monitoring the IoT market size is forecast to grow from US$ 8,575 million in 2021 to US$ 18,525.0 million at a CAGR of 8.01% in 2031 [41].

It is certainly not new that connected sensors and IoT can play a role in animal husbandry. Precision livestock farming (PLF) as a trendier term adopts an innovative production system approach [42] playing a key role in Industry 4.0 [43]. More efficient production of quality food at lower cost will be an important tool to improve sustainability and respond to the imminent energy crisis and food shortage we are facing today. Physical vitals of livestock such as temperature are

monitored for farmers to take early actions before potential diseases cause severe economic loss. The body temperature of livestock is a precise health indicator and changes in body temperature are often the first sign of an acute illness. For these purposes, the target data acquisition process of animal body temperatures is not latency-sensitive, rarely an acquisition interval down to minute level is needed. Also, the needed network coverage is usually local. As this use case addresses industrialized smart pig farming, the difference from existing NB-IoT/(e)MTC (i.e. they target at providing long-range communication while achieving long battery life time) is coverage intended for pig barns is indoor.

As the adoption of PLF continues, feedback is often received from livestock farmers at industry conferences. In EU the EU-PLF conference was held as early as 2016. Feedback includes that livestock farmers have been considering replacement of monitoring IoT devices (with battery-powered transponders) with cheaper ear tags. This is partly because of economical drawbacks with increasing pig herd sizes e.g. observed in EU [51]. More importantly, the thick and heavy neck- or leg-mounted devices can cause discomfort, so that they are often rubbed off against enclosures or damaged by animals.

These above aspects in smart livestock farming can be better addressed by Ambient IoT. Figure 5.23.1-1 illustrates a typical pig farm consisting of a number of pig barns. For intensive piggeries the typical surface area of a pig barn is 4000 ~ 6000 m^2 [52].



Figure 5.23.1-1: An example pig farm (upper) consisting of pig barns (lower)

Attached to each pig is a small, thin and light tag (a type of Ambient IoT device) that includes a basic temperature sensor. These devices power themselves by harvesting ambient energy (e.g. solar).

The base stations provide to the tags random access and data transmission over the radio interface. The tags are capable of storing tags' identifiers and small sized-data captured by sensors. The 5G system provides base station capability (e.g., "tag reading" functionality), tag operation and management. The monitored data is collected remotely according to the health-analyzing applications.

5.23.2 Pre-conditions

Pig farm BIO-Duurzaamheid is modernizing their hog health management process to improve efficiency and productivity. They attach to each pig a very light-weight wireless temperature sensor (tag), a form of Ambient IoT device, for the remote livestock health management application to retrieve pigs' temperatures to detect early signs of ailment or

infections that could take days or weeks to develop symptoms. The temperature rad-out takes places several times a day, interval time is in the order of tens of minutes. The remote livestock health management application analyses the collected sensor values to identify potential illness of certain animals before symptoms appear.

BIO-Duurzaamheid has a service level agreement with GroenTEL to deploy 5G network to enable the communication of Ambient IoT devices with the network. Inside each industrialized warehouse-like pig barn, a gNB is installed to provide the radio coverage. As part of the service level agreement, GroenTEL provides efficient 5G coverage and energy-efficient communication within each pig barn managed by BIO-Duurzaamheid. This includes:

- Interfacing with BIO-Duurzaamheid's remote livestock health management system;
- Providing energy-efficient mechanisms for Ambient IoT devices' network access
- Providing efficient communication between the network and Ambient IoT with the required communication performance
- Ensuring the long lifespan of Ambient IoT devices without human intervention of any energy storage component possibly used in Ambient IoT devices

5.23.3 Service Flows

1. The tags (type of Ambient IoT devices with wireless sensors) measure pigs' body temperatures. The type of temperature sensors are very simple, typical sampling rate is less than 10 Hz with sample size of 32 bits [39], thus the data generated per tag is less than 320 bit/s. Assuming tag ID length is 96 bits, and it is transmitted together with sensor data, then the total throughput is < 500 bit/s. BIP-Duurzaamheid obtain pigs' body surface temperature daily with varying monitoring intervals depending on the needs, but the monitoring frequency is not higher than once per 15 minutes, or on half-hourly basis.

2. Upon the request from the livestock health management application, the 5G network starts inventory process via the selected gNB(s). By detecting the signals from the gNB, the tags respond to the command.

3. The 5G core network, based on the requests issued by the application function, performs operations (i.e. "inventory", "read", etc.) on tags correspondingly. "Inventory" operation is to read the tag identifier. "Read" operation is to read temperature sensor data.

4. The 5G core network then sends the results of the operations to the livestock health management application. The application function includes analytics functions that detect the anomaly and notifies the farmers of BIO-Duurzaamheid when necessary.

5. In some additional situations, BIO-Duurzaamheid's livestock management application requests the 5G core network to perform sensor data read-out operation on specific tags attached to particular individual livestock (e.g. pregnant sows, recovering boars). The corresponding tags respond to the operation and report the temperature data.

5.23.4 Post-conditions

The 5G network enables efficient communication for Ambient IoT devices, the livestock management application is enabled by the 5G system to retrieve temperature sensor data collected by sensors in the Ambient IoT devices. Depending on the needs, the livestock management application is enabled by 5G system to retrieve the sensor data for either certain individual pigs or an entire drove.

Owing to the thin and light-weight Ambient_IoT devices, livestock are more receptive of wearing them without feeling discomfort. This reduces asset damage and increases livestock welfare. As a result of data analysis done by the livestock management application, early signs of ailment of livestock are identified. This increases BIO-Duurzaamheid's pig farm production. Thanks to the efficient Ambient_enabled IoT communication enabled by the 5G system, BIO-Duurzaamheid continue to use Ambient_enabled IoT for daily monitoring once per 15 minutes (or per half hour) without worrying about energy drain leading to replacement or manual recharging of Ambient_IoT devices throughout the planned production lifespan of their livestock.

5.23.5 Existing features partly or fully covering the use case functionality

SA1 has performed various studies on IoT in previous releases, where related normative stage 1 requirements are introduced in TS 22.011 [9], TS 22.278 [7], TS 22.368 [6], and TS 22.261 [8].

TS 22.011 introduces access control for MTC, examples of periodic network selection attempts are:

For UEs only supporting any of the following, or a combination of, NB-IoT, GERAN EC-GSM-IoT [18], and Category M1[13] of E-UTRAN enhanced-MTC, the UE shall interpret the interval value to be between 2 and 240 hours, with a step size of 2 hours between 2 and 80 hours and a step size of 4 hours between 80 and 240 hours.

In the absence of a permitted value in the SIM/USIM, or the SIM/USIM is phase 1 and therefore does not contain the datafield, then a default value of 60 minutes, shall be used by the UE except for those UEs only supporting any of the following, or a combination of: NB-IoT, GERAN EC-GSM-IoT [18], and Category M1 [17] of E-UTRAN enhanced-MTC. For those UEs a default value of 72 hours shall be used.

NOTE: Use of values less than 60 minutes may result in excessive UE battery drain.

TS 22.368 addresses features of MTC communication and service requirements related to MTC device triggering, addressing, identifiers, low mobility, small data transmission, infrequent MT communication, security, remote MTC device management, group-based MTC features including policing and addressing, etc. Example requirements are:

The system shall provide mechanisms to lower power consumption of MTC Devices.

The system shall provide mechanisms for the network operator to efficiently manage numbers and identifiers related to MTC Subscribers.

TS 22.261 captures some important service requirements for IoT, e.g.

The 5G system shall support a secure mechanism for a home operator to remotely provision the 3GPP credentials of a uniquely identifiable and verifiably secure IoT device.

The 5G system shall support a secure mechanism for the network operator of an NPN to remotely provision the non-3GPP identities and credentials of a uniquely identifiable and verifiably secure IoT device.

An IoT device which is able to access a 5G PLMN in direct network connection mode using a 3GPP RAT shall have a 3GPP subscription.

The 5G system shall allow the operator to identify a UE as an IoT device based on UE characteristics (e.g. identified by an equipment identifier or a range of equipment identifiers) or subscription or the combination of both.

An IoT device which is able to connect to a UE in direct device connection mode shall have a 3GPP subscription, if the IoT device needs to be identifiable by the core network (e.g. for IoT device management purposes or to use indirect network connection mode).

The 5G system shall support operator-controlled alternative authentication methods (i.e. alternative to AKA) with different types of credentials for network access for IoT devices in isolated deployment scenarios (e.g. for industrial automation).

In these specifications, albeit the service requirements addressing traits for IoT in terms of low device power consumption, small and infrequent data transmissions, long service lifetime, and resource efficiently, the IoT devices considered in 3GPP have been assumed to be powered by at least batteries up till now. To enable extremely small, light-weight, battery-less Ambient IoT devices that engage in basic IoT data transaction and appropriate level of operator management and charging suitable for the target scenarios, new challenges to the 5G system are foreseen and need to be addressed.

5.23.6 Potential New Requirements needed to support the use case

[PR.5.23.6-1] The 5G system shall support energy efficient communication mechanisms (i.e. minimizing the device communication power consumption) for Ambient IoT devices, while meeting the communication performance requirements.

[PR 5.23.6-2] The 5G system shall provide a mechanism for a 3rd party application to write user data to and to read user data from an Ambient IoT device.

[PR 5.23.6-3] The 5G system shall be able to collect charging information for a large group of closely located Ambient IoT devices in an efficient way.

NOTE: for example, the efficiency could be reduced total number of charging data related to a group of Ambient IoT devices, the reduction is compared with already specified 3GPP technologies

[PR.5.23.6-4] The 5G system shall provide the network connection to address the following KPIs for the use of Ambient IoT devices on smart pig farms.

Table 5.23.6-1: Potential key performance requirements for the use of Ambient IoT devices industrialized smart pig farming

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Smart livestock farming (pig barns)	>10 s (Note 1)	NA	NA	<500 bit/s	Typically, [< 100 bytes] (note 2)	850 000 devices / km^2 (note 4)	250 m Indoor	6000 m^2 (note 5)	Quasi-stationary	15 minutes to half an hour (note 3)	NA	NA	NA

NOTE 1: Latency is not critical for this use case.
NOTE 2: Electronic Product Code standard [5], this size is the payload size.
NOTE 3: The livestock health management application monitors pigs' temperature on a half-hourly basis, sometimes even down to once per 15 minutes [53].
NOTE 4: Stocking density of 1.2 m^2/pig is considered high and 2.4 m^2/pig considered low [54]. In [55] the stockinggrazing density range from 0.82 m^2/pig and 2.46 m^2/pig is studied and 1.23 m^2/pig proofs suitable stocking density for growing pigs. The device density of 850000 is from a real farm and translates to stocking density of 1.17 m^2/pig.
NOTE 5: For a relatively large-sized industrialized smart pig farm, the surface area of a barn is typically 6000 m^2.

5.24 Use case on smart manhole cover safety monitoring using Ambient IoT

5.24.1 Description

Manholes date back to the mid-19th century. Sanitary sewer manholes are primarily used for joining or/and changing the direction of the sewer. As part of the underground infrastructure, manholes provide access for maintenance and for potentially installing additional sewer lines. Worldwide, there have been increasingly a large number of manholes in villages and cities (e.g. parks, sidewalks, parking lots, streets, etc.) as modern civilization grows. For example, U.S. EPA estimates the number of sewer manholes nationwide to be 12 million. Most coincide with the typical lengths of city and suburban blocks, about 100 to 500 feet apart [56]. The total count of utility manholes (adding onto sewer manholes) in the United States is approximately 20 million. A typical large Chinese city has around 1 million manholes. For example, more than 123 million manholes are in use in Wuhan alone [57], which the local government has plans to digitally identify and register each of them. Usually local authorities such as municipalities monitor and inspect assets of city infrastructure. At times manned manhole inspection is carried out with cameras underground to verify degrees of deterioration and provide rehabilitation recommendations. In that process, surface features such as manhole cover and pavement can be reported if restoration is needed [58].

However, manhole covers alone are rarely monitored frequently enough. Often gone unnoticed are they doing their job to keep traffic (e.g. motorist, cyclists) and pedestrians safe, until one falls. In and outside busy cities, falling into manholes causes potential danger of severe injuries or death is lurking, not only where poorly lit.

A recent incident in October 2022 downtown Saint-Nazaire (west coast France) concerns a 39-year-old man found dead early Saturday morning, drowned head down leaving lower legs hanging upright outside the manhole [59]. There are also many stories where children fall into manholes, parents desperately trying to pull them outside but often deadly tragedy

prevails [60]. As a matter of fact, accidental fall due to damaged or missing manhole covers has become a silent killer around us. But this is preventable thanks to the use of Ambient IoT.

In Q1 2022, China published intelligent manhole cover national standard GB/T 41401 [61] for municipalities to digitally manage this important asset. The standard requires the manholes to be identifiable by the asset management application, displacement of manhole cover (due to e.g. accidental damage or theft) can be detected by tilt sensors. Additionally, underground water level sensors, vibration sensors (e.g. to detect shock events), and temperature sensors could be deployed.

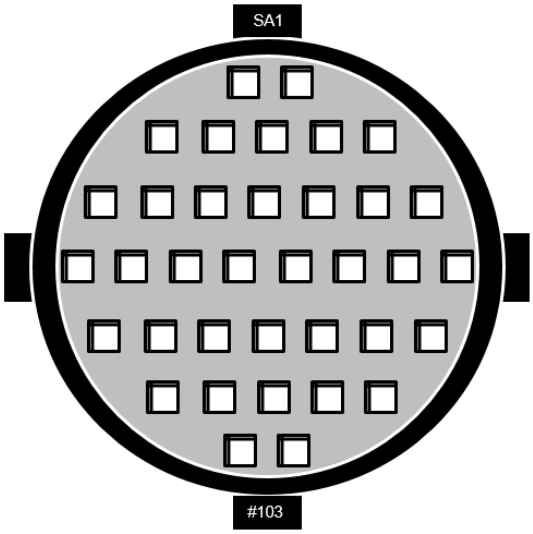


Figure 5.24.1-1: Manhole where the fatal accident took place , incident as in [59]

In general, manhole use case depends on network deployment. This use case assumes relatively dense network deployment.

For the manhole cover use case, a large number of sensors (a type of Ambient IoT devices) need to be efficiently connected, particularly because they have very limited power source. The communication power consumption of such Ambient IoT devices are expected to be less than 1 mW [86] [87]. As manhole covers are stationary and deployed in outdoor public areas. And because this use case concerns road safety, the communication service availability with sufficient 5G network coverage is important. The data acquisition process of these sensor data is not latency critical. The sensor data (tilt, underground water level, shock) is needed once every 15 minutes. The acquisition of detected abnormality is required within 30 seconds [61]. All these sensors should be durable and maintenance free, as sending technicians to solely replace sensors at each manhole location would be an extra process adding to municipalities OPEX and environmental impact.

In this use case, the municipality M responds to a recent tragic incident similar to [59] by implementing the smart manhole cover management programme “La Bouche d'égout sans Souci”. M has service level agreement with La-Tel-Verte to provide 5G network coverage and enable communication of Ambient IoT devices with the 5G network. La-Tel-Verte per service level agreement provides energy-efficient communication and management services to the municipality M:

- interfacing with municipality M’s manhole cover management platform where application “La Bouche d'égout sans Souci” runs;

- providing energy efficient device management for the Ambient IoT devices based on the instructions from the manhole cover management platform;

- providing energy efficient operation (e.g. inventory, read) the Ambient IoT devices based on the instructions from the manhole cover management platform;

- providing energy efficient security mechanisms for the communication between Ambient IoT devices and the network.

5.24.2 Pre-conditions

Municipality M has a service level agreement with La-Tel-Verte that ensures Ambient IoT communication service availability by providing sufficient 5G network coverage. This enables the communication of Ambient IoT devices with the 5G network. Municipality M's manhole project team has installed wireless sensors, a form of Ambient IoT devices, onto all manhole covers within its responsible area to monitor the corresponding parameters (e.g. water level [62], tilt of manhole cover [63], vibration [64]). Water level sensor information can be used to forecast potential flooding. Based on data from tilt sensors and/or vibration sensors, displacing or missing manhole covers could be detected for safety intervention.

5.24.3 Service Flows

1. The 5G core network receives the request from the application function ("La Bouche d'égout sans Souci" in municipality M's manhole cover management platform) to operate on the Ambient IoT devices in a certain area. The 5G network starts to operate on these devices accordingly. Once detecting the signals from the 5G network these Ambient IoT devices can respond to the command.

2. Since each of these Ambient IoT devices are uniquely identifiable, in response they send the identification information to the 5G core network and complete the authentication procedure.

3. The Ambient IoT devices (wireless sensors) measure the parameters, such as water level, tilt, and shock/vibration.

   In this use case, sample size for water level sensor and tilt sensor is respectively 8 bits [62] and 32 bits [63]. With typical sampling rate being 10 Hz, the data generation is less than 400 bit/s. For vibration measurement, typical sampling rate is 10 Hz with sample size of max 48 bits [64] (e.g. 3-axis, 8 bit per axis; possibly per axis two measurement values to register acceleration). Thus, in total the data generation per Ambient IoT device is up to 880 bit/s.

4. The 5G core network, based on the requests issued by the application function, performs operations such as "inventory" and "read" on the Ambient IoT devices correspondingly. "Inventory" operation is to read the Ambient IoT device identifier. "Read" operation is to read sensor data.

5. The 5G core network then sends the results of the operations to the application function. The application function includes diagnostic functions that analyze the sensor data and detect the anomaly and trigger maintenance actions when necessary. Typically, the sensor data is collected once per 15 minutes.

6. When the application function "La Bouche d'égout sans Souci" receives sensor data that is considered by the application function to be related to potential problem of a specific manhole cover, the application function can request the information of the Ambient IoT device on that manhole cover more frequently. The max allowed latency is 30 seconds [61].

5.24.4 Post-conditions

The 5G network enables efficient communication for Ambient IoT devices. Thanks to that, municipality M's intelligent manhole cover management application function "La Bouche d'égout sans Souci" is enabled by the 5G system to remotely monitor the large number of manholes in its responsible area for safety and maintenance reasons. Depending on the needs/logic of the application function, the application function is enabled by 5G system to retrieve the sensor data from a specific or a given group of Ambient IoT devices.

5.24.5 Existing features partly or fully covering the use case functionality

SA1 has performed various studies on IoT in previous releases, where related normative stage 1 requirements are introduced in TS 22.011 [9], TS 22.278 [7], TS 22.368 [6], and TS 22.261 [8].

TS 22.011 introduces access control for MTC, examples of periodic network selection attempts are:

> *For UEs only supporting any of the following, or a combination of, NB-IoT, GERAN EC-GSM-IoT [18], and Category M1[13] of E-UTRAN enhanced-MTC, the UE shall interpret the interval value to be between 2 and 240 hours, with a step size of 2 hours between 2 and 80 hours and a step size of 4 hours between 80 and 240 hours.*

In the absence of a permitted value in the SIM/USIM, or the SIM/USIM is phase 1 and therefore does not contain the datafield, then a default value of 60 minutes, shall be used by the UE except for those UEs only supporting any of the following, or a combination of: NB-IoT, GERAN EC-GSM-IoT [18], and Category M1 [17] of E-UTRAN enhanced-MTC. For those UEs a default value of 72 hours shall be used.

NOTE: Use of values less than 60 minutes may result in excessive UE battery drain.

TS 22.368 addresses features of MTC communication and service requirements related to MTC device triggering, addressing, identifiers, low mobility, small data transmission, infrequent MT communication, security, remote MTC device management, group-based MTC features including policing and addressing, etc. Example requirements are:

The system shall provide mechanisms to lower power consumption of MTC Devices.

The system shall provide mechanisms for the network operator to efficiently manage numbers and identifiers related to MTC Subscribers.

TS 22.261 captures some important service requirements for IoT, e.g.

The 5G system shall support a secure mechanism for a home operator to remotely provision the 3GPP credentials of a uniquely identifiable and verifiably secure IoT device.

The 5G system shall support a secure mechanism for the network operator of an NPN to remotely provision the non-3GPP identities and credentials of a uniquely identifiable and verifiably secure IoT device.

An IoT device which is able to access a 5G PLMN in direct network connection mode using a 3GPP RAT shall have a 3GPP subscription.

The 5G system shall allow the operator to identify a UE as an IoT device based on UE characteristics (e.g. identified by an equipment identifier or a range of equipment identifiers) or subscription or the combination of both.

An IoT device which is able to connect to a UE in direct device connection mode shall have a 3GPP subscription, if the IoT device needs to be identifiable by the core network (e.g. for IoT device management purposes or to use indirect network connection mode).

The 5G system shall support operator-controlled alternative authentication methods (i.e. alternative to AKA) with different types of credentials for network access for IoT devices in isolated deployment scenarios (e.g. for industrial automation).

Additional consideration needs to be given in support of Ambient IoT devices that are battery-less or with limited energy storage capability, which present new challenges to the 5G system.

5.24.6 Potential New Requirements needed to support the use case

[PR.5.24.6-1] The 5G system shall support energy efficient communication mechanisms (i.e. minimizing the device communication power consumption) for Ambient IoT devices, while meeting the communication performance requirements.

[PR 5.24.6-2] The 5G system shall provide a mechanism for a 3rd party application to write user data to and to read user data from an Ambient IoT device.

[PR 5.24.6-3] The 5G system shall be able to collect charging information for a large group of closely located Ambient IoT devices in an efficient way.

NOTE: for example, the efficiency could be reduced total number of charging data related to a group of Ambient IoT devices, the reduction is compared with already specified 3GPP technologies.

[PR.5.24.6-4] The 5G system shall provide the network connection to address the KPIs for the use of Ambient IoT devices for smart manhole cover monitoring, see table 5.24.6-1.

Table 5.24.6-1: Potential key performance requirements for the use of Ambient IoT devices for smart manhole cover monitoring

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Smart manhole cover remote monitoring	10 s - 30 s (note 1)	99 %	NA	<1 kbit/s	Typically, [< 100 bytes] (note 2)	<1000 devices / km^2 (note 3)	300 m - 500 m Outdoor(note 6, note 7)	City wide including rural areas (note 4)	Stationary	15 min (note 5)	NA	NA	NA

NOTE 1: Latency is not critical. Per GB/T 41401 [61], the max latency is smaller than 30 seconds.
NOTE 2: This size is the payload size, compatible with allowed business data length by Electronic Product Code standard [5]. Considering EPC for identification is 96 bits, the total message size is < 100 bytes.
NOTE 3: Assuming there is one manhole every 200 feet [56]. Referring to data from the United States [56], sewer manholes are about 100 to 500 feet apart, additionally there are other utility manholes present. According to Wuhan data [57], 123 million manholes are known per 8569 km^2.
NOTE 4: As local authority is the customer, the service should be available for all the utility manholes within the responsible area of that municipality.
NOTE 5: For manhole cover remote monitoring, per 15-minutes data acquisition is sufficient to largely increase road traffic safety.
NOTE 6: The value is dependent on the actual network deployment.
NOTE 7: This communication range implies a relatively dense network deployment.

5.25 Use case on smart bridge health monitoring using Ambient IoT

5.25.1 Description

Bridges as public infrastructure characterize city formation. To facilitate transportation, more bridges are built as cities form and evolve. Apart from obvious roles bridges play in transortation of people and goods, its importance also lies in its being the social infrastructure and cutural assets [65]. As the number of bridges grow, longer and bigger bridges are put in use. Incidents of bridge collapsing usually lead to catastrophy. In the course of city expansion and population growth, numerous deaths due to bridge collapse have been recorded, such as recent events reported in [66], [67] and [68]. The most heart-breaking tragic to date is the Indian pedestrian bridge collapse in Guikarat in the midst of Diwali religious celebration, where death toll rises above 130 [69]. As awareness continues to rise, various actors including local governements are starting to take on a more active role in monitoring and maintenance of those vital infrastructure [70].

Ambient IoT can be used for smart bridge health monitoring to contribute to prevention of disasters. As bridges are parts of outdoor public infrastructure and the use case is about safety, the communication service availability with sufficient 5G network coverage are important. Additionally, as more health data of a bridge is collected, the corresponding actors (e.g. local governmement) have deeper knowledge in its health state to better control the maintenance work and eventually control the cost [71]. In 2022, China published national specification GB/T 39339.2 that hightlights safety monitoring of bridges as part of transit facilities in cities [72].

In this use case, local government of city Philario responds to a recent tragic incident by implementing the smart bridge health monitoring programme “SAFE”. Philario has service level agreement with O-Tel to provide 5G network coverage and enable communication of Ambient IoT devices with the 5G network O-Tel per service level agreement provides energy-efficient communication and management services to the local government of Philario:

- interfacing with local government Philario’s smart bridge health monitoring platform;

- providing energy efficient device management for the Ambient IoT devices based on the instructions from Philario's smart bridge health monitoring platform;

- providing energy efficient operation (e.g. inventory, read) the Ambient IoT devices based on the instructions from Philario's smart bridge health monitoring platform;

- providing energy efficient security mechanisms for the communication between Ambient IoT devices and the network.

The communication power consumption of such Ambient IoT devices are expected to be less than 1 mW [86] [87].

5.25.2 Pre-conditions

Philario's local government's bridge safety maintenance project team X has installed wireless sensors, a form of Ambient IoT devices, onto selected bridges within its responsible area to monitor the corresponding parameters (e.g. tilt sensors to monitor inclination of bridge deck or pier [73], vibration [74]). X has a service level agreement with service provider Y that deploys sufficient 5G network coverage to ensure communication service availability. This enables the communication of Ambient IoT devices with the 5G network, where needed. Based on per 15-minute data from tilt sensors and vibration sensors, health status of bridges is recorded and analyzed for safety intervention.

5.25.3 Service Flows

1. The 5G core network receives the request from the application function (request sent from Philario's smart bridge health monitoring platform) to operate on the Ambient IoT devices installed on a given bridge or a given set of bridges. The 5G core network starts to operate on these devices accordingly. Once detecting the signals from the 5G network these Ambient IoT devices can respond to the command.

2. Since each of these Ambient IoT devices are uniquely identifiable, in response they send the identification information to the 5G core network and complete the authentication procedure.

3. The Ambient IoT devices (wireless sensors) measure the parameters, such as water level, tilt, and vibration.

   The sample size for tilt sensor 32 bits [73]. With typical sampling rate being 10 Hz, the data generation is less than 400 bit/s. For vibration measurement, typical sampling rate is 10 Hz with sample size of max 48 bits [74] (e.g. 3-axis, 8 bit per axis; possibly per axis two measurement values to register acceleration). Thus, in total the data generation per Ambient IoT device is up to 880 bit/s.

4. The 5G core network, based on the requests issued by the application function, performs operations such as "inventory" and "read" on the Ambient IoT devices correspondingly. "Inventory" operation is to read the Ambient IoT device identifier. "Read" operation is to read sensor data.

5. The 5G core network then sends the results of the operations to the application function. The application function includes diagnostic functions that analyze the sensor data and detect the anomaly and trigger maintenance actions when necessary. Typically, the sensor data is collected once per 15 minutes.

6. When the bridge health monitoring application receives sensor data, by analysis it may decide to request the information of the Ambient IoT device associated with a specific bridge more frequently (e.g. per 10-minute) on-demand.

5.25.4 Post-conditions

The 5G network enables efficient communication for Ambient IoT devices. Thanks to that, local government Philario's bridge health monitoring application function is enabled by the 5G system to remotely monitor the large number of sensors (Ambient IoT devices) on the target bridges within Philario's responsible area. Depending on the needs/logic of the application function, the application function is enabled by 5G system to retrieve the sensor data from a specific or a given group of Ambient IoT devices.

5.25.5 Existing features partly or fully covering the use case functionality

SA1 has performed various studies on IoT in previous releases, where related normative stage 1 requirements are introduced in TS 22.011 [9], TS 22.278 [7], TS 22.368 [6], and TS 22.261 [8].

TS 22.011 introduces access control for MTC, examples of periodic network selection attempts are:

For UEs only supporting any of the following, or a combination of, NB-IoT, GERAN EC-GSM-IoT [18], and Category M1[13] of E-UTRAN enhanced-MTC, the UE shall interpret the interval value to be between 2 and 240 hours, with a step size of 2 hours between 2 and 80 hours and a step size of 4 hours between 80 and 240 hours.

In the absence of a permitted value in the SIM/USIM, or the SIM/USIM is phase 1 and therefore does not contain the datafield, then a default value of 60 minutes, shall be used by the UE except for those UEs only supporting any of the following, or a combination of: NB-IoT, GERAN EC-GSM-IoT [18], and Category M1 [17] of E-UTRAN enhanced-MTC. For those UEs a default value of 72 hours shall be used.

NOTE: Use of values less than 60 minutes may result in excessive UE battery drain.

TS 22.368 addresses features of MTC communication and service requirements related to MTC device triggering, addressing, identifiers, low mobility, small data transmission, infrequent MT communication, security, remote MTC device management, group-based MTC features including policing and addressing, etc. Example requirements are:

The system shall provide mechanisms to lower power consumption of MTC Devices.

The system shall provide mechanisms for the network operator to efficiently manage numbers and identifiers related to MTC Subscribers.

TS 22.261 captures some important service requirements for IoT, e.g.

The 5G system shall support a secure mechanism for a home operator to remotely provision the 3GPP credentials of a uniquely identifiable and verifiably secure IoT device.

The 5G system shall support a secure mechanism for the network operator of an NPN to remotely provision the non-3GPP identities and credentials of a uniquely identifiable and verifiably secure IoT device.

An IoT device which is able to access a 5G PLMN in direct network connection mode using a 3GPP RAT shall have a 3GPP subscription.

The 5G system shall allow the operator to identify a UE as an IoT device based on UE characteristics (e.g. identified by an equipment identifier or a range of equipment identifiers) or subscription or the combination of both.

An IoT device which is able to connect to a UE in direct device connection mode shall have a 3GPP subscription, if the IoT device needs to be identifiable by the core network (e.g. for IoT device management purposes or to use indirect network connection mode).

The 5G system shall support operator-controlled alternative authentication methods (i.e. alternative to AKA) with different types of credentials for network access for IoT devices in isolated deployment scenarios (e.g. for industrial automation).

Additional consideration needs to be given in support of Ambient IoT devices that are battery-less or with limited energy storage capability, which present new challenges to the 5G system.

5.25.6 Potential New Requirements needed to support the use case

[PR.5.25.6-1] The 5G system shall support energy efficient communication mechanisms (i.e. minimizing the device communication power consumption) for Ambient IoT devices, while meeting the communication performance requirements.

[PR 5.25.6-2] The 5G system shall provide a mechanism for a 3rd party application to write user data to and to read user data from an Ambient IoT device.

[PR 5.25.6-3] The 5G system shall be able to collect charging information for a large group of closely located Ambient IoT devices in an efficient way.

NOTE: for example, the efficiency could be reduced total number of charging data related to a group of Ambient IoT devices, the reduction is compared with already specified 3GPP technologies.

[PR.5.25.6-4] The 5G system shall provide the network connection to address the KPIs for the use of Ambient IoT devices for smart bridge health monitoring as in the table below.

Table 5.25.6-1: Potential key performance requirements for the use of Ambient IoT devices for smart bridge health monitoring

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Smart bridge health monitoring	10 s (note 1)	99%	NA	<1 kbit/s	Typically, [< 100 bytes] (note 2)	<1000 devices / km^2 (note 4)	300 m - 500 m Outdoor(note 5)	Along the bridge	Stationary	15 min (note 3)	NA	NA	NA

NOTE 1: Latency of smart bridge health monitoring is not critical, as the sensor information is needed by the monitoring application on a per-15 minutes basis.
NOTE 2: This size is the payload size, compatible with allowed business data length by Electronic Product Code standard [5]. Considering EPC for identification is 96 bits, the total message size is < 100 bytes.
NOTE 3: For bridge health monitoring, per 15-minutes data acquisition is sufficient.
NOTE 4: For bridge health monitoring applications, the distances among sensors are larger than 10 meters. Big data analysis-based bridge health monitoring deploys 369 sensors of various types on Changjiang bridge, in Shanghai [75]. In Beijing, around 150 sensors are deployed on Dongsha bridge of 1360 meter length [76]. In busy sections of mega cities, concentration of bridges can be considerate.
NOTE 5: The value is dependent on the actual deployment.

5.26 Use case on Elderly Health Care

5.26.1 Description

As elderly population is increasing, close, daily monitoring of their health is acquiring more relevance in health care systems. Chronic diseases that require continuous attention are common among this age group. Moreover, autonomy of the elderly has an important role in active aging, as it is strongly associated with longevity, good self-assessed health, and the prevention of depression and cognitive deterioration. Autonomy is an essential concept because it relates directly to dignity, regardless of health circumstances.

Due to extreme conditions, like local epidemics, pandemic or natural disasters, health care systems can be overwhelmed. Relying on technology such as automated assistance using mobile communications system infrastructure can lower pressure on the health system, as chronic diseases are appropriately followed up. This improves the overall health system, freeing resources for other critical, life-threatening situations.

This use case presents a scenario where an elder is aided to quickly locate medicines both indoors and outdoors using Ambient IoT devices (Ambient IoT tags). These Ambient IoT tags are very small, battery-less powered IoT devices that use an energy harvesting mechanism to produce a limited amount of power, at microwatts level. After some initial setup, we assume that the Ambient IoT device can harvest the energy necessary from RF signals to be able to operate.

This use case is about an elder, Scott. He is 85 and had heart attack at 80. Since then, he needs to take a variety of medicines daily prescribed by Rachel, his doctor. During his last check-up, his heart condition has worsened because he forgets sometimes to take his medicines. Scott's memory is not what used to be. He remembers he must take medicines but forgets which ones.

To help Scott to be autonomous on his own, and improve his heart condition, Rachel decided to prescribe him medicines from a new supplier: DontForgetYourMed. DontForgetYourMed offers a personalised service that allows the patient's doctor to set up the quantity and frequency of the medicines prescribed. Using medicines packets with Ambient IoT tags attached, DontForgetYourMed reaches the patient to take those medicines at specific times, making the availability of the service on demand.

In the case of Scott, his heart rate is monitored with his smartwatch, and it is notified regularly to DontForgetYourMed service. This way, Rachel can modify the frequency of some medicines depending on a configured threshold of Scott's heart rate.

5.26.2 Pre-conditions

The following pre-conditions and assumptions apply to this use case:

- A subscription to DontForgetYourMed service that establishes a communication with the UE of the elder to send alerts and request confirmations related to the medicines prescribed.
- A smart watch (UE) to receive and confirm alerts and monitor elder's heart rate continuously.
- 5G network to support the communication with Ambient IoT devices.
- Medicine packets equipped with an Ambient IoT tag and a small led. When the medicine packet is received by the elder, some pre-settings will give energy to the Ambient IoT device, and it will also establish the location of the Ambient IoT device.
- Each Ambient IoT tag is connected to the 5G network, and the communication is enabled
- Communication will be established on demand, not continuously.

5.26.3 Service Flows

Indoors scenario is more focused in habit: medicine is taken daily at specific time of the day. Outdoors scenario is more focused as an exception of the normal routine: the service reacts to the elder's heart rate. Both scenarios could be combined.

Indoors scenario

1. Each morning, Scott receives an alert in his smart watch where the medicine dose is displayed. This alert recommends him to take one pill from 'packet Blue' and one pill from 'packet Pink'.
2. Scott confirms the reception of the message and then, DontForgetYourMed requests the 5G system to establish a downlink communication with the ambient IoT tag and switches on a small LED embedded within the same tag.
3. Scott walks to his medicine cabinet to retrieve the medicines. Inside, 2 packets are lit: 'packet Blue' and 'packet Pink'. After taking both pills, he returns the packets to the medicine cabinet.
4. After a pre-configured timer expires, Scott receives another alert to confirm he has taken each of the requested medicines. He confirms them both, and starts a new, healthy day.
5. After the confirmation, DontForgetYourMed requests the 5G system to establish another downlink communication with the ambient IoT tag, and the small LED is switched off.

Outdoors scenario

1. Scott feels far better so he decides to go to play tennis with Oliver and informs Rachel. She prescribes an additional medicine to take during the game, depending on Scott's heart rate. When the game is on, Scott's heart rate is too high, so he receives an alert to take one pill from 'packet Blue'.
2. Scott confirms the reception of the message and then, DontForgetYourMed requests the 5G system to establish a downlink communication with the ambient IoT tag and switches on a small LED embedded within the same tag.
3. He starts to feel bad but as the medicine packet is in his backpack, Oliver helps him to retrieve it. He looks for Scott's backpack to see one medicine blue packet lit: 'packet Blue'. He gets the packet to Scott and helps him to get his medicine.

4. After the pre-configured timer expires, Scott receives another alert to confirm he has taken the medicine. Oliver helps Scott to confirm it, and rests until he feels better.

5. After the confirmation, DontForgetYourMed requests the 5G system to establish another downlink communication with the ambient IoT tag, and the small LED is switched off.

5.26.4 Post-conditions

5G communication has been established to a UE to prescribe medicines, and request for confirmation of medicine intake.

Several downlink communications have been established from the gNB to one or several medicine packets. This downlink communication has been low power, and for a small period, and the amount of data sent has been minimal.

5.26.5 Existing features partly or fully covering the use case functionality

None.

5.26.6 Potential New Requirements needed to support the use case

[PR.5.26.6-001] The 5G system shall be able to support mechanisms to communicate efficiently with Ambient IoT devices.

[PR.5.26.6-002] When setting up communication to an Ambient IoT device the 5G system shall be able to handle the unavailability of Ambient IoT devices either due to lack of power or due to power saving mechanisms of the Ambient IoT device.

[PR.5.26.6-003] The 5G system shall be able to provide an Ambient IoT service with following KPIs:

Table 5.26.6-1: Ambient IoT service KPI for elderly health care

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Indoors elderly health care scenario	1 s	NA	NA	<1 kbit/s	<100 bits	<20 per 100 m^2	20 m	<250 m^2 (note 1)	Static	NA	NA	NA	NA
Outdoors elderly health care scenario	1 s	NA	NA	<1 kbit/s	<100 bits	<20 per 100 m^2 (note 2)	200 m	City wide including rural areas	Static	NA	NA	NA	NA

NOTE 1: Average size of a big house.

NOTE 2: For the outdoor scenario the device density is expected to be generally lower than indoor[88]. For example, based on outdoor tennis court sizes (four players in a 23.77m x 10.97m doubles matches court), assuming one medicine box per elderly consumer, the device density is not higher than 5/100m^2.

5.27 Use case on end-to-end logistics

5.27.1 Description

In a logistics scenario, Ambient IoT devices can be used to track specific goods (e.g. TVs) from the factory where they are made until delivery at the final end-customer. E.g. an Ambient IoT device could be attached to the packaging of a TV, which can then be used for tracking in the factory, in one or more warehouses on the way from factory to the customer and finally to the track the delivery at the end customer.

In this end-to-end logistics trajectory, an Ambient IoT device will encounter multiple different networks (NPNs, PLMNs) in different regions and countries. Each of these networks have to follow different frequency regulations and have different frequency licenses. The Ambient IoT device needs to be able to deal with these differences.

Limitations in range for Ambient IoT can imply that some additional infrastructure needs to be deployed. However, it is not assumed that a network needs to be deployed specifically for Ambient IoT support. It should be possible to integrate Ambient IoT communication in networks that are also used for other 5G communication.

The assumption in this use case is that the Ambient IoT devices need to be triggered to send their identification. When the Ambient IoT devices are not triggered they remain in a suspended state and cannot send or receive information.

The owner of a network (e.g. in a factory or warehouse) can decide which part of the network (e.g. which base stations) to use to trigger Ambient IoT devices. This way, e.g. Ambient IoT devices in only part of the factory are triggered. It should also be possible to trigger only a selected group of Ambient IoT devices (e.g. through broadcasting a group ID), or to trigger the Ambient IoT devices for specific actions (e.g. receive information instead of broadcasting an ID).



Figure 5.27.1-1: Tracking a cardboard box a TV is transported in by attaching an Ambient IoT Device to the box

Where the network itself does not have the support to trigger the Ambient IoT device (e.g. in the PLMN where the TV is delivered to the door of the customer), a handheld device can be used. This handheld device can, when needed, relay the communication from the Ambient IoT device towards the network. The handheld device can collect several messages from one or more Ambient IoT devices before forwarding these messages to the network. This 'store-and-forward' messaging also allows messaging from an Ambient IoT device when there is no end-to-end connectivity from Ambient IoT device to the network.

5.27.2 Pre-Conditions

Factory A has deployed a standalone NPN.

TV_company has an agreement with the NPN in Factory A and with PLMN_X to use a third party interface to trigger Ambient IoT devices.

The Ambient IoT device has credentials with PLMN_X that can also be used for the standalone NPN in factory A. Furthermore, PLMN_X has roaming agreements with operators in other countries.

Warehouse B deploys a NPN implemented as a slice on PLMN_Y.

5.27.3 Service Flows

1. TV_company uses Factory A to manufacture its TVs. After manufacturing, the TVs are packaged in a cardboard box and the box is then equipped with an Ambient IoT device (in the form of a sticker) for tracking/tracing of the TV.

2. Within the Factory A, the TVs are stored for a while in internal storage. Tracking of the TVs in storage is possible through Ambient IoT. The TV_company instructs the NPN that is deployed in the Factory A to trigger the TVs in the internal storage only. TVs that are still in the production area do not need to be triggered. The Ambient IoT devices are triggered to send their ID. This way the TV_company can create an inventory of all TVs that are still in store at the factory. The Ambient IoT devices can also be triggered to receive information (e.g. instructions).

3. A shipment of TV is sent to a warehouse in a Country_X. The warehouse uses a NPN implemented as a slice on PLMN_Y. Because of roaming agreements between PLMN_Y and PLMN_X, the Ambient IoT devices can access the NPN network in the warehouse. Note for the NPN in Country_X different regulations apply than for the NPN in Factory A.

4. TV_company wants to create an inventory of the TVs that are stored in the warehouse. However, the warehouse also stores products from other companies. Therefore the TV_company instructs the NPN to trigger only the TVs from TV_company through the use of a group ID that identifies these TVs. Ambient IoT devices on other products will not be triggered to respond.

5. Finally the TV is delivered to the end-customer. The delivery driver uses a handheld device to trigger the Ambient IoT device on the box to send its ID to the network. If the network is within range, the Ambient IoT device can communicate directly to the network. Otherwise the handheld device can relay the communication. Even if the handheld device is not

connected to the network either, the Ambient IoT devices can still send messages to the handheld device. The handheld device can forward these messages to the network when it connects to the network.

5.27.4 Post-Conditions

TV_Company has full traceability of its products from the factory in one country, to the delivery at the end-customer in another country.

5.27.5 Existing features partly or fully covering the use case functionality

3GPP TS22.101 Clause 29.2 contains the following requirements:

Support of 3rd party requested broadcast

- *The 3GPP Core Network shall enable a 3rd party service provider to request sending a broadcast message in a specified geographic area (as specified in TS 22.368 [52]) expecting to reach a group of devices that are served by the 3rd party service provider.*

A 3rd party requested broadcast can be used to trigger a group of IoT devices in a specific geographic area. But this requirement does not include the feature of waking up Ambient IoT devices.

3GPP TS22.368 Clause 7.1.2 contains the following requirements on MTC Device triggering:

- *The network shall be able to trigger MTC Devices to initiate communication with the MTC Server based on a trigger indication from the MTC Server.*

- *The system shall provide a mechanism such that only trigger indications received from authorized MTC Servers will lead to triggering of MTC Devices.*

- *Upon receiving a trigger indication from a source that is not an authorized MTC Server, the network shall be able to provide the details of the source (e.g. address) to the MTC User.*

- *The system shall provide a mechanism to the MTC User to provide a set of authorized MTC Server(s).*

- *Upon receiving a trigger indication, if the network is not able to trigger the MTC Device, the 3GPP system may send an indication to the MTC Server that triggering the MTC Device has been suppressed.*

Note: Suppression of triggering could be due to system conditions such as network congestion.

- *A MTC Device shall be able to receive trigger indications from the network and shall establish communication with the MTC Server when receiving the trigger indication. Possible options may include:*

- *Receiving trigger indication when the MTC Device is not attached to the network.*

- *Receiving trigger indication when the MTC Device is attached to the network, but has no data connection established.*

- *Receiving trigger indication when the MTC Device is attached to the network and has a data connection established.*

Based on the requirements in 3GPP TS22.368, 3GPP TS23.682 Clause 4.5.1 contains the following description:

Device Triggering is the means by which a SCS sends information to the UE via the 3GPP network to trigger the UE to perform application specific actions that include initiating communication with the SCS for the indirect model or an AS in the network for the hybrid model. Device Triggering is required when an IP address for the UE is not available or reachable by the SCS/AS.

5.27.6 Potential New Requirements needed to support the use case

[PR.5.27-001] The 5G system shall be able to support an Ambient IoT device to function in different countries in accordance with local regulations.

[PR.5.27-002] The 5G system shall support Ambient IoT device triggering to get one or more Ambient IoT devices to perform specific actions including to initiate communication with the network.

[PR.5.27-003] The 5G network shall be able to trigger Ambient IoT devices. The 5G system shall enable an authorized 3rd party to instruct the 5G network in which area, which group of Ambient IoT devices needs to be triggered and which action the Ambient IoT devices need to perform when triggered (e.g. send ID, receive further information, send measurement value).

[PR.5.27-004] The 5G system shall support relaying communication from an Ambient IoT device to the network.

[PR.5.27-005] The 5G system shall support authenticated, encrypted and integrity protected store-and-forward communication from an Ambient IoT device to the network via a single UE when there is no end-to-end connection between the Ambient IoT device and the network

NOTE: It is assumed that the UE and the Ambient IoT device are authorized by the operator to communicate with each other.

5.28 Use case on pressure powered switch

5.28.1 Description

A pressure powered switch can harvest energy from the kinetic energy of the pushing on the switch. Existing non-3GPP products can e.g. connect wirelessly to a controller over approximately 25 m indoor and 150 m outdoor. These products do not adhere to 3GPP protocols for signalling and user communication.



Figure 5.28.1-1: A pressure powered switch

The registration procedure for 5G according to TS 23.502 [77] shows 25 different steps which 8 involve the UE. Several of these steps involve multiple interactions. This registration procedure does not even include the actual data communication. Compared to the minimal information transfer requirement for a button switch (only its ID), it is clear that the registration involves far more information than the actual data transport. Furthermore, a registration procedure will likely take longer than the time the switch has power available. This runs the risk that the switch can start a registration (or other) signaling procedure, but cannot successfully complete it. Furthermore, it is important to note that some procedures are timer-driven. For example, the UE is configured with a periodic registration timer and can be implicitly de-registered if the UE is not able to perform a registration procedure when the timer expires. It cannot be assumed that switch will be pressed frequently enough to avoid an implicit de-registration.

The purpose of the use case is to propose a requirement to optimize Ambient IoT signaling to reduce the amount of signaling interactions and thus save power and time.

Note: The pressure powered switch is an example for other Ambient IoT devices that only have a very limited amount of energy for a very short amount of time (e.g. door/window sensors, vibration sensors).

5.28.2 Pre-conditions

None

5.28.3 Service Flows

1. The switch is being pushed.

2. The switch harvests the energy from the push, wakes up and communicates to the network.

3. Through optimized protocols only a minimal number of signalling interactions is needed. This allows the switch to complete the signalling procedure. As part of the signalling, the switch also transmits its identity.

4. The network transfers the identity information from the switch to an application server, which e.g. determines that a light needs to be turned on.

5.28.4 Post-conditions

The light is successfully switched on.

5.28.5 Existing features partly or fully covering the use case functionality

Existing signaling procedures and data transfer procedures for 5G are specified in TS 23.502 [77]. It is clear that these procedures are not optimized for a minimal number of signaling interactions. This runs the risk that an Ambient IoT device cannot complete these procedures before running out of power.

5.28.6 Potential New Requirements needed to support the use case

[P.R.5.1.6-001] The 5G system shall support more efficient procedures for Ambient IoT control and user data transmission compared to earlier 3GPP technologies; in terms of a reduced number of interactions between the network and the Ambient IoT device.

NOTE: In this context each interaction is a single instance of control or user data transmission to or from the Ambient IoT device

[P.R.5.1.6-002] The 5G system shall support procedures that take into account the specific nature of Ambient IoT devices (e.g. some Ambient IoT devices will not be able to initiate procedures periodically or after a specific time).

5.29 Use case on Device Permanent Deactivation

5.29.1 Description

This use case illustrates the need to define capabilities that allows the end user or a third party to remotely manage the permanent deactivation of an Ambient IoT device.

The scenario describes a production manager who oversees the manufacture of Integrated Circuits (IC) wafers. The environmental conditions under which the wafers are produced may be considered as industrial secrets, as the production process may require a precise combination of pressure, temperature and humidity to produce IC wafers of optimal quality. To assist in quality control, the production process includes the use of Ambient IoT devices to record the environmental conditions under which the wafers are produced. The production manager may use the sensor data recorded by the Ambient IoT device following the completion of the manufacturing process to verify that environmental conditions were maintained as required.

5.29.2 Pre-conditions

The production manager has inactive Ambient IoT devices in storage that can collect sensor data, record the data and transmit the recorded data.

5.29.3 Service Flows

Device activation

1. As a new batch of IC wafers is about to be manufactured, the production manager removes from storage the Ambient IoT devices and adds an Ambient IoT device to each pre-production wafer.

2. The production manager accesses an application that is used to manage the connectivity and operations of Ambient IoT devices. Via this application, the production manager activates an Ambient IoT device to enable the operations of the device (e.g., take and record sensor data).

Device operation

3. During the manufacture process, the Ambient IoT device records the data collected by its sensors.

4. Following completion of the manufacturing process, the production manager wants to access the sensor data that has been recorded on the Ambient IoT device.

5. The production manager accesses an application to manage the connectivity and operations of the Ambient IoT devices. Via this application, the production manager triggers the Ambient IoT device to upload the recorded sensor data to the network.

6. Following completion of the manufacturing process, the production manager wants to clear the sensor data that has been recorded on the Ambient IoT device.

7. The production manager accesses an application to manage the connectivity and operations of the Ambient IoT devices. Via this application, the production manager triggers the Ambient IoT device to delete the recorded sensor data.

Device deactivation

8. The production manager wants to deactivate Ambient IoT devices to disable their operation while the devices are not being used in the manufacturing process.

9. The production manager accesses an application to manage the connectivity and operation of the Ambient IoT devices. Via this application, the production manager deactivates and disables the operation of the device.

10. The production manager removes the deactivated Ambient IoT devices from the finished wafers and returns the devices to storage for possible re-use.

Device end of life cycle

11. The production manager determines that the Ambient IoT devices have reached the end of their life cycle and should no longer be used during the wafer manufacturing process.

12. The production manager accesses an application to manage the connectivity and operation of the Ambient IoT devices. Via this application, the production manager permanently configures an Ambient IoT Device such that all transmissions of the Ambient IoT device are permanently deactivated.

13. The production manager discards the permanently deactivated Ambient IoT device as it is no longer possible to activate, enable its operation or access recorded data.

5.29.4 Post-conditions

Device permanently deactivated

Two-way communications by an Ambient IoT device are permanently deactivated.

5.29.5 Existing features partly or fully covering the use case functionality

TS 22.261 clause 6.14.1 describe the following:

During their life cycle these IoT devices go through different stages, ..., the activation of the IoT device by the preferred operator, a possible change of operators, etc. These stages need to be managed securely and efficiently.

Clause 6.14.2 defines the following requirement:

Based on operator policy, the 5G system shall provide means for authorised 3rd parties to request changes to UE subscription parameters for access to data networks, e.g., static IP address and configuration parameters for data network access.

The requirement above covers remote *UE subscription* activation and subscription suspension / deactivation.

If the subscription of an Ambient IoT device has been suspended or terminated, the device can still continually harvest energy and therefore may continue to attempt accessing the network or be accessed by a network. This can result in a security risk where a discarded Ambient IoT Device may be obtained by unauthorized users such that data or other parameters of the device may be subject to unauthorized access.

Therefore, there is a need to permanently deactivated the device such that the device may never be activated and accessed again.

5.29.6 Potential New Requirements needed to support the use case

[PR 5.29.6-1] Based on operator policy, the 5G system shall provide suitable mechanism to permanently disable the capability of an Ambient IoT device or a group of Ambient IoT devices to transmit RF signals.

[PR 5.29.6-2] Based on operator policy, the 5G system shall provide means for a trusted third party to request the deletion of any digitally stored information of an Ambient IoT device.

5.30 Use case on Ambient IoT device acting as a controller in smart agriculture

5.30.1 Description

In agricultural production, there are many factors that affect the growth of crops, such as soil, climate, water, species, pests and weeds, etc. The yield of crops is the result of the combined influence of these factors. The smart agriculture can increase production, expand planting range and planting cycle by using sensors to monitor the growing environment of crops, and using some controller to correspondingly periodically control the equipment (e.g., the pesticide spraying equipment and irrigation equipment) in the farmland based on the sensed results.

It's hard to provide a stable and continuous power supply for the controller deployed in the outdoor farmland. The Ambient power-enabled IoT devices, which can obtain and/or store energy from the environment, can be attached in the smart agriculture equipment and used as the controller to control the equipment in the farmland. The ambient IoT devices can harvest the energy from the environment to support its communication implementation.

Considering the operations of the pesticide spraying and irrigation equipment in the farmland is periodically, the ambient IoT controller can be activated periodically to save its energy. After the 5G network receives the demand from the Farm Management Platform, it can periodically activate the ambient IoT controller and then the ambient IOT controller will receive operation information and operate accordingly. The operation information includes e.g., the period to switch on and switch off the irrigation equipment, the amount of sprayed pesticide in each time, the spraying direction of each pesticide spraying equipment and etc.

Following is an example of service flow to describe the ambient IoT device is activated periodically and acting as a controller to control the equipment in the farmland.

5.30.2 Pre-conditions

The farmer "FF" owns a huge farmland, and installed several pesticide spraying and irrigation equipment in the different location of the farmland to cover the whole farmland. Each pesticide spraying or irrigation equipment is attached with an ambient IoT device, which is an "Ambient IoT controller" to control the operation of pesticide spraying and irrigation equipment (e.g., to control the on/off of pesticide spraying and irrigation system, the spraying direction operation of pesticide spraying system, and etc.) .

Some sensors are also deployed in the farmland to sense and monitor the condition and environment of the farmland.

The operator M has deployed 5G network to provide the "Green Farm comm." communication service for farmers. The farmer "FF" has subscribed the service.

5.30.3 Service Flows

1. The FMP collects the sensing results from the sensors in the farmland. Based on the analysis of the current condition of farmland, the FMP decides which pesticide spraying and irrigation equipment should be put into use.

2. Considering the growth regulation of farmland crops, irrigation and pesticide spraying can work periodically. Thus, the FMP decides the communication patterns of ambient IoT controllers and ask the 5G network to periodically activate the ambient IoT controllers.

3. According to FMP's command, the 5G network periodically trigger the ambient IoT controllers to be activated. After the ambient IoT controllers have been activated, they receive the operation information via 5G network. The operation information includes e.g. the period to switch on and switch off the irrigation equipment, the period to switch on and switch off the pesticide spraying equipment, the amount of sprayed pesticide in each time, the spraying direction of each pesticide spraying equipment, the period to report the status information etc.

4. After receiving the operation information from the 5G network, the ambient IoT controllers can control the corresponding pesticide spraying and irrigation equipment accordingly, and report the operation status information to the FMP via the 5G network. The status information can include the feedback information about whether to receive the operation information and to operate successfully, and can include the current status of the controlled pesticide spraying and irrigation equipment). In the configured switch off period, the ambient IoT controllers can keep inactive to save the power.

5. The FMP continuously collects the sensing results from the sensors in the farmland. When the FMP observes that the farmland condition varies, the FMP decides to update the operation of the pesticide spraying and irrigation equipment, e.g., change to another set of pesticide spraying and irrigation equipment, change the switch on period, adjust the direction of the pesticide spraying equipment etc.

6. Then the FMP can ask the 5G network to periodically trigger the corresponding new ambient IoT controllers associated with the updated pesticide spraying and irrigation equipment and begin new operation.
7. Additionally, if there is an emergency order from FMP, e.g. high temperature warning to trigger the irrigation operation at once, the message need to be delivered with as short the latency as possible.

5.30.4 Post-conditions

The ambient IoT controller can periodically activated and control the pesticide spraying and irrigation equipment accordingly.

The on-demand control of all the pesticide spraying and irrigation equipment can be realized to adapt the variation of farmland condition, then the crops in the farmland can grow normally and the yield of crops can be improved.

5.30.5 Existing features partly or fully covering the use case functionality

None.

5.30.6 Potential New Requirements needed to support the use case

[P.R.5.30.6-001] The 5G system shall provide means for a trusted third-party to trigger an ambient IoT device or group of ambient IoT devices to communicate periodically.

[P.R.5.30.6-002] The 5G system shall be able to provide ambient IoT service with following KPIs:

Table 5.30.6-1: KPI Table of Ambient IoT controller in smart agriculture

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Ambient IoT controller in smart agriculture	Several seconds	99 %	N/A	NA	128bit (DL) (note 1)	NA	500m outdoors	40,000m^2 to 4,000,000 m^2 (note 2)	Static	NA	NA	NA	NA

NOTE 1: this size refers to the payload size of the control information sent by the 5G network to the ambient IoT controller.

NOTE 2: This is typical outdoor farmland size range in China.

6 Traffic Scenarios

6.1 Traffic scenario on flower auction

6.1.1 Description

In the Netherlands, there is extensive logistics industry for flowers and vegetables. A specific case are the auctions where flowers from all over the world are brought in by the growers, then auctioned and subsequently distributed to buyers all over the world.

Flowers are transported on four-wheel containers that can be rented and are that used throughout the logistic chain. These containers are now equipped with a RFID tag. It would be beneficial if Ambient IoT tags could be used. RFID tags are scanned when containers with flowers arrive or leave the auction; tracking and tracing with Ambient IoT could get regular reports from all containers anywhere at the auction.



Figure 6.1.1-1: Logistics at a flower auction

Shipments of flowers can be tracked and traced based on the containers they are on. This is of interest for growers, buyers and the auction. There is also logistics of empty containers, where also the company that owns the containers can benefit from Ambient IoT. Finally, the auction has an interest in managing its space. Companies that own or rent containers are charged for leaving containers on the auction grounds overnight.

Communication service availability is important. Tracking and tracing is mainly done to find out where things have gone wrong in the logistics, e.g., missing containers. If the communication from tags is not significantly more reliable than the logistics itself, then tracking and tracing does not provide a benefit.

Some numbers:

- There are multiple flower auctions in The Netherlands.
- The size of the flower auction location in Aalsmeer is 1 732 769 m^2.
- 44 million flowers are auctioned per day

6.1.2 Assumptions

We assume every four wheeled container is equipped with a tag in the form of an Ambient IoT device.



Figure 6.1.2-1: Container with flowers (Photo: Container Centralen)

Density of containers can be estimated based on the dimensions of the containers. A container is (lxwxh) 1350 mm x 565 mm x 1900m. Packing these containers closely together gives a density of 740 x 1770 = 1,3 million containers per km^2.



Figure 6.1.2-2: Density of flower containers (Photo: Royal FloraHolland)

Ceiling in the flower auction is at 9 meters. We assume that base stations are attached to the ceiling. Number of base stations that is needed to cover the flower auction is dependent on the communication range and on the number of devices per base station. Here we assume a base station spacing of one base station for every 50 m x 50 m of ceiling. This gives a maximum range of approx. 35 meters from ceiling to container. The number of containers in that 50 m x 50 m area is approximately 3000.

The assumption is that the Ambient IoT devices are woken up and triggered for communication on demand by the 5G network, where e.g., the flower auction or a flower grower can decide when to wake up and trigger the Ambient IoT devices for communication. When woken up and triggered the Ambient IoT devices respond by transmitting information, e.g., their identity numbers, to the network.

The flower auction can decide in which parts of the flower auction the Ambient IoT devices are woken up. This can be done by e.g. only providing the wake up via some of the base stations. The flower auction can also decide to trigger only part of the Ambient IoT devices.

Assumption is that an identity can be provided within 96 bits (is EPC length used for identification).

Assumption is that probability of errors in logistics handling (e.g., a container is left behind) is <1%. A communication service availability of 99,99% would imply that the chance that communication for tracking a container is approximately 2 orders of magnitude better than the logistics handling reliability.

There is no strict latency requirement when large amounts of containers are triggered. A latency in the order of 10 seconds is acceptable.

6.1.3 Potential Functional Requirements

None identified.

6.1.4 Potential Key Performance Requirements

[PR 6.1-001] The 5G system shall be able to provide Ambient IoT service with the following KPIs:

Table 6.1.4-1: KPIs for Flower Auction scenario

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy

Container logistics in a flower auction	<10 s	99,99% (note 1)	NA	<1 kbit/s (note 3)	96 bits (note 2)	< 1,3Million/km^2 (note 4)	35 m Indoors	1 700 000 m^2 (note 5)	NA	NA	NA	NA	NA

NOTE 1: Chance of communication service unavailability needs to be significantly lower than chance of errors in logistics handling. This communication service availability applies at application level, the communication service availability at radio level can be different.

NOTE 2: Only an identifier for the tag is sent (Electronic Product Code (EPC) lengths used for identification is 96 bits).
NOTE 3: This value is calculated as the instant data rate for transmitting 96 bits within 100 ms time period. The need for data transmission is infrequent.
NOTE 4: Based on closely packing containers.
NOTE 5: Size of the flower auction location in Aalsmeer.

6.2 Traffic Scenario on cow stable

6.2.1 Description

In the dairy industry, there is an increase in the scale of farms. Whilst the number of farms decreases the number of cows per farm increases. The largest dairy farms in the US have over 15000 cows. A more typical dairy farm has around 200 cows.



Figure 6.2.1-1: A typical dairy farm

Because of the increase of the number of cows, there is a lot of automation in dairy farming. For example cows are milked using a milk robot, and manure is removed with robots.



Figure 6.2.1-2: Dairy farm automation with milk robot (left) and manure robot (right)

In order to identify individual cows (e.g. in the milk robot), the cows have tags. These tags also perform measurements of the cows vital signs (e.g. temperature, movement) to e.g. determine the fertility cycle of the cow and monitor health.



Figure 6.2.1-3: Cow sensor (sensor is just behind the ear)

It is also possible to create a cow sensor in the form of a capsule (or bolus) that can be swallowed by cows and remains in the first of the cow's four stomachs. The advantage of this is that the sensor is smaller, providing more freedom to the cow, and that more internal data from within the cow can be measured.

Many cows spend their whole life in the stable. However, data in the Netherlands shows that more than 80% of cows also can graze outside in the meadows around the farm.

6.2.2 Assumptions

We assume a typical farm with 200 cows. Every cow is equipped with a sensor.

The dimensions of the stable vary with the interior design of the stable, but 30 m x 60 m is a realistic dimension. For simplicity we assume that cows can go everywhere within the stable. The assumption is that there is an indoor base station on the ceiling in the middle of the stable.

The sensors provide measurement data every hour, but can also provide immediate alerts in case the cow is in distress (e.g. during calving). Assumption is that each measurement (identifier, plus 4 measured values) can be provided within 500 bits.

The cow sensor can harvest energy from the movement and temperature of the cow. A continuous power scenario is assumed where the sensor continuously has power available.

Maximum range for the sensor to base station is approximately 35 meters. Note that when the cows are outside, the distance to outside base stations can be much larger (kilometres). It is not assumed that the sensors can communicate over such long range. Even though the sensor may detect the base station, it should not attempt to connect to the base station when it is out of range.

A typical size of a capsule / bolus for cows is approximately 30 mm by 100 mm. For smaller ruminants, such as sheep or goats, this size is not suitable and smaller capsules will have to be made.

6.2.3 Potential Functional Requirements

Ambient IoT devices shall not transmit (creating interference) when the network is outside communication range.

6.2.4 Potential Key Performance Requirements

[PR 6.1-001] The 5G system shall be able to provide Ambient IoT service with the following KPIs:

Table 6.2.4-1: KPIs for dairy stable scenario

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy

Cows in dairy stable	1 s (note 4)	99,9%	NA	< 0.5 kbit/s (note 2)	500 bits (note 1)	< 1 /km^2 (note 3)	35 m Indoors	1 800m^2 (note 5)	NA	NA	NA	NA	NA

NOTE 1: An identifier and four measurement values (e.g. temperature, movement, …).
NOTE 2: This value is calculated as the instant data rate for transmitting 500 bits within 1s transmission time. The need for data transmission is infrequent (e.g. once per hour).
NOTE 3: 200 cows in the stable
NOTE 4: There is no great urgency with cow monitoring that requires a lower latency.
NOTE 5: Assuming a 30 m x 60 m stable.

6.3 Traffic Scenario on Electronic Shelf Label

6.3.1 Description

Recently, a growing number of electronic shelf labels (ESLs) have been deployed in retail industry. The technology has freed the retail stores from labour-intensive paper label replacement work, where a typical grocery store needs to replace about 10,000 price tags per week [78]. The ESLs also help retailers and manufactures to use real-time inventory data to optimize inventory and automate shelf refilling. More importantly, the ESLs and smart shelves system enable retailers to build a better, friendly, and smarter shopping experience to improve shopper engagement. For example, some retailers have integrated electronic shelf label with smart shopping cart to bring better shopper engagement by guiding customers in the store to their shopping items quickly and reduce customer line-up time to zero [79]. Additionally, data collected from ESL system provide good insights on consumer shopping behaviours, which allows retailers to implement a dynamic pricing strategy.

The ESL device is usually equipped with battery, RF module, microcontroller, plastic housing, electronic paper display. Although, most recent electronic paper technologies allow ESL units to draw minimum to zero energy from battery to keep static text and images [80], from time to time, battery replacement still make environment-friendly concern and maintenance cost two pain points for most retailers when the mass deployment of these labels becomes mainstream.



Figure 6.3.1-1: Electronic Shelf Labels deployed in a retail store

In the coming years, Ambient-powered ESL is going to gain momentum in the market. It will dramatically reduce the cost of ESLs. And battery change, from both environment-friendly and labour-saving perspectives, is no longer a concern for retail stores. The Ambient-powered ESL helps to automate several store routines, e.g. price tag change, stock refilling, dynamic price adjustment, customer behaviour analysis, etc. Integrated with a rich set of 5G features, the Ambient-powered IoT ESL is potentially going to drive the next wave of retail industry ESL system upgrade.

In addition to benefits gained from battery-less deployment, thanks to the 5G Ambient IoT system in the retail store, now the store manager does not need to worry about the long-lasting issue where some items' paper price labels do not match advertised flyer prices. He can just feed the store management portal with the same source pricing data used for weekly flyer promotion. The Ambient IoT store management system will then automatically update the affectedly Ambient IoT price tags based on the same pricing strategy as the store flyer. This process can be done remotely without presence of store clerks. It usually completes in just a few minutes. Previously, to get similar task done, the process might require four night-shift clerks on-site to complete for a large retail store.

In another scenario, the Ambient IoT ESL unit, equipped with temperature sensor, could be used to monitor anomaly in the frozen food area, and report to the store management platform periodically. An anomaly alert then triggers the store management platform to page store clerks to check the status of the freezer. On the smart shelf, the Ambient IoT system could report that items on the self are now out of stock. Store clerks will then be notified to refill the shelf in a timely manner.

6.3.2 Assumptions

We assume a retailing giant starts to deploy 5G network to provide Ambient IoT communication service for ESL system in the store. The retail store, average 15,800 square meters, offer more than 100,000 different items. The ESL devices are installed on the 6-shelf racks with dimension of 1.8mx1.2mx0.5m. In addition to the Ambient IoT label, some items on the shelf are also attached with trackable tags. Averaging over effective display area in the store, it is estimated that the device density is less than 1.5 million/km². It is also noted that in certain area of the retail store where small items are stocked as illustrated in Figure 6.3.2-1, the shelf could accommodate about 90 labels per square meter. In the other example illustrated in Figure 6.3.2.1-1, the device density of ESLs equipped with temperature and humidity sensor inside refrigerators is estimated to be less than 10 per 100 square meters.



Fig. 6.3.2-1: display shelfs in a retailing super market

In an electronic shelf label (ESL) system, product pricing and descriptions on the labels can be updated upon request, without applying a fixed transfer interval. However, the temperature and humidity conditions of the shelf or refrigerators are typically reported at predetermined intervals via sensors integrated into the ESL. These intervals are commonly determined in accordance with food safety guidelines that vary across regions and organizations. For example, the United States Department of Agriculture (USDA) recommends that food should not be left outside of refrigeration for more than two hours, and if the temperature exceeds 32°C, this time limit is reduced to one hour [90]. Meanwhile, the Australian Food and Grocery Council (AFGC) stipulates that chilled food should not be left outside of refrigeration for more than 20 minutes [91]. Based on these industry guidelines, we propose employing a transfer interval key performance indicator (KPI) of “20 minutes to 2 hours” for this particular traffic scenario. These transfer intervals can be provisioned into Ambient IoT devices or the management system by the Ambient IoT service providers or 5G operators.

In the other assumption, we assume the Ambient IoT system has a communication service availability of 99%. This implies there might be 1% Ambient IoT devices could not be reachable even after multiple retransmissions. For a retail store with 2,000 different items to update price tag, this means about 20 ESL price tags need to be updated with human intervention. This workload is acceptable for such a store.

6.3.3 Potential Key Performance Requirements

[PR.6.3.3-1] The 5G system shall be able to support Ambient IoT devices with the following KPIs

Table 6.3.3.1 – Potential key performance requirements for Electronic Shelf Label use case

Scenario	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range	Service area dimension	Device speed	Transfer interval	Positioning service latency	Positioning service availability	Positioning Accuracy
Electronic Shelf Label	1s	99%	NA	0.8kbit/s DL (note 1)	100 Bytes (note 2)	less than 1.5 million/km2 (inventory), less than 0.1 million/km2 (temperature sensor)	50m indoors	15,800 square meters	stationary	20 minutes to 2 hours [Note 3]	NA	NA	NA

Note 1: the user experience data rate is estimated based on transmission of 100 Bytes within 1 second.

Note 2: the message payload size is calculated based on the capacity of 50 Unicode characters for item description and pricing on the electronic shelf label.

Note 3: These are the typical intervals where electronic shelf labels equipped with sensors report temperature and humidity conditions of the shelf or refrigerators in accordance with food safety guidelines that vary across regions and organizations.

7 Consolidated potential requirements and KPIs

7.1 Consolidated potential requirements

Mapping table is used per each subclause for consolidated potential requirements.

7.1.1 Communication aspects of Ambient IoT devices

Table 7.1.1-1 Consolidated Requirements for communication aspects of Ambient IoT devices

CPR #	Consolidated Potential Requirement	Original PR #	Comment
CPR 7.1.1-1	The 5G system shall be able to support mechanisms to communicate: • between an Ambient IoT device and the 5G network using Ambient IoT direct network communication or Ambient IoT indirect network communication. • between an Ambient IoT device and Ambient IoT capable UE using Ambient IoT device to UE communication.	PR.5.2.6-001 PR.5.6.6-001 PR.5.11.6-001 PR.5.12.6-002 PR.5.13.6-001 PR.5.1.6-001 PR.5.4.6-001 PR.5.2.6-002 PR.5.12.6-002 PR.5.8.6-003 PR.5.27-004 PR.5.13.6-004 PR.5.3.6-001 PR.5.22.6-1, PR5.23.6-1, PR.5.24.6-1 PR.5.26.6-001	CPR 7.1.1-1 is updated to include communication modes. **Note 1**: This requirement applies to the 5G network and only UEs with the capability to communicate with an Ambient IoT device. **Note 2:** Examples of the communication between 5G network/Ambient IoT capable UE and Ambient IoT devices can include periodic sensor reporting or network-initiated inventory.
CPR 7.1.1-2	The 5G system shall be able to support 5G network or an Ambient IoT capable UE to communicate with a group of Ambient IoT devices simultaneously.	PR 5.2.6-001 PR.5.6.6-001 PR.5.11.6-001 PR.5.12.6-002 PR.5.13.6-001 PR.5.1.6-001 PR.5.4.6-001 PR.5.2.6-002 PR.5.12.6-002 PR.5.8.6-003 PR.5.27-004 PR.5.13.6-004 PR.5.3.6-001 PR.5.22.6-1, PR.5.23.6-1, PR.5.24.6-1 PR.5.26.6-001	
CPR 7.1.1-3	The 5G network shall support a mechanism to authorize an Ambient IoT capable UE to communicate with an Ambient IoT device.	PR.5.8.6-002 PR.5.12.6-001 PR.5.14.6-001 PR.5.21.6-003 PR.5.21.6-001	

7.1.2 Positioning/location of Ambient IoT devices

Table 7.1.2-1 Consolidated Requirements for positioning/location of Ambient IoT devices

CPR #	Consolidated Potential Requirement	Original PR #	Comment
CPR 7.1.2-1	The 5G system shall support location services for Ambient IoT devices (e.g., to locate Ambient IoT devices using absolute or relative positioning methods) Note: The intention is not to use Ambient IoT devices to locate other Ambient IoT devices.	PR.5.9.6-002 PR.5.12.6-004 PR.5.2.6-004 PR.5.8.6-001 PR 5.10.6-001 PR.5.21.6-002	

7.1.3 Management of Ambient IoT devices

Table 7.1.3-1 Consolidated Requirements for management of Ambient IoT devices

CPR #	Consolidated Potential Requirement	Original PR #	Comment
CPR 7.1.3-1	The 5G network shall support suitable management mechanisms for an Ambient IoT device or a group of Ambient IoT devices.		
CPR 7.1.3-2	The 5G system shall support a mechanism to: - disable the capability to transmit RF signals for one or more Ambient IoT device that is / are currently able to transmit RF signals - enable the capability to transmit RF signals for one or more Ambient IoT device that is / are currently disabled to transmit RF signals	PR.5.1.6-005 PR.5.16.6-003 PR.5.17.6-001	
CPR 7.1.3-3	Based on operator policy, the 5G system shall provide a suitable mechanism to permanently disable the capability of an Ambient IoT device or a group of Ambient IoT devices to transmit RF signals.		
CPR-7.1.3-4	Subject to operator policy and regulatory requirements, the 5G system shall support suitable mechanisms for the Ambient IoT device to move between one or more networks and countries.	PR.5.27-001	

7.1.4 Collected information and network capability exposure

Table 7.1.4-1 Consolidated Requirements for "collected information" and network capability exposure

CPR #	Consolidated Potential Requirement	Original PR #	Comment
CPR 7.1.4-1	Subject to user consent, operator policy and 3rd party request, the 5G system shall be able to get information from Ambient IoT devices (e.g. sensor data) and provide it to a trusted 3rd party via the 5G network.	PR.5.1.6-002 PR.5.12.6-003 PR.5.13.6-005 PR.5.14.6-003 PR.5.16.6-004 PR.5.18.6-1 PR.5.21.6-004 PR 5.2.6-003	
CPR 7.1.4-2	Subject to user consent, operator's policy and 3rd party request, the 5G system shall provide information about an Ambient IoT device or a group of Ambient IoT devices (e.g. position) to the trusted 3rd party via the 5G network.		
CPR 7.1.4-3	The 5G system shall enable an authorized 3rd party to instruct the 5G network to trigger a group of Ambient IoT devices in a specific area and which action the Ambient IoT devices need to perform when triggered (e.g. send ID, receive further information, send measurement value).	PR.5.3.6-003 PR.5.19.6-001 PR.5.27-003	
CPR 7.1.4-4	The 5G system shall provide suitable mechanisms to support communication between a trusted and authorized 3rd party and an Ambient IoT device or group of Ambient devices.	PR.5.20.6-001 PR.5.22.6-2, PR.5.23.6-2, PR.5.24.6-2 PR.5.30.6-001	

7.1.5 Charging

Table 7.1.5-1 Consolidated Requirements for charging

CPR #	Consolidated Potential Requirement	Original PR #	Comment
CPR. 7.1.5-1	The 5G system shall be able to collect charging information in a suitable way for using Ambient IoT services on per Ambient IoT device basis or a group of Ambient IoT devices (e.g., total number of communications per charging period).	PR.5.3.6-004 PR.5.3.6-005 PR.5.22.6-3, PR.5.23.6-3, PR.5.24.6-3	

7.1.6 Security and privacy

Table 7.1.6-1 Consolidated Requirements for security and privacy

CPR #	Consolidated Potential Requirement	Original PR #	Comment
CPR 7.1.6-1	The 5G system shall be able to provide a mechanism to protect the privacy of information (e.g., location and identity) exchanged during communication between an Ambient IoT device and the 5G network or an Ambient IoT capable UE.	PR.5.1.6-003 PR 5.3.6-002 PR.5.6-002 PR.5.13.6-002 PR.5.21.6-005 PR.5.8.6-004 PR.5.1.6-004 PR.5.8.6-005 PR.5.12.6-006 PR.5.13.6-003 PR.5.14.6-005 PR.5.16.6-001 PR.5.20.6-002	
CPR 7.1.6-2	The 5G system shall enable security protection suitable for Ambient IoT, without compromising overall 5G security protection.	PR.5.1.6-003 PR.5.3.6-002 PR.5.6-002 PR.5.13.6-002 PR.5.21.6-005 PR.5.8.6-004 PR.5.1.6-004 PR.5.8.6-005 PR.5.12.6-006 PR.5.13.6-003 PR.5.14.6-005 PR 5.16.6-001 PR.5.20.6-002	
CPR 7.1.6-3	Based on subscription and operator policies, the 5G system shall authorize an Ambient IoT capable UE to communicate with a specific Ambient IoT device.	PR.5.12.6-005 PR.5.14.6-004	

7.2 Consolidated potential KPIs

7.2.1 KPIs for inventory

Table 7.2.1-1 KPIs for inventory

Scenarios	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range (Note 1)	Service area dimension	Device speed	Transfer interval	Positioning service latency

inventory or asset management	Typically, seconds level	99%	NA	<2 Kbit/s	96/256 bits	<1,5 Million/km² indoor only (Note 2)	30~50m indoor, 200m~400m outdoor	1-10km²	3~10km/h	NA	NA

NOTE 1: The communication range is the communication distance between the ambient IoT device and the 5G network or between the ambi capable UE.
NOTE 2: The device density is much lower outdoors as only a subset of assets (e.g. stored indoors) will be in transit, and a much larger area

7.2.2 KPI for sensor data collection

Table 7.2.2-1 KPIs for sensor data collection

Deployment	Scenarios	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range (Note 1)	Service area dimension	Device speed	Transfer interval
Indoor	**Room envirionment monitoring (e.g. domicile, machinery)**	20-30 s	99 %	NA	<1 kbit/s	< 100 bits	1.5/m²	10-30m	NA	Stationary	NA
	Indoor agriculture and husbandry	>10 s	99.9%	NA	<1 kbit/s	Typically, < 1000 bits	1 per m²	30 - 200 m	6000 m²~30,000m²	Quasi-stationary	15 minutes to half an hour
Outdoor	**Smart grid**	1 s	99%	NA	< 1kbit/s	Typically, < 100 bytes	< 10,000 /km²	Typically 50-200 meters	[several km² up to 100 000 km²]	Stationary	5-15 min
	Outdoor husbandry and logistics	Typically, > tens of seconds	99%	NA	<500 bit/s	Typically, [< 100 bytes]	<5200 devices / km²	[300 m - 500 m]	430000 m²	Up to 3 km/h	15 min
	Smart city	10 s - 30 s	99%	NA	<1 kbit/s	Typically, < 100 bytes	<1000 devices / km²	300 m - 500 m	City wide including rural areas	Stationary	15 min

NOTE 1: The communication range is the communication distance between the ambient IoT device and the 5G network or between th capable UE.

7.2.3 KPI for tracking

Table 7.2.3-1 KPIs for tracking

Deployment	Scenarios	Max. allowed end-to-end latency	Communication Service Availability	Reliability	User-experienced data rate	Message Size	Device density	Communication Range (Note 1)	Service area dimension	Device speed	Transfer interval

Indoor	**Indoor tracking**	1s	99.9%	NA	<1 kbit/s	<1k bits	25/100 m²	10m indoor	200 m²	up to 3km/h	1 per hour	
Outdoor	**Outdoor tracking**	1s	99.9%	NA	<1 kbit/s	<1 kbits	<10 per 100 m²	500m	Up to the whole PLMN	up to 10 km/h	1 per hour	
NOTE 1: The communication range is the communication distance between the ambient IoT device and the 5G network or between t capable UE.												

7.2.4 KPI for actuator control

Table 7.2.4-1 KPIs for actuator control

Deployment	**Scenarios**	**Max. allowed end-to-end latency**	**Communication Service Availability**	**Reliability**	**User-experienced data rate**	**Message Size**	**Device density**	**Communication Range (Note 1)**	**Service area dimension**	**Device speed**	**Transfer interval**	**Po se la**
Indoor	**Indoor actuator control**	Several seconds	99%	NA	2 kbit/s	<100 Bytes	less than 1.5 million/km²	50m indoors	<250 m² for home, and 15,800 square meters for supermarket	stationary	20 minutes to 2 hours	NA
Outdoor	**Outdoor actuator control for large coverage**	Several seconds	99%	N/A	NA	128bit (DL)	NA	[500]m outdoors	40,000~4,000,000m²	Static	NA	NA
	Outdoor actuator control for medium coverage	Several seconds	99%	NA	<2 kbit/s	<200 bits	<20per 100 m²	200 m	City wide including rural areas	Static	NA	NA
NOTE 1: The communication range is the communication distance between the ambient IoT device and the 5G network or between th capable UE.												

8 Conclusion and recommendations

This technical report identifies use cases and potential new requirements related to ambient IoT. The resulting service requirements have been consolidated in clause 7.

It is recommended to consider the consolidated requirements identified in this TR for the subsequent normative work.

Annex A:
Ambient IoT availability scenarios

The Ambient IoT devices foresee to have different kind of communication pattern that could be dependent on power available for communication e.g. harvesting and the availability of storage capability or the specific use case. Following pattern are foreseen:

- **Normal operation**: In this scenario, Ambient IoT devices have power available continuously or at least for signicant amounts of time, either because there is continuous power harvesting or possibly in combination with limited energy storage (e.g. in a capacitor) to overcome momentary variations in power harvesting. The main effect of this scenario is that the processor and communications module in the Ambient IoT device can be continuously active. The communications module can listen to network at regular intervals to determine if there is mobile terminated traffic (e.g. trigger messages) and can transmit data when relevant.

- **Device triggered operation**: In this scenario, devices have power available only intermittently. The main effect of this scenario is that the Ambient IoT device can only be active for the short periods of time. And it is the Ambient IoT device that decides when to communicate with the network. It is possible that the Ambient IoT device is not able to listen to the network for mobile terminated traffic for very long periods of time. This has an impact on service aspects such as provisioning.

- **On demand operation**: in this scenario 5G network will wake up and trigger the device to communicate in a relevant manner. This scenario only considers the network to trigger the communication and the Ambient IoT device cannot determine when to communicate. Waking up of the Ambient IoT device can be combined with a trigger to perform a specific action (e.g. do measurement) or to communicate (e.g. send an identifier). Waking up can also imply that the Ambient IoT device starts listening to the network for further instructions.

These scenarios are not intended as definitions for device categories. The purpose is to enable discussion of service aspects that are associated with different scenarios.

Annex B:
Ambient power and energy storage

B.1 Typical ambient power

For ambient IoT devices, energy can be harvested from different types of ambient power sources. Some examples of which could be ambient power includes radio waves, solar energy/light, thermal energy and mechanical vibration etc.

RF Energy

RF-based energy can be harvested from radio waves ranging from 3 kHz to 300 GHz using a single-stage or multistage converter (i.e., rectifier circuit, as shown in Figure B-1). The amount of power that can be harvested depends on the source power, antenna gain, and the distance from the RF source. Ambient RF energy has a relatively low energy density, e.g., from several microwatts to tens of microwatts.

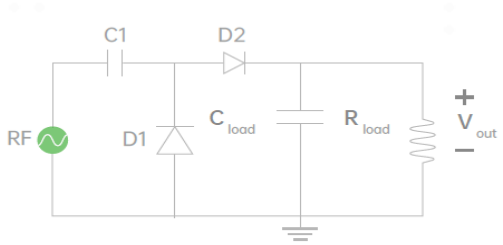


Figure B.1-1: rectifier circuit For RF energy harvesting

Based on the current state of art, the minimum RF power can be harvested is around -30dB 00. The conversion efficiency for RF Energy is listed in the table below.

Table B.1-1: Conversion efficiency for RF Energy 0

Efficiency（%）	Input power（dBm）	Center frequency（MHz）	Reflector unit
1.2	-14	950	0.3-μm CMOS convertor
5.1	-14.1	920	0.18-μm CMOS convertor
10	-22.6	906	0.25-μm CMOS convertor
11	-14	915	90-μm CMOS convertor
12.8	-19.5	900	0.18-μm CMOS , $CoSi_2$ - Si Schottky
13	-14.7	900	0.35-μm CMOS convertor
16.4	-9	963	0.35-μm COMS convertor
18	-19	869	0.5-μm CMOS convertor
26.5	-11.1	900	0.18-μm CMOS convertor
36.6	-6	963	0.35-μm CMOS convertor
47	-8	915	0.18-μm CMOS convertor
49	-1	900	Skyworks SMS7630 Si Schottky

The main advantage of RF-based energy harvesting is its availability in deployed environments and the fact that RF power is controllable (e.g., power can be sent by a transmitter on demand or periodically). Potential applications include logistics/warehouse, manufacturing, smart homes, health monitoring, and environmental monitoring etc.

Solar Energy/Light

Solar power/light can be transformed into electrical power using photovoltaic cells and it uses photovoltaic effect for energy harvesting with conversion efficiency of 10-40% 0. For the outdoor case, solar energy is one of the most common ambient power, it can supply inexhaustible clean energy and has high power density of up to 100 mW/cm^2 0.

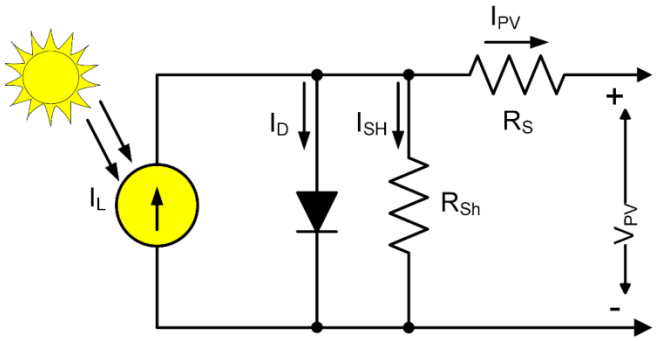


Figure B.1-2: The equivalent electrical circuit of a single diode solar PV cell [21]

Solar power is unstable, inconsistent, and intermittent. It is highly dependent on the atmospheric condition, surrounding obstructions, etc. It is available during daytime but inefficient on a cloudy day or during the night. Solar energy harvesting can be mainly used for outdoor environmental monitoring, agriculture, husbandry, transportation, etc.

For the indoor cases, light from the lighting equipment can be used. Although the power density is lower than solar, e.g., 100uw/cm^2, it is much stable and controllable. Energy harvested from light can be used for manufacturing, indoor environmental monitoring etc.

Thermal Energy

Thermal energy is another ambient power source that are available for lots of use cases. Electrical power is directly generated by exploiting the temperature difference in thermoelectric devices taking advantage of thermoelectric effects, such as the Seebeck effect or the Thomson effect. Thermoelectric generators have low efficiency (only about 5–6%) 0. The power density is 25~1000uw/cm^2 depending the environment condition.

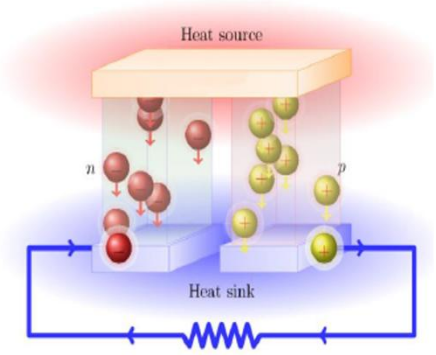


Figure B.1-3: Seebeck effect

Although with low conversion efficiency, thermal energy can be used in many outdoor applications or indoor cases as long as temperature difference or temperature fluctuation can be expected in the environment. For example, outdoor environmental monitoring, smart grid, agriculture, husbandry etc.

Mechanical Vibration

The piezoelectric effect generates electrical voltages or currents from mechanical strains, such as vibration or deformation. Typical piezoelectric-based energy harvesters keep creating power when there is a continuous mechanical motion, such as acoustic noises and wind, or they sporadically generate power for intermittent strains, such as human motion (walking, clicking a button, etc.). The volume of the piezoelectric power generators is relatively small and typical output power

density values of usual piezoelectric materials are around 250 μW/cm^3 but they can create more power when a motion or deformation is intense 0**Error! Reference source not found.**.

Figure B.1-4: Piezoelectric energy harvesting generator [25]

B.2 Energy storage

From the discussion above, it can be seen that kinds of ambient power have the following characteristics:

- For typical ambient power, it can be observed the power harvested is very limited, e.g. from 1uW to 100mW (per cm^2/cm^3).
- For some ambient power from artificial power source (e.g., light, RF waves), the power can be stable and constant. But for some other kind of ambient power such as solar, heat or vibration, the ambient power will be unstable (intermittent, not constant). It is impossible to use the ambient power as a direct power source for electronic devices.

Therefore, Energy storage element is needed, at least for some ambient IoT devices due to the following reasons:

- The energy storage element is able to stabilize and control the power output, smooth the fluctuation.
- It is able to collect the weak harvested power (e.g., in the level of micro Ampere or even nano Ampere) and provide the required higher peak discharge current (e.g., in the level of tens of micro ampere to hundreds of micro ampere) for the ambient IoT devices.
- Therefore, it makes it possible to use more kinds of ambient power sources for ambient IoT by using the energy storage element.

Note: It is still necessary to have no power storage for some types of ambient IoT devices (e.g. using energy from radio waves).

Capacitor can be considered as the basic energy storage elements for ambient IoT devices. Capacitor have limited power perseverance time and storage capacity, which can restrict the ambient IoT application.

For example, with a fully charged capacitor of 24uF, it can drive the ambient IoT devices for 3.6k bits communication (1.5V, 10uA and 1kbit/s are assumed).

- 24uF*1.5V= 36uC = 36 uAs

The communication is depending on the power consumption. Capacitor could be used in case power source are stable and constant.

The printed solid-state battery can be considered as an additional power storage with similar durability and higher capacity. With a solid-state battery of 1uAh@1.5V as example, it can drive the ambient IoT devices for even 360k bits communication (1.5V and 10uA).

Annex C:
Considerations when choosing harvesting source

When using energy harvesting as the main source of energy for low power devices there are a lot of design parameters to understand.

The harvesting sources needs to match the total energy consumption of the Ambioent IoT device.

The available energy needs to balance the energy demand for the communication, the computation and other present elements like sensors.

Other factors that matters are if it is possible to harvest continously or only part of the time, e.g. if solar cells are used as harvesting method it is only possible to harvest during day time.

Below there is a schematic picture of the different parts of the system for an Ambient IoT device. Not all parts are mandatory in an Ambient IoT device, for instance sensors and energy storage are not always present.

The power management will ensure that the correct voltage will be available for the sensor, communication and compute parts.

Ambient IoT devices can be both with and without some kind of energy storage and there are many types of storage components, from capacitors, super capacitors to more battery types of storages.

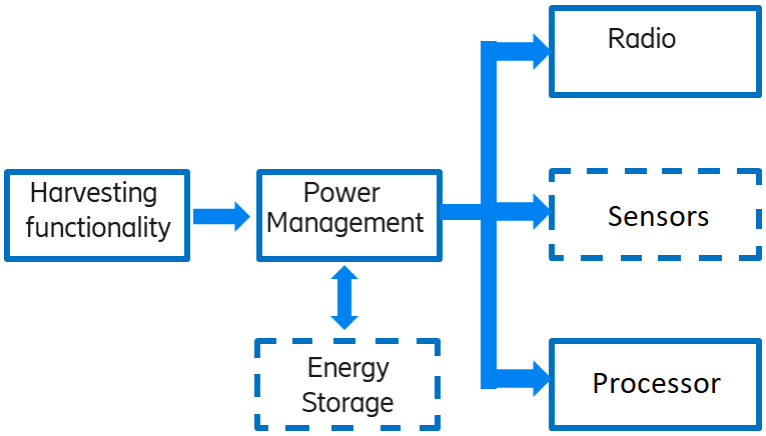


Figure C-1 Example of parts in an Ambient IOT device

The following example assumes that some storage capability is available in the device

Below shows two examples with same type of harvesting but with different length of the active time periods that the sensor/compute/communication system needs to be active. In the first example, the Ambient IoT device transmits in shorter bursts and with a higher power compared to second example. In the second example the time for the sensor/compute/communication is active longer. This would mean that, given everything else is equal, the available power for the active period in example two is lower.

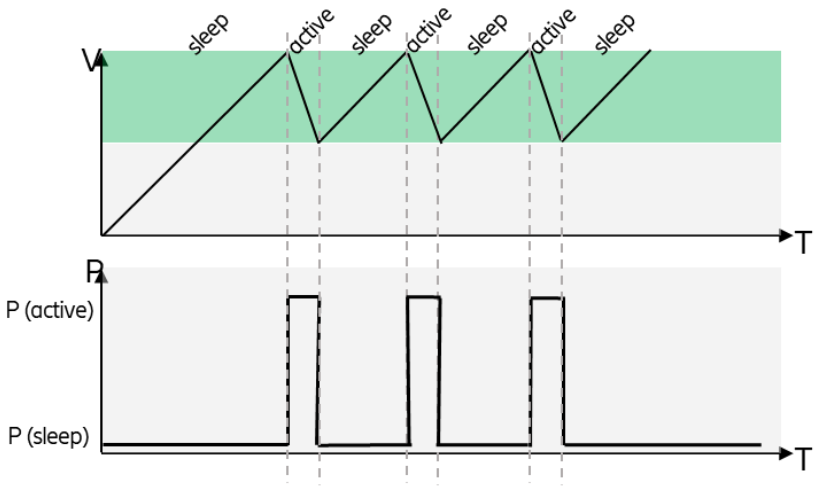


Figure C-2 Example where the active period for the device is relative short

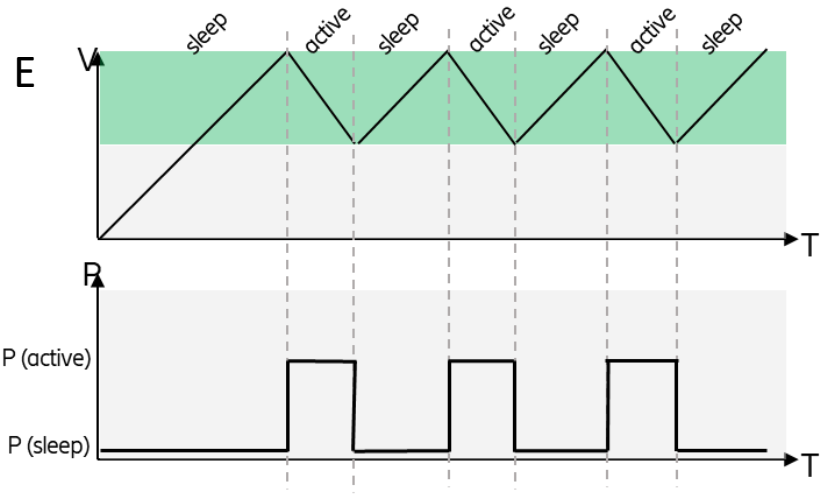


Figure C- 3 Example where the active period for the device is relative long

There is therefore not a simple formula to determine the available power for the sensor/compute/communication functionsfrom the characteristics of the energy harvesting method.

Annex D:
Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2022.05	SA1#98e	S1-221254	-	-	-	Initial Skeleton	0.0.0
2022.05	SA1#98e					Inclusion of: S1-221255; S1-221256; S1-221257; S1-221258; S1-221259; S1-221260; S1-221261; S1-221262	0.1.0
2022.09	SA1#99e		-	-	-	Add agreed use cases, power scenarios etc. Inclusion of: S1-222362; S1-222363; S1-222364; S1-222365; S1-222366; S1-222367; S1-222368; S1-222369; S1-222370; S1-222371; S1-222372; S1-222373; S1-222374; S1-222375; S1-222376; S1-222377; S1-222378; S1-222379; S1-222380; S1-222122	0.2.0
2022.11	SA1#100		-	-	-	Update of names, formats. Corrections. Inclusion of: S1-223207	0.2.1

2022.11	SA1#100		-	-	-	Update of requirements, more use cases, annex and others. The KPI tables are updated in unified format base on S1-223799. Inclusion of: S1-223207; S1-223321; S1-223698; S1-223545; S1-223546; S1-223356; S1-223357; S1-223360; S1-223361; S1-223480; S1-223481; S1-223363; S1-223364; S1-223583; S1-223556; S1-223582; S1-223235; S1-223571; S1-223700; S1-223573; S1-223562; S1-223702; S1-223703; S1-223704; S1-223705; S1-223555; S1-223706; S1-223707; S1-223708; S1-223229; S1-223699; S1-223166; S1-223737	0.3.0
2023.02	SA1#101		-	-	-	New use cases, updates of the existing use cases, clear up of FFS etc Inclusion of: S1-230654; S1-230655; S1-230757; S1-230758; S1-230662; S1-230663; S1-230526; S1-230123; S1-230759; S1-230609; S1-230610; S1-230760; S1-230761; S1-230231; S1-230613; S1-230614; S1-230615; S1-230238; S1-230616; S1-230621; S1-230665; S1-230763; S1-230619; S1-230800; S1-230765	1.1.0

2023.05	SA1#102		-	-	-	updates of the existing use cases, clear up of FFS/Editor note etc Inclusion of: S1-231400; S1-231293; S1-231401; S1-231292; S1-231403; S1-231405; S1-231812; S1-231682; S1-231407; S1-231459; S1-231409; S1-231402; S1-231410; S1-231460; S1-231411; S1-231300; S1-231413; S1-231227; S1-231414; S1-231461; S1-231813; S1-231464	1.2.0
2023.08	SA1#103		-	-	-	updates of the existing use cases, clear up of FFS/Editor note, remove the brackets of KPI value, CPR and KPI consolidation, conclusion etc Inclusion of: S1-232178; S1-232033; S1-232315; S1-232379; S1-232349; S1-232303; S1-232279; S1-232304; S1-232305; S1-232149; S1-232157; S1-232355; S1-232388; S1-232359; S1-232387; S1-232310; S1-232380; S1-232172; S1-232381; S1-232389; S1-232313; S1-232391; S1-232644; S1-232619; S1-232392; S1-232393; S1-232645; S1-232647; S1-232648; S1-232352	1.3.0
2023.09	SA#101	SP-231015				MCC Clean-up for presentation for SA approval	2.0.0
2023.11	SA1#104					Final KPI consolidation etc. Inclusion of: S1-233401; S1-233402; S1-233403; S1-233404	2.1.0
2023.12	SA#102	SP-231404				MCC clean-up for 2nd attempt for SA approval	2.2.0

2023.12	SA#102	-				Approved by SA#102	19.0.0